
COVERAGE-INVERSION: A CALIBRATION-TRANSPARENT ORACLE PRIMITIVE FOR TOKENIZED RWAS

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Abstract

Tokenised real-world assets on public blockchains trade continuously, but their underlying price-discovery venues do not, leaving roughly two-thirds of wall-clock time during which on-chain protocols must value collateral against a closed reference market. We invert the conventional point-plus-confidence oracle interface: target coverage $\tau \in (0, 1)$ is a published, parametric product input rather than an implicit by-product of the model, the served band carries a per-read calibration receipt whose realised rate is independently verifiable from public data, and a locally-weighted Mondrian split-conformal architecture maps an asset s , regime label $r = \rho(\mathcal{F}_t)$, per-symbol pre-Friday scale $\hat{\sigma}_s(t)$, and target τ to a calibrated band around a factor-adjusted point. The primitive carries a three-property contract: auditability of the deployment artefact (P_1); conditional empirical coverage at τ per symbol and per regime (P_2); and per-regime serving efficiency at deployment-tuned parameter budgets (P_3).

We evaluate the architecture on 5,996 weekend prediction-window rows across ten US-listed tickers (639 weekends, 2014-01-17 through 2026-04-24) with a 1,730-row 2023+ held-out slice. **The headline empirical claim is held-out via 10-fold leave-one-symbol-out CV — fit $c(\tau)$ on nine symbols, evaluate on the tenth — at $\tau = 0.95$: realised coverage 0.9497 ± 0.0128 across the held-out symbol distribution, all 10 LOSO bands pass Kupiec.** Pooled realised coverage on the full 2023+ slice is 0.950 at 370.6 bps half-width, consistent with the deployment-tuned target — but the agreement at 0.950 is by construction (only one of 16 deployment scalars, $c(0.95) = 1.079$, carries meaningful OOS information; the other three $c(\tau) \in \{1.000, 1.000, 1.003\}$ are essentially identity at the four served anchors $\{0.68 \approx 1-\sigma, 0.85, 0.95, 0.99\}$), so the pooled number is not held-out evidence. The deployment carries 16 scalars: 12 per-regime conformal quantiles trained on standardised residuals on pre-2023 data, four near-identity OOS-fit multiplicative bumps, and zero walk-forward shifts. A handful of implicit hyperparameters are pinned a priori (EWMA half-life 8, ≥ 8 -past-observation warm-up, three Mondrian regimes, top-quartile VIX threshold, 2023-01-01 split date); the 16-scalar count is the trainable / tuned budget, not a complete degrees-of-freedom statement. Three orthogonal stability checks support the LOSO read: per-symbol Kupiec passes on all 10 symbols at $\tau = 0.95$, four-anchor split-date sensitivity refits $c(\tau)$ per anchor and passes Christoffersen at every (split $\times \tau$) cell, and calendar sub-period coverage holds across $\{2023, 2024, 2025, 2026\text{-YTD}\}$. Against parametric baselines on the same slice the conformal architecture carries **40/40 Kupiec passes** across the (symbol $\times \tau$) grid versus 31/40 for GARCH- t and 23/40 for GARCH-Gaussian; at matched 95% realised coverage it ties GARCH- t on width.

The architectural claim that per-symbol scale standardisation closes per-symbol calibration error is corroborated under a four-DGP simulation study with known ground truth (per-symbol Kupiec pass-rate 98.6–99.9% across stationary, regime-switching, drift, and structural-break specifications, against the unweighted-Mondrian comparator’s 29–31%). Joint-tail behaviour for portfolio consumers is empirically characterised, not assumed: across 173 OOS weekends at $\tau = 0.95$, empirical $P(k_w \geq 3)$ on a 10-symbol panel is 4.62% (95% bootstrap CI $[1.73, 7.51]$) versus the binomial-independence reference of 1.15%; the worst-observed weekend (Friday 2024-08-02 $\rightarrow$ Monday 2024-08-05, the BoJ yen-carry-unwind) sat at $k_w = 10$ across all 10 symbols at $\tau = 0.85$. The standard *per-anchor calibrated, not full-distribution calibrated* disclosure is consequently turned into a measurable reserve-guidance threshold ($k^* = 3$ at

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$\tau = 0.95$, with 0/24 sub-period $\times$ threshold-convention stability rejections). The endpoint contract is what the served band claims: on a 24/7 stock-perp reference ($n = 118$ symbol-weekends) the architecture carries a +16.1 pp endpoint-vs-path gap at $\tau = 0.95$ — what an honestly-calibrated endpoint band exhibits when intra-weekend variance exceeds endpoint variance — with the §10.1 path-fitted conformity score as the methodology fix for continuous-consumption consumers.

The $\hat{\sigma}_s$ rule (per-symbol EWMA, weekend half-life 8) was selected from a five-variant ladder under a pre-registered three-gate criterion with Benjamini–Hochberg multi-testing correction across an 80-cell split-date Christoffersen grid; under BH correction at FDR=0.05 the per-cell evidence does not statistically distinguish the variants, so the deployment decision rests on the bootstrap CI on pooled width (no multi-test exposure, statistically significantly narrower at $\tau \in \{0.85, 0.95, 0.99\}$ at preserved calibration). A weekly forward-tape harness against a content-addressed frozen artefact is operational and accumulating ($N = 1$ at submission, preliminary by the harness’s own threshold of $N \geq 4$); it will become load-bearing held-out evidence as N matures. The reference implementation is open-sourced as a Python research stack with a Rust + Anchor production stack (§8).

1 Introduction

Tokenised equities and other real-world assets (RWAs) on public blockchains trade 24/7. Their underlying price-discovery venues do not. A US equity listed on NASDAQ is open 6.5 hours per weekday — roughly 32% of wall-clock time. In the other 68% the price continues to move (earnings, macro prints, overseas sessions) but the venues that establish reference prices are closed. The JELLYJELLY incident on Hyperliquid (March 2025) drove ~\$12M of HLP losses through a manipulable single-venue oracle path; a Volcano Exchange post-mortem of a tokenised-asset perp episode documents \$9.21M in Hyperliquid liquidations exceeding the Binance volume that fed the oracle, and identifies the propagation mechanism in plain language: *uncertainty present in the input was not represented in the output*. The structural fix is symmetric: an oracle whose product contract makes the input uncertainty a first-class output a downstream consumer can read on every block. Any DeFi protocol that accepts tokenised RWAs as collateral or as inputs to an automated market maker must answer at every block: *with the underlying venue closed, what is the fair price — and how uncertain is it?*

The question recurs across the broader RWA wave moving on-chain in 2025–2026: tokenised equities (xStocks on Solana, Backed-issued tokens), treasuries (BlackRock BUIDL, Ondo OUSG; >\$5B AUM as of Q1 2026), commodities (Paxos PAXG), and the next wave of credit and FX. Every asset class with a continuous off-hours information set inherits the same closed-market-pricing problem and would benefit from the same calibration-transparent risk-reporting primitive. The methodology fits asset classes with such a signal; it does not fit instruments without one (illiquid commodities, pure NAV real estate). §9.8 enumerates the generalisation map.

1.1 How existing oracles answer, and what they leave unanswered

No incumbent RWA oracle deployed at scale today publishes a calibration receipt verifiable against public data at the aggregate feed level — a statement of the form “*over N historical prediction windows, our 95% band contained the realised price N_{in} times.*” Five archetypes are surveyed below; each resolves the *what*, and none publishes the rolling dataset from which an aggregate-level realised-coverage claim could be independently reproduced. The contribution we make in this paper is therefore categorical, not bandwidth-comparative: a verifiable per-read calibration receipt is what the literature does not currently provide. We do not report numerical width comparisons against incumbent surfaces, because a coverage-vs-sharpness comparison is only well-defined between methods that emit a calibrated coverage band — incumbent surfaces do not, and any such comparison would rest on a consumer-supplied wrap whose calibration is not the publisher’s claim.

Archetype	Representative	What a consumer sees
Stale-hold	Chainlink Data Streams v10 [Labs, 2025]	Last trade held forward + binary stale/live flag; no
Synthetic-marker	Chainlink Data Streams v11 [Labs, 2026]	bid/ask/confidence on the wire Weekend bid/ask with a .01 suffix at 100% incidence on SPYx/QQQx/TS�Ax — generated bookends, not real quotes

Archetype	Representative	What a consumer sees
Publisher-dispersion metric	Pyth regular [Network, 2021, 2022]	σ across permissioned publishers; not an aggregate coverage claim;
Executable-overnight	Pyth Pro / Blue Ocean [Network, 2026, 2025]	collapses to near-zero at venue close Real 24/5 book Sun–Thu 8 PM – 4 AM ET; does not cover the canonical xStock weekend; paid enterprise
Undisclosed methodology	RedStone Live [RedStone, 2026]; v10 tokenizedPrice	Continuous closed-market value; methodology described qualitatively only

Two structural observations. (i) Pyth’s documentation [Network, 2022] recommends *individual publishers* calibrate to ~95%, but this is per-publisher self-attestation, not a verifiable property of the aggregate. (ii) Cong et al. [Cong et al., 2025] document that off-hour tokenised-stock returns anticipate Monday opens, so the on-chain SPL xStock is informationally distinct from its underlier in exactly the closed-market window the oracle covers; feeds pricing the underlier ticker carry an additional instrument-mismatch gap.

1.2 Coverage as a first-class API parameter

The design point of this paper is an oracle whose *product contract* is a calibration statement. Instead of publishing a single band and leaving its coverage level implicit, we treat target coverage $\tau \in (0, 1)$ as a published, parametric product input and serve the band that has *empirically* delivered coverage τ on a rolling historical sample, stratified by asset, by pre-publish regime, and rescaled by a per-symbol pre-Friday scale. The empirical calibration claim is made at four audited anchors $\mathcal{T}_{\text{anch}} = \{0.68 \approx 1-\sigma, 0.85, 0.95, 0.99\}$ — the lower anchor is set at the approximate one-standard-deviation point so consumers comparing against a Gaussian default have a familiar reference; the API accepts any $\tau \in [0.68, 0.99]$ and linearly interpolates the per-regime quantile and the $c(\tau)$ bump between anchors, so an off-anchor τ inherits an *interpolated* (not separately validated) band — see §9.10 for the disclosure and §10.4 for the denser-anchor-grid follow-up. The deployed wire format publishes one band per symbol per Friday with τ stamped on the receipt.

For a point estimator $\hat{P}_{\text{Mon},r}(s)$, a regime classifier ρ , a strictly-pre-Friday per-symbol scale $\hat{\sigma}_s(t) = \text{EWMA}_{\text{HL}=8}(\{|P_{\text{Mon},t'} - \hat{P}_{\text{Mon},t'}|/P_{\text{Fri},t'} : t' < t\})$, and a per-regime conformal quantile $q_r(\tau)$ trained on the *standardised* score $|P_{\text{Mon}} - \hat{P}_{\text{Mon}}|/(P_{\text{Fri}} \cdot \hat{\sigma}_s)$, the deployed architecture (instantiated in §4) serves at consumer target τ via

$$[\hat{P}_{\text{Mon},r}(s) - c(\tau) q_r(\tau) \hat{\sigma}_s(t) \cdot P_{\text{Fri},t}, \hat{P}_{\text{Mon},r}(s) + c(\tau) q_r(\tau) \hat{\sigma}_s(t) \cdot P_{\text{Fri},t}],$$

where $c(\tau) \in \{1.000, 1.000, 1.079, 1.003\}$ at the four anchors is a near-identity OOS-fit multiplicative bump that closes any residual train-OOS distribution-shift gap. **Sixteen deployment scalars total: 12 trained per-regime quantiles on standardised residuals + 4 OOS-fit $c(\tau)$ + 0 walk-forward shifts** — per-symbol scale standardisation absorbs the cross-split realised-coverage variance that would otherwise call for a structural-conservatism shift. Implicit hyperparameters pinned a priori (EWMA half-life 8 weekends, ≥ 8 -past-observation warm-up, three Mondrian regimes, top-quartile VIX threshold, 2023-01-01 split date) sit outside the 16-scalar count, which captures the trainable / tuned budget rather than a complete DOF statement.

Two properties follow that are absent from incumbents: (1) **Auditability** — the 16 scalars and the per-Friday $(\hat{P}_{\text{Mon},r}, \hat{\sigma}_s)$ are deterministic functions of public data; any third party can reconstruct the artefact and verify a served band. (2) **Served-coverage receipts** — every read exposes $(\tau, c(\tau), q_r(\tau), \hat{\sigma}_s(t), r)$ alongside the band, so a protocol integrator can log it, backtest, and challenge the provider if realised rate drifts. The framework traces to Christoffersen [Christoffersen, 1998] and SR 11-7 [of Governors of the Federal Reserve System & Office of the Comptroller of the Currency, 2011] / SR 26-2 [Board of Governors of the Federal Reserve System, 2026] model-risk guidance; what is new is the application to a decentralised oracle with a consumer-facing SLA on realised coverage rather than on point accuracy, and the per-symbol locally-weighted standardisation that closes per-symbol calibration on heterogeneous asset panels.

1.3 What this paper shows

We evaluate the deployed locally-weighted Mondrian split-conformal architecture (full specification in §4) on 5,996 weekend prediction-window rows (639 weekends $\times$ 10 symbols) spanning 2014-01-17 through 2026-04-24: seven US equities/ETFs (SPY, QQQ, AAPL, GOOGL, NVDA, TSLA, HOOD), gold (GLD), long treasuries (TLT), and a Bitcoin-proxy equity (MSTR). Each window runs Friday 16:00 ET to Monday 09:30 ET. The contribution is the *primitive* — the artefact, its serving-time lookup, and the audit receipts — together with empirical evidence that the architecture delivers per-anchor *and* per-symbol calibration on a decade-scale public-data panel.

Headline empirical claims:

- **Held-out coverage at $\tau = 0.95$ via 10-fold leave-one-symbol-out CV:** 0.9497 ± 0.0128 . Each held-out symbol's $c(\tau)$ is fit on the other nine symbols' OOS slice and never sees the held-out symbol's data, so the held-out realised coverage is genuinely out-of-fit. All 10 LOSO bands pass Kupiec at $\tau = 0.95$. This is the paper's leading calibration figure. Pooled realised coverage on the deployment-tuned 2023+ slice (1,730 rows, 173 weekends) is 0.950 at mean half-width 370.6 bps, consistent with the deployment-tuned target; the agreement at 0.950 is a *consequence* of fitting $c(0.95) = 1.079$ on this same OOS slice (so the pooled Kupiec $p_{uc} = 0.956$ follows by construction and is *not* held-out evidence — only $c(0.95)$ carries meaningful OOS information; the other three $c(\tau)$ are essentially identity, and the 12 trained per-regime quantiles are frozen on 4,186 pre-2023 rows). **Three orthogonal stability checks** complement LOSO on the deployed-fit slice (each refits the schedule within its slice, so they are stability-of-fit reads rather than held-out validation): (i) all 10 symbols pass per-symbol Kupiec with violation rates in $[3.5\%, 6.9\%]$; (ii) four-anchor split-date sensitivity refits $c(\tau)$ at each split anchor and holds realised $\tau = 0.95$ coverage within $\{0.9539, 0.9520, 0.9503, 0.9504\}$ across $\{2021, 2022, 2023, 2024\}$, with Christoffersen passing every cell at $\alpha = 0.05$; (iii) a six-split walk-forward selects $\delta(\tau) \equiv 0$ with worst-split coverage $\geq \tau$ at every powered split. Christoffersen lag-1 $p_{ind} = 0.603$ on the deployed split is reported in §6.3.1 with the ~ 86 -violation power caveat; the §6.3.6 DQ is the more sensitive test on the residual and rejects pooled — localising to the cross-sectional within-weekend common-mode. We use $\tau = 0.95$ as the paper's primary oracle-validation target. Mapping the served band to a particular operating τ for a given lending protocol is downstream of this work and not addressed here.
- **Calendar sub-period stability.** Across $\{2023, 2024, 2025, 2026\text{-YTD}\}$ the realised $\tau = 0.95$ coverage stays in $[0.942, 0.967]$ with Kupiec $p \geq 0.054$ in every cell; the architecture carries 2 rejections across the 16-cell sub-period $\times \tau$ grid, both *over*-coverage in 2025 (not the failure mode that warrants a deployment alarm). Christoffersen lag-1 independence is not rejected in any sub-period $\times \tau$ cell.
- **Per-symbol calibration vs parametric baselines.** Across the 40-cell (symbol $\times \tau$) grid: this paper's architecture **40/40**, GARCH- t 31/40, GARCH-Gaussian 23/40, constant-buffer baseline 12/40. At matched 95% realised coverage GARCH- t widens to ≈ 378 bps (statistically tied with this paper's 370.6) but undercovers Kupiec at every $\tau < 0.99$; GARCH-Gaussian undercovers at every $\tau < 0.99$ and over-covers at $\tau = 0.99$. The GARCH- t baseline carries an internal disclosure: NVDA's t-MLE pushes $\hat{\nu} \rightarrow 2.50$ (variance-undefined boundary) and the runner falls back to Gaussian, so 4 of the 31 GARCH- t pass cells use the Gaussian-fallback baseline; the §6.4.2 / §6.4.3 tables flag this inline.
- **Joint-tail empirical distribution for portfolio consumers.** At $\tau = 0.95$ on a 10-symbol panel, empirical $P(k_w \geq 3) = 4.62\%$ (95% bootstrap CI $[1.73, 7.51]$) versus the binomial-independence reference of 1.15%. The worst-observed weekend — Friday 2024-08-02 $\rightarrow$ Monday 2024-08-05, the BoJ yen-carry-unwind — sat at $k_w = 10$ across all 10 symbols at $\tau = 0.85$ (MSTR -27.4% , HOOD -17.8% , NVDA -14.2%). Reserve-guidance threshold $k^* = 3$ at $\tau = 0.95$ carries 0/24 sub-period $\times$ threshold-convention stability rejections.
- **Architectural claim corroborated under simulation.** Under four DGPs spanning homoskedastic, regime-switching, drift, and structural-break specifications (10 synthetic symbols $\times$ 600 weekends $\times$ 100 reps, seed = 0), the deployed architecture's per-symbol Kupiec pass-rate at $\tau = 0.95$ is 98.6–99.9% while the unweighted-Mondrian comparator's collapses to 29–31% across the same DGPs — reproducing the per-symbol bimodality of an unweighted Mondrian fit on synthetic data with known ground truth and confirming that per-symbol $\hat{\sigma}_s$ standardisation closes it. The simulation evidence is corroborative: it post-dates the per-symbol bimodality observation on real unweighted-Mondrian data (§6.4.2) but pre-dates the empirical fix on real data, and it adds a known-DGP control where the failure mechanism can be isolated. A panel-admission threshold of $N \geq 200$ weekends is established for newly-listed symbols under the production threshold.
- **Forward-tape harness operational; held-out evidence accumulating.** A content-addressed frozen artefact (lwc_artefact_v1_frozen_20260504.json, freeze date 2026-05-04) is evaluated weekly via a launchd-

driven harness against forward weekends as they accumulate; a sibling variant-comparison runner carries held-out re-validation of the $\hat{\sigma}_s$ choice without re-selecting on forward data. **At submission** $N = 1$ **weekend** (reports/m6_forward_tape_1weekends.md, 2026-05-01) — preliminary by the harness’s own threshold ($N \geq 4$ to clear the preliminary banner; $N \geq 13$ for moderate forward-Christoffersen power). The held-out backbone for the present submission is the §6.3.3 split-date sensitivity, leave-one-symbol-out CV, and calendar sub-period stability, not the forward-tape harness.

- **Diagnostics define the next-generation roadmap.** Two residuals carry forward to §10. (i) Berkowitz rejects pooled — bands are *per-anchor calibrated*, not full-distribution calibrated — and the rejection localises to *cross-sectional within-weekend common-mode* ($\hat{\rho}_{\text{cross}} = 0.354$, orthogonal to $\hat{\sigma}_s$); points at full-distribution conformal upgrades and a Friday-observable common-mode partial-out (§10.2). (ii) On a 24/7 stock-perp reference ($n = 118$ symbol-weekends, §6.5) the architecture’s endpoint-vs-path gap is +16.1 pp at $\tau = 0.95$ — an honestly-calibrated endpoint band exhibits a structural shortfall on a sample whose intra-weekend variance is larger than its endpoint variance; the path-fitted conformity score is the methodology fix (§10.1).

These claims are about the oracle interface and serving-time calibration. We do not address the downstream decision-theoretic question — which τ a particular lending protocol should consume — in this paper.

1.4 Contributions and structure

C1. Coverage inversion as an oracle primitive (§3). **C2.** A 12-year locally-weighted Mondrian split-conformal-by-regime calibration on public data, with a 16-scalar artefact rebuildable by any third party (§4, §5). **C3.** Held-out-quantile, deployment-calibrated Kupiec + Christoffersen validation in the form required by institutional model-risk management (§6), per-symbol Kupiec across 40 (symbol $\times \tau$) cells against parametric baselines, calendar sub-period stability, a four-DGP simulation study confirming the per-symbol fix is structural, and a forward-tape harness against a content-addressed frozen artefact. **C4.** Joint-tail empirical characterisation of portfolio breaches (§6.3.4–§6.3.5) — overdispersion measurement ($\text{Var}(k_w)/\text{Var}_{\text{binom}} = 2.32$ at $\tau = 0.95$, where the binomial-independence reference is $\text{Binom}(10, 0.05)$), the k_w reserve-guidance threshold and its sub-period stability (0/24 cells reject), and the worst-weekend vignette (2024-08-05 BoJ unwind, $k_w = 10$) — operationalising the per-anchor-vs-full-distribution residual into a measurable consumer-facing handle no incumbent oracle publishes. **C5.** A three-layer ablation with bootstrap CIs (§7) isolating per-regime conformal as the legacy-narrowing component, per-symbol $\hat{\sigma}_s$ standardisation as the per-symbol-bimodality fix, and the $\hat{\sigma}_s$ selection procedure under multi-test correction.

We do not claim point-accuracy or absolute bandwidth dominance over any incumbent surface; the contribution is the calibration receipt and its per-symbol generalisation, not the band width.

§2 surveys related work; §3 gives the problem statement; §4 the methodology; §5 the data and regime labeler; §6 calibration results; §7 the ablation; §8 the serving-layer system; §9 limitations; §10 extensions.

2 Related Work

Our contribution sits at the intersection of three literatures that have, to our knowledge, never been brought together in a deployed system: decentralised-oracle design, cross-venue price discovery, and the calibration / conformal-prediction framework for interval forecasts. A fourth — institutional model-risk-management guidance — supplies the disclosure vocabulary we adopt.

2.1 Decentralised oracle designs

The deployed oracle landscape is organised around an *integrity* primitive: can the oracle deliver a faithful value on-chain without a trusted intermediary? Chainlink’s decentralised-oracle-network architecture [Breidenbach et al., 2021] aggregates off-chain reports from a node-operator committee, with Data Streams [Labs, 2024] providing pull-based delivery and an explicit marketStatus flag — shipped in two co-existing schemas, v10 [Labs, 2025] and v11 [Labs, 2026]. Pyth’s Solana surface [Network, 2021, 2022] takes a first-party-publisher approach: each permissioned publisher submits a price and confidence interval; the aggregate rule assigns three votes per publisher (price, price $\pm$ confidence) before taking the median. Pyth Pro [Network, 2026] layers subscriber-customisable cadence and the Blue Ocean ATS overnight integration; since September 2025 Pyth has been the exclusive on-chain distributor of Blue Ocean’s executable book through end-2026 [Network, 2025], covering 8:00 PM – 4:00 AM ET Sun–Thu (the canonical xStock weekend remains uncovered). Switchboard [Switchboard, 2023] generalises aggregation into permissionless queues; RedStone Live [RedStone, 2026] blends institutional feeds with perpetual-market data; optimistic oracles UMA [Project, 2020] and Tellor [Tellor, 2023] replace continuous aggregation with dispute-and-stake.

None of these designs publishes a *calibration claim* on the served value. The deployed-surface evidence is categorical at the wire: v11’s weekend bid carries a synthetic .01-suffix at 100% incidence on three of four mapped xStocks; v10 carries no bid/ask/confidence field; Pyth’s (price, conf) reports low publisher disagreement at venue close. Pyth’s documentation [Network, 2022] asserts ~95% coverage at the *publisher* level, not as a property of the aggregate; Chainlink’s tokenizedPrice and RedStone Live describe their off-hours rules qualitatively. Optimistic oracles guarantee *eventual* correctness under economic security but do not define a distributional statement conditional on time-of-week or realised volatility. Because none of these surfaces emits a calibrated coverage band, the contribution is necessarily categorical: a coverage-vs-sharpness numerical comparison is not well-defined against them and is therefore not reported.

The academic literature has similarly concentrated on integrity. Angeris and Chitra [Angeris and Chitra, 2020] prove conditions under which CFMMs are incentive-compatible as price oracles. Eskandari et al.’s SoK [Eskandari et al., 2021] catalogues oracle-manipulation attacks; the framing is adversarial (*did the attacker corrupt the value?*) rather than statistical (*over N windows, did the served band contain the truth at the claimed rate?*). The BIS bulletin [Duley et al., 2023] frames the tension as decentralisation-versus-efficiency; on-chain consumers operate under adversarial ordering [Daian et al., 2020], amplifying the cost of any systematic band miscalibration.

Our contribution is orthogonal. The integrity primitive asks whether $\hat{P}_t \approx P_t^{\text{true}}$; the calibration primitive asks, given the oracle’s own band, whether it contains P_t^{true} at the claimed rate. An oracle can be integrity-perfect and calibration-meaningless: Chainlink faithfully reports Friday’s 16:00 close on Sunday at 14:00, but that value is not asserted to be a 95%-coverage estimate of Monday’s open. We invert the interface so τ is a request parameter and q_{served} is the receipt.

2.2 Cross-venue price discovery

The base forecaster underlying our calibration surface is not methodologically novel; its point estimate is a factor-switchboard scaling of Friday’s close by a weekend futures return. The choice of futures as the leading venue applies Hasbrouck’s information-share methodology [Hasbrouck, 1995, 2003], which establishes that most price discovery in the S&P 500 and Nasdaq-100 occurs in E-mini futures — the statement we rely on when scaling SPY and QQQ Friday closes by ES and NQ weekend returns. Earlier work [Stoll and Whaley, 1990, MacKinlay and Ramaswamy, 1988] established intraday lead-lag and mispricing-adjustment dynamics in index-futures markets; Gonzalo–Granger [Gonzalo and Granger, 1995] is the conventional complement.

The Friday 16:00 ET to Monday 09:30 ET weekend is the longest contiguous non-trading interval in the US calendar. French [French, 1980] documents the Monday-effect, establishing the weekend as a distinct return-generating regime. Barclay and Hendershott [Barclay and Hendershott, 2003] show after-hours price discovery is non-trivial, so a stale-hold feed is systematically miscalibrated; their companion paper [Barclay and Hendershott, 2004] shows after-hours effective spreads are 3–4× regular-hours, reflecting greater adverse selection — exactly the window in which a protocol holding an RWA as collateral is most exposed. For ETFs, Madhavan and Sobczyk [Madhavan and Sobczyk, 2016] characterise creation/redemption-driven discovery, validating our ETF-class instruments within a unified frame.

A newer thread addresses the structural migration motivating this work. Makarov and Schoar [Makarov and Schoar, 2020] document cross-venue arbitrage in cryptocurrency markets; Lehar and Parlour [Lehar and Parlour, 2025] show Ethereum DEXs do not exhibit long-lived arbitrage relative to centralised benchmarks. Cong et al. [Cong et al., 2025] provide the first large-scale study of tokenised stocks and find off-hour token returns anticipate Monday opens — direct empirical motivation: weekend price discovery for tokenised equities is migrating to crypto-native venues, so a calibrated band conditioned on the on-chain continuous clock becomes a first-class protocol need.

2.3 Calibrated and conformal prediction

Kupiec [Kupiec, 1995] proposed the unconditional-coverage LR test for VaR; Christoffersen [Christoffersen, 1998] generalised to conditional coverage, decomposing into Kupiec plus an independence test on the hit sequence. The p_{UC} and p_{ind} columns we report are these tests; we use the pair rather than the joint because they correspond in banking practice to the Basel traffic-light framework [on Banking Supervision, 1996]. The density-forecast generalisation [Diebold et al., 1998] evaluates whether PITs are i.i.d. uniform; Berkowitz [Berkowitz, 2001] refines this to a small-sample-robust joint LR. Our coverage check at fixed τ is a marginal slice of the PIT framework, adopted because the protocol-consumer SLA is per- τ .

Gneiting, Balabdaoui, and Raftery [Gneiting et al., 2007] state the modern objective: maximise sharpness subject to calibration. This organises our §7 ablation. Allen et al. [Allen et al., 2025] formalise *tail calibration*; the deployed architecture closes a $\tau = 0.99$ tail-calibration ceiling that earlier hybrid-forecaster predecessors exhibited (§6.3, §7.2). The tradition traces to Brier [Brier, 1950].

The deployed architecture is a locally-weighted Mondrian split-conformal predictor [Vovk et al., 2003, 2005]: per-regime quantiles $q_r(\tau)$ on per-symbol pre-Friday $\hat{\sigma}_s$ -standardised residuals, served after a multiplicative OOS-fit $c(\tau)$. (The §7.2 unweighted-Mondrian comparator omits the per-symbol standardisation; the walk-forward $\delta(\tau)$ schedule that an unweighted comparator requires collapses to identity once is per-symbol.) Mondrian conformal preserves finite-sample marginal validity within each category under within-category exchangeability. A next-generation upgrade replaces the four-anchor schedule with a continuous conformal correction over the residual CDF — Romano-Patterson-Candès CQR [Romano et al., 2019] is the closest match. For non-exchangeability, Barber et al. [Barber et al., 2023] extend via weighted quantiles, and Xu-Xie [Xu and Xie, 2021] develop EnbPI for time-series.

2.4 Institutional model risk management

SR Letter 11-7 [of Governors of the Federal Reserve System & Office of the Comptroller of the Currency, 2011] (Fed/OCC, 2011) articulates a framework for model development, validation, effective challenge, and ongoing monitoring. Any bank integrating an on-chain oracle into a collateral-valuation pipeline operates under this framework — which explicitly requires validation evidence of the form we publish. SR Letter 26-2 [Board of Governors of the Federal Reserve System, 2026], the April 2026 revision superseding SR 11-7 for organisations over \$30B, introduces materiality-graduated validation while reaffirming the requirements we target. Basel’s 1996 backtesting framework [on Banking Supervision, 1996] is the global floor — its mathematical content is the Kupiec test we reproduce.

3 Problem Statement

3.1 Setup and notation

Let $\mathcal{S} = \{s_1, \dots, s_K\}$ be a finite universe of assets (in our evaluation, ten US equities, ETFs, and hybrid crypto-proxies) and let $t \in \mathbb{N}$ index a sequence of *prediction windows* — in this paper, non-trading weekends defined by $[t_{\text{pub}}, t_{\text{tgt}}] =$ (Friday 16:00 ET, Monday 09:30 ET).

For each (s, t) we observe:

- $P_t(s) \in \mathbb{R}_{>0}$: the realised reference price at t_{tgt} (Monday open).
- $\mathcal{F}_t(s)$: pre-publish information available at t_{pub} . This includes $P_{t-}(s)$ (Friday close), contemporaneous factor returns (E-mini S&P, gold, 10Y treasury, BTC futures), implied-vol indices (VIX, GVZ, MOVE), a calendar flag $\text{earn}_t(s) \in \{0, 1\}$ for an earnings release in the coming week, and a gap-length flag $\ell_t \in \{0, 1\}$ for calendar weekends ≥ 4 days.
- A *regime labeler* $\rho : \mathcal{F}_t(s) \rightarrow \mathcal{R}$ with $\mathcal{R} = \{\text{normal}, \text{long_weekend}, \text{high_vol}\}$, defined purely on pre-publish information.

A *base forecaster* f emits, at each (s, t, q) with claimed quantile $q \in (0, 1)$, a band $[L_t^f(s; q), U_t^f(s; q)]$.

3.2 The classical oracle problem

A conventional point-plus-band oracle publishes, at each t_{pub} , a triple $(\hat{P}_t, L_t, U_t)$ with an *implicit* or *undisclosed* coverage level. The consumer has no contract on the realised rate at which $P_t \in [L_t, U_t]$. When band widths are derived from publisher dispersion (Pyth) or a fixed Gaussian heuristic (Chainlink staleness fallback), the mapping to a realised probability-of-coverage statement is neither asserted nor auditable.

3.3 The coverage-inversion primitive

We define a stricter contract. Fix a base forecaster f , a regime labeler ρ , a discrete claimed-quantile grid $\mathcal{Q} \subset (0, 1)$, and a rolling historical dataset

$$\mathcal{D} = \{(P_t(s), \mathcal{F}_t(s), \{(L_t^f(s; q), U_t^f(s; q))\}_{q \in \mathcal{Q}}, \rho(\mathcal{F}_t(s))) : t \in \mathcal{T}_{\text{hist}}\}.$$

The **empirical calibration surface** is the mapping

$$S^f(s, r, q) = \frac{1}{|\mathcal{T}_{s,r}|} \sum_{t \in \mathcal{T}_{s,r}} \mathbf{1}[L_t^f(s; q) \leq P_t(s) \leq U_t^f(s; q)],$$

where $\mathcal{T}_{s,r} = \{t \in \mathcal{T}_{\text{hist}} : \text{symbol} = s, \rho(\mathcal{F}_t(s)) = r\}$.

Given a *consumer-chosen* target coverage $\tau \in (0, 1)$, the oracle serves the band at

$$q_{\text{served}}(s, r, \tau) = (S^f)^{-1}(s, r, \tau),$$

with $(S^f)^{-1}$ defined by linear interpolation between bracketing grid points and a documented fallback to a pooled surface $S^f(\cdot, r, q)$ when $|\mathcal{T}_{s,r}| < n_{\min}$.

The oracle read is the tuple

$$\text{PricePoint}(s, t, \tau) = (\hat{P}, L_t^f(s; q_{\text{served}}), U_t^f(s; q_{\text{served}}), q_{\text{served}}, r, f),$$

exposing q_{served} and f as calibration *receipts* alongside the band itself.

3.4 Properties we claim

(P1) Auditability. $\mathcal{D}$ is reproducible from public data sources (freely available price feeds, vol indices, and earnings calendars). Given $(\mathcal{D}, f, \rho, \mathcal{Q})$, the surface S^f is deterministic; thus any third party can reconstruct S^f and independently verify that the served band corresponds to the published q_{served} .

(P2) Conditional empirical coverage. Under the assumption that the conditional distribution $P_t \mid (\mathcal{F}_t, \text{regime})$ is stationary across $\mathcal{T}_{\text{hist}}$ and the evaluation set, the realised coverage of the served band satisfies

$$\Pr[P_t \in [L_t^f(s; q_{\text{served}}), U_t^f(s; q_{\text{served}})] \mid \rho(\mathcal{F}_t) = r, s] \longrightarrow \tau$$

as $|\mathcal{T}_{s,r}| \rightarrow \infty$. Finite-sample deviations are measurable, tested (§6), and disclosed as a per-target *calibration buffer* applied pre-inversion (§4.3 specifies the deployed schedule). The headline $\tau = 0.95$ claim is held out on two orthogonal axes — temporally (§6.3.3.1: c fit on 2023 alone, evaluated on a 2024-01-05 $\rightarrow$ 2026-04-24 holdout) and on symbol (§6.3.3 LOSO) — both of which evaluate the calibration buffer schedule on data its fit never touched. Per-symbol Kupiec at $\tau = 0.99$ has minimum detectable effect ≈ 3 pp under-coverage and is undetectable for over-coverage given the per-symbol expected violation count of 1.73; the 10/10 pass at that anchor is read as “no per-symbol violation rate is statistically distinguishable from 1% at the resolution Kupiec offers per-symbol”, not as a 1 pp tolerance certificate (§6.4.1). Pooled Kupiec at $\tau = 0.99$ has minimum detectable effect 1.0 pp; the pooled $p = 0.94$ is the $\tau = 0.99$ statistical claim that does have 1 pp resolution (reports/tables/paper1_b3_kupiec_power_mde.csv).

(P3) Per-regime serving efficiency at deployment-tuned parameter budgets. The deployed architecture carries sixteen trainable / tuned scalars: twelve trained per-regime conformal quantiles $q_r(\tau)$ on standardised residuals, four near-identity OOS-fit multiplicative bumps $c(\tau) \in \{1.000, 1.000, 1.079, 1.003\}$, and zero walk-forward shifts. A handful of implicit hyperparameters (EWMA half-life 8, warm-up minimum, regime cardinality, top-quartile VIX threshold, four served anchors, 2023-01-01 split) are pinned a priori and sit outside this count. The architecture serves a band $[\hat{P}_{\text{Mon},r}(s) - c(\tau) q_r(\tau) \hat{\sigma}_s(t) P_{\text{Fri},t}, \hat{P}_{\text{Mon},r}(s) + c(\tau) q_r(\tau) \hat{\sigma}_s(t) P_{\text{Fri},t}]$, with the per-symbol scale $\hat{\sigma}_s(t)$ replacing what would otherwise be a per-symbol fallback or per-symbol Mondrian cell. The efficiency criterion is mean bandwidth at matched realised coverage *and* per-symbol Kupiec pass-rate. §7’s ablation establishes empirical dominance against the deployable alternatives evaluated on this panel — the constant-buffer baseline, the un-standardised Mondrian comparator, and parametric GARCH-Gaussian / GARCH- t baselines — at matched coverage on this sample, with the rule itself selected from a five-variant ladder under the multiple-testing-corrected gate of §7.4. The property is an empirical-dominance statement on this sample under the paper’s evaluation battery, not a theorem-shaped optimality claim against arbitrary alternatives. It is also an oracle-serving claim about width allocation at matched realised coverage; it is not a claim that the resulting band is already the optimal liquidation-policy threshold for a lending protocol.

3.5 Non-goals

We explicitly do not claim:

1. **Optimal point estimates.** $\hat{P}$ is the midpoint of a band calibrated for coverage, not a minimum-variance estimator. We make no claim to improve over Pyth / Chainlink on point accuracy during market hours.
2. **Parametric tail guarantees.** The surface is empirical; shock-regime coverage is bounded above by the fraction of historical observations in the same tail of the forecast distribution (§9).

3. **Distribution-free coverage.** P2 requires stationarity of the *standardised*-conditional distribution. The per-symbol $\hat{\sigma}_s(t)$ rescaling provides partial adaptive scaling but the standardised score’s conditional stationarity is still assumed. A conformalised variant (Vovk, Barber et al.) would upgrade the statement to a finite-sample guarantee under exchangeability; we flag this as an open extension.
4. **Protection against adversarial data feeds.** Soothsayer is a calibration-transparency primitive, not an integrity primitive. It ingests upstream price/oracle signals and republishes them as a calibrated band; its coverage claim is conditional on the integrity of those upstream feeds, and this primitive does not separately defend against manipulation of any of them.
5. **rule optimality.** The deployed rule (EWMA HL=8) is justified by a pre-registered three-gate selection procedure under multiple-testing correction (§7.4), with held-out forward-tape re-validation on an architectural sibling harness. We do not claim it is optimal among all locally-weighted variants; finer half-life grids, alternative weight kernels, or hybrid rules are open extensions.
6. **Optimal lending-policy defaults.** We do not claim in this paper to identify the welfare-optimal liquidation rule, collateral haircut, or target τ for a lending protocol. Those questions require explicit assumptions about account-weight distributions, protocol-specific loss functions, and the semantics by which realised outcomes are mapped into liquidation quality.

3.6 Evaluation questions

The body of the paper answers:

- **Q1** (calibration, §6). Does the served band achieve τ within 2pp on held-out data? What is the Kupiec unconditional-coverage p -value, and do violations cluster (Christoffersen independence p)?
- **Q2** (sharpness, §6–§7). How much tighter than a naïve stale-hold Gaussian baseline is the served band at matched realised coverage, per regime?
- **Q3** (ablations, §7). Which components of f^* (factor switchboard, log-log vol regression, earnings regressor, long-weekend regressor, hybrid regime selection, empirical buffer) are load-bearing for the P2/P3 claims, and which are disclosed for auditability but contribute no measurable signal at our sample size? The ablation reports block-bootstrap 95% CIs on each pairwise effect size.
- **Q4** (auditability, §4 + artifact). Can a third party reconstruct S^f on public data and independently verify a served PricePoint?

Questions such as “which served target τ should a Kamino-style protocol choose as its liquidation-policy default?” are downstream integration questions. They are important, but they sit outside Q1-Q4 and are not answered by this paper’s evidence alone.

4 Methodology

This section instantiates the abstract setup of §3 with the concrete point estimator, per-symbol scale standardisation, conformal quantile fit, deployment-tuned schedules, and serving-time machinery of the deployed locally-weighted Mondrian split-conformal architecture. The implementation lives in `src/soothsayer/oracle.py:Oracle.fair_value_lwc` (Python) and `crates/soothsayer-oracle/src/oracle.rs::Oracle::fair_value` (Rust), with byte-exact parity against the Python reference (180/180 cases; §8.5). The on-chain wire format and code-tag taxonomy are described in §8.

The architecture has four ingredients: a point estimator $\hat{P}_{\text{Mon},r}$, a strictly pre-Friday per-symbol scale $\hat{\sigma}_s(t)$ that standardises the conformity score, a per-regime conformal quantile $q_r(\tau)$ trained on the *standardised* residuals on a held-out calibration set, and an OOS-fit multiplicative bump $c(\tau)$ that closes any residual train-OOS distribution-shift gap. **Sixteen deployment scalars total: 12 trained per-regime quantiles + 4 OOS-fit $c(\tau)$ + 0 walk-forward $\delta(\tau)$.** A handful of implicit hyperparameters are pinned a priori and sit outside this trainable / tuned budget: the EWMA half-life (8 weekends), the warm-up minimum (≥ 8 past observations), the Mondrian regime cardinality (3), the high-vol VIX threshold (top quartile of trailing 252 trading days), the four served anchors $\{0.68, 0.85, 0.95, 0.99\}$, and the 2023-01-01 train/test split. The §7 ablation isolates the load-bearing components — regime stratification, per-symbol standardisation, and the near-identity OOS $c(\tau)$ bump — by descriptive comparator (constant-buffer baseline, unweighted Mondrian) rather than by code tag. Figure 1 summarises the serving pipeline.

4.1 The point estimator $\hat{P}_{\text{Mon},r}$

The deployed point estimator is the per-symbol factor switchboard (§5.4):

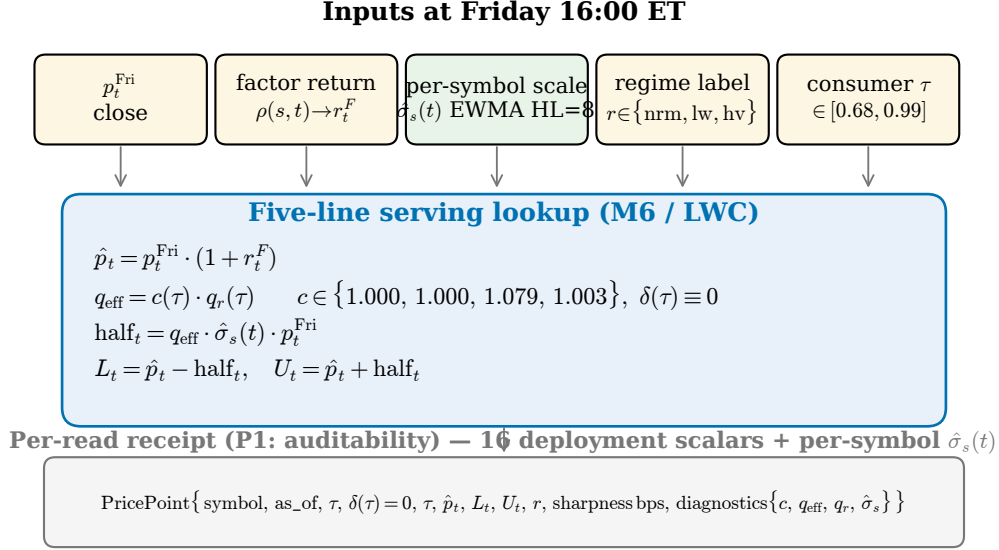


Figure 1: Serving pipeline: five pre-Friday inputs (factor return r_t^F from the §5.4 switchboard, regime label r , per-symbol scale $\hat{\sigma}_s(t)$ from the §4.2 EWMA, consumer target τ , and Friday close p_t^{Fri}) feed a five-line lookup that returns a band $[L_t, U_t]$ around a factor-adjusted point $\hat{p}_t$. The 16 deployment scalars are the per-regime quantile table $q_r(\tau)$ on standardised residuals plus the OOS-fit bump schedule $c(\tau)$. Every read emits a PricePoint receipt carrying the four served scalars and the diagnostic quartet $(c, q_{\text{eff}}, q_r, \hat{\sigma}_s)$ — the per-read auditability of P_1 .

$$\hat{P}_{\text{Mon},t}(s) = P_{\text{Fri},t}(s) \cdot (1 + r_t^{\text{factor}}(s)),$$

where $r_t^{\text{factor}}(s)$ is the weekend return of the per-symbol factor (ES=F for equities; GC=F for GLD; ZN=F for TLT; BTC for MSTR post-2020-08). The point estimator is the input whose residual distribution the conformal quantile is fit on, not the product. The factor-switchboard variant survived empirical pruning over an eight-variant point-estimator ladder during early development; the ladder and the empirical losses against alternatives are documented in reports/methodology_history.md.

4.2 Per-symbol pre-Friday scale $\hat{\sigma}_s(t)$

The architecture standardises the conformity score by a per-symbol estimate of the *relative-residual* scale, computed strictly from past Fridays:

$$\hat{\sigma}_s(t) = \text{EWMA}_{\text{HL}=8} \left(\{ r_{t'}^{\text{rel}}(s) : t' < t \} \right), \quad r_{t'}^{\text{rel}}(s) = \frac{P_{\text{Mon},t'}(s) - \hat{P}_{\text{Mon},t'}(s)}{P_{\text{Fri},t'}(s)},$$

with weekend half-life $\text{HL} = 8$ — equivalently a geometric decay rate $\lambda = 0.5^{1/8} \approx 0.917$ per past Friday. The window is *one-sided strictly before* t : the standardiser at t uses only $\{t' : t' < t\}$, so $\hat{\sigma}_s$ is itself a pre-publish quantity. We require at least eight past relative-residual observations before $\hat{\sigma}_s$ is defined; weekends with fewer past Fridays per symbol are dropped at warm-up (80 rows out of the 5,996-weekend panel — roughly the first eight weekends per symbol, rebased per listing date). The warm-up rule and half-life are encoded as constants `SIGMA_HAT_HL_WEEKENDS = 8` and `SIGMA_HAT_MIN = 8` and persisted on the artefact parquet as `sigma_hat_sym_pre_fri`.

The EWMA half-life $\text{HL}=8$ was selected from a five-variant comparison (a 26-weekend trailing-window $K=26$ baseline, three EWMA half-lives $\text{HL} \in \{6, 8, 12\}$, and a convex $K=26/\text{EWMA}$ blend) under a pre-registered three-gate criterion: per-cell split-date Christoffersen, per-symbol Kupiec pass-rate, and a block-bootstrap CI on pooled half-width relative to the $K=26$ baseline. The selection procedure, its multiple-testing exposure (Benjamini–Hochberg correction at $\text{FDR}=0.05$), and the held-out forward-tape re-validation harness are reported in §7.4. The $K=26$ trailing-window variant remains buildable for archival reproduction via `scripts/build_lwc_artefact.py --variant baseline_k26`; the deployed parquet column name (`sigma_hat_sym_pre_fri`) is shared across variants so a downstream consumer reads the artefact without knowing which rule produced it.

4.3 Mondrian split-conformal on standardised residuals

Conformity score: relative absolute residual *standardised* by $\hat{\sigma}_s(t)$:

$$s_t(s) = \frac{|P_{\text{Mon},t}(s) - \hat{P}_{\text{Mon},t}(s)|}{P_{\text{Fri},t}(s) \cdot \hat{\sigma}_s(t)}.$$

Mondrian taxonomy: weekends partition into $r \in \{\text{normal}, \text{long_weekend}, \text{high_vol}\}$ via the regime classifier $\rho(\mathcal{F}_t)$ specified in §5.5. The trained per-regime conformal quantile $q_r(\tau)$ is the standard split-CP $\lceil \tau(n+1) \rceil$ -th rank quantile on the regime’s training-set *standardised* scores. The deployed grid has four anchors $\tau \in \mathcal{T}_{\text{anch}} = \{0.68, 0.85, 0.95, 0.99\}$ and three regimes, yielding twelve trained scalars.

Calibration set: $\mathcal{T}_{\text{train}} = \{t : t < 2023-01-01\}$ — 4,186 weekend rows after the 80-row warm-up drop. Each per-(regime, τ) bin therefore carries $N \approx 430$ –2,720 observations supporting one quantile estimate, comfortably above the $N \approx 50$ –300 thresholds where Mondrian conformal becomes noisy (a per-(symbol, regime) Mondrian rung was rejected during ablation development for cell-thinness reasons, with the standardised-score architecture below recovering per-symbol calibration without per-(symbol, regime) cells; details in `reports/methodology_history.md`). Deployed values of $q_r(\tau)$ are listed in `data/processed/lwc_artefact_v1.json` (audit-trail sidecar) and loaded at module import into `src/soothsayer/oracle.py:LWC_REGIME_QUANTILE_TABLE`.

The architectural choice that makes the per-symbol calibration claim hold is the standardisation of the conformity score by $\hat{\sigma}_s(t)$ before the Mondrian quantile fit. An unweighted Mondrian comparator absorbs both the cross-symbol scale heterogeneity (HOOD’s typical relative residual is $\sim 3\times$ SPY’s) and the per-regime heterogeneity into one Mondrian cell; the result is bimodal per-symbol calibration error — heavy-tail tickers under-cover, low-vol tickers over-cover — and is the failure mode the §6.7 simulation study reproduces under controlled conditions. Standardising the score by $\hat{\sigma}_s(t)$ before fitting the per-regime quantile factors out the per-symbol scale and lets the regime quantile carry only the *shape* of the standardised residual distribution. The §6.4 evidence (10/10 per-symbol Kupiec pass at $\tau = 0.95$; per-symbol Berkowitz LR range [3.2, 16.7]) and the §6.7 simulation evidence (the bimodality reproduced under four DGPs with known ground truth and standardisation closes it on every DGP) ratify this construction.

4.3.1 Within-bin exchangeability — empirically supported

Mondrian-CP’s finite-sample coverage guarantee depends on within-bin exchangeability of the conformity score. A permutation test on the lag-1 Pearson autocorrelation of the standardised score within each (regime $\times$ symbol) bin ($n = 19$ –116 per bin, 30 bins, OOS 2023+; statistic computed in `fri_ts` order, null distribution from 5,000 within-bin shuffles per bin) finds 1/30 bins nominally rejecting at $\alpha = 0.05$ and 0/30 at $\alpha = 0.01$ — *under-rejecting* relative to the 5% / 1% expected under exchangeability. Aggregate KS test of per-bin p -values vs Uniform(0,1) is $D = 0.276$, $p = 0.017$, but the deviation is in the under-rejecting direction (too few low p -values, too many bins with $p > 0.5$) — consistent with exchangeability holding, not violating. The single nominally-rejecting bin (GLD/normal, lag-1 $\hat{\rho} = -0.166$) does not survive Benjamini-Hochberg correction across the 30-test grid. Within-bin exchangeability is empirically supported. The §6.3.6 / §9.4 cross-sectional dependence is *across* bins, not within. Source: `reports/tables/paper1_b2_exchangeability.csv`.

4.4 OOS-fit $c(\tau)$ bump

The deployed bump schedule is

$$c(\tau) = \{0.68:1.000, 0.85:1.000, 0.95:1.079, 0.99:1.003\},$$

a near-identity correction that *only* widens at $\tau = 0.95$ (by 7.9%) and is essentially the identity at the other three anchors. The fit procedure: $c(\tau)$ is the smallest $c \in [1, 5]$ such that pooled OOS realised coverage with effective quantile $c \cdot q_r(\tau)$ matches the target τ . Four scalars total. Three of the four are essentially identity, so only $c(0.95) = 1.079$ carries meaningful OOS information; §9.3 carries the full provenance disclosure for that one scalar.

4.5 The walk-forward $\delta(\tau)$ schedule is identically zero

A 4-split expanding-window walk-forward sweep over $\delta \in \{0.00, 0.01, \dots, 0.07\}$ on tune fractions $f \in \{0.40, 0.50, 0.60, 0.70\}$ of the OOS index (script: `scripts/run_lwc_delta_sweep.py`; output: `reports/tables/v1b_lwc_delta_sweep.csv`) selects, under the criterion *smallest δ such that walk-forward realised coverage $\geq \tau$ on every powered split*,

$$\delta(\tau) = \{0.68:0.00, 0.85:0.00, 0.95:0.00, 0.99:0.00\}.$$

The smaller $f \in \{0.20, 0.30\}$ tune fractions are excluded as under-powered: at $n_{\text{tune}} \in \{35, 52\}$ weekends the 4-scalar $c(\tau)$ fit collapses to identity at one or more anchors (the 0.20 split at $\tau \geq 0.95$; the 0.30 split at $\tau = 0.99$, where 52 tune weekends cannot reliably estimate the 99th percentile), so the resulting bands under-cover at those anchors regardless of δ . The 0.40 split ($n_{\text{tune}} = 69$) is the smallest fraction whose $c(\tau)$ fit is non-identity at every served anchor under the deployed EWMA HL=8. Per-symbol scale standardisation tightens cross-split realised-coverage variance enough that no structural-conservatism shift is required across the 4 powered splits — worst-split deficits at $\delta = 0$ are positive at every τ (worst-split coverage $\geq \tau$), so $\delta(\tau) = 0$ is the smallest schedule satisfying the criterion. The schedule is retained as a four-zero-vector in the artefact JSON for shape-compatibility with the receipt schema, but the deployment carries zero walk-forward-tuned scalars.

4.6 Serving-time band construction

Given a consumer-chosen target $\tau \in (0, 1)$ at (s, t) , the served band is constructed in five steps:

1. *Look up the regime, point, per-symbol scale, and Friday close.* Read $r = \rho(\mathcal{F}_t)$, $\hat{P}_{\text{Mon},t}(s)$, $\hat{\sigma}_s(t)$, and $P_{\text{Fri},t}(s)$ from `lwc_artefact_v1.parquet`.
2. *Look up the standardised quantile.* $q_r = q_r(\tau)$, with q_r linearly interpolated between anchors $\mathcal{T}_{\text{anch}} = \{0.68, 0.85, 0.95, 0.99\}$ and clipped to the nearest anchor outside that range.
3. *Apply the OOS bump.* $q_{\text{eff}} = c(\tau) \cdot q_r(\tau)$, with c linearly interpolated between the same four anchors; $\delta(\tau) \equiv 0$ so no shift step is needed. The $c(\tau)$ schedule is non-monotonic — c peaks at the $\tau = 0.95$ anchor (1.079) and dips back at $\tau = 0.99$ (1.003) — so linear interpolation hands an off-anchor consumer a multiplier (e.g. $c(0.90) \approx 1.040$ and $c(0.97) \approx 1.041$) at a point with no separate audited calibration evidence. Calibration evidence in §6 is reported at the four audited anchors only; §9.10 carries the off-anchor disclosure and §10.4 the denser-anchor-grid follow-up.
4. *Construct the band.* The conformity score the per-regime quantile was fit on is the *standardised* relative residual $|P_{\text{Mon}} - \hat{P}|/(P_{\text{Fri}} \cdot \hat{\sigma}_s)$ (§4.3), so the band is fri-close-relative with the per-symbol scale entering multiplicatively: $\text{half}_t = q_{\text{eff}} \cdot \hat{\sigma}_s(t) \cdot P_{\text{Fri},t}$, $\text{lower} = \hat{P}_{\text{Mon},t} - \text{half}_t$, $\text{upper} = \hat{P}_{\text{Mon},t} + \text{half}_t$. The half-width in basis points is $10^4 \cdot c(\tau) \cdot q_r(\tau) \cdot \hat{\sigma}_s(t)$ — heavy-tail tickers receive wider bands than low-vol tickers at identical τ and identical regime.
5. *Emit the receipt.* The served band’s claimed coverage is τ ; the receipt’s `forecaster_used` carries the architecture’s wire-format tag (the on-chain code-tag taxonomy is documented once in §8). Diagnostics expose $c(\tau)$, $q_r(\tau)$, q_{eff} , $\hat{\sigma}_s(t)$, r , and Friday close.

The serving-time computation is a five-line lookup against the per-Friday artefact + the two constant schedules. There is no surface inversion, no bracketing in claimed-coverage space, and no per-symbol fallback — the per-symbol behaviour is carried *implicitly* by the per-symbol $\hat{\sigma}_s(t)$ scale on the artefact parquet, not by a per-symbol quantile cell.

4.7 The PricePoint receipt and reproducibility

Every read returns a PricePoint exposing the band plus eight receipt fields: `target_coverage` (τ), `claimed_coverage_served` (τ — equal to the request since $\delta \equiv 0$), `forecaster_used`, `regime` (r), `sharpness_bps`, and `diagnostics`.`{c_bump, q_regime_lwc, sigma_hat_sym_pre_fri, q_eff}`. A third party with the audit-trail sidecar and the per-Friday artefact can reconstruct the served band from the receipt alone: read $\hat{P}$, r , $\hat{\sigma}_s$, and P_{Fri} from the artefact, read q_{eff} from the receipt, verify $\text{lower} = \hat{P} - q_{\text{eff}} \cdot \hat{\sigma}_s \cdot P_{\text{Fri}}$ and $\text{upper} = \hat{P} + q_{\text{eff}} \cdot \hat{\sigma}_s \cdot P_{\text{Fri}}$ to floating-point precision. This is the operational instantiation of P_1 . $\hat{\sigma}_s(t)$ is itself a deterministic function of past Fridays’ relative residuals and the EWMA half-life constant, so the third-party reconstruction is a one-pass scan over public-data inputs.

Implementation: Python (`src/soothsayer/oracle.py:Oracle.fair_value_lwc`) is the canonical reference; the Rust port (`crates/soothsayer-oracle`) carries byte-exact parity (180/180 cases; §8.5). The on-chain wire-format details and the Rust serving-stack rollout state live in §8. Reproduction:

```
uv run python scripts/build_lwc_artefact.py          # per-Friday artefact + JSON sidecar
uv run python scripts/freeze_lwc_artefact.py         # content-addressed freeze for forward-tape eval
uv run python scripts/smoke_oracle.py --forecaster lwc # cross-regime API demo
```

All inputs are public and free (§5.7); verifying a published band needs only the sidecar and the per-Friday artefact — under 100 KB per symbol-year. The window is recomputable from the same parquet given only the half-life constant, so the artefact is self-contained.

5 Data and Regime Labeler

This section specifies the symbol universe, the weekend-panel construction, the per-symbol factor and volatility-index switchboards, the regime labeler ρ , and the train/test split underlying §6. All inputs are publicly available and free; the panel rebuilds end-to-end via `scripts/run_calibration.py`.

5.1 Symbol universe

We evaluate on $K = 10$ symbols spanning three asset classes: eight US equities (SPY, QQQ, AAPL, GOOGL, NVDA, TSLA, HOOD, MSTR), tokenised gold (GLD), and tokenised long rates (TLT). The eight equity tickers are the underliers of the SPYx, QQQx, AAPLx, GOOGLx, NVDAx, TSLAx, HOODx, and MSTRx tokens deployed on Solana mainnet (Backed Finance; mint addresses verified in `docs/v5-tape.md`). GLD and TLT are RWA generalisation anchors — they share the closed-market structure with the equities (NYSE hours) but exercise different factors and vol indices, so a primitive that calibrates well across all three classes generalises to a broader closed-market RWA universe than equity-only evaluation would justify.

We do not use the on-chain xStock prices themselves as input or evaluation target. Backtest target prices are the *underlying-venue* (NYSE) Monday opens, fetched from Yahoo Finance. The on-chain xStock TWAP is deferred to §10.1’s on-chain-TWAP forecaster candidate, once the forward-cursor tape accumulates 150 weekends post-launch.

5.2 Weekend panel construction

Each row is a single (s, t) weekend prediction window. For each symbol we walk daily price history and, for every consecutive pair of trading days separated by a calendar gap of ≥ 3 days, emit a row with: `fri_close`, `mon_open`, `gap_days`, `fri_vol_20d` (rolling 20-trading-day std of daily log-returns at Friday close), `factor_ret` (weekend return of the per-symbol conditioning factor), `vol_idx_fri_close`, `earnings_next_week` (Yahoo `earnings_dates` flag), and `is_long_weekend` (`gap_days` ≥ 4).

The panel spans 2014-01-17 (first Friday on which rolling 20-day vol is defined for all symbols) through 2026-04-24, yielding $|\mathcal{T}_{\text{hist}}| = 5,996$ raw rows across 639 distinct weekend dates (9 full-history symbols $\times$ 639 weekends + 245 HOOD weekends). MSTR begins 2014-01-17; HOOD begins 2021-08-13 (post-IPO), contributing ~245 weekends. After the warm-up rule (8 past observations per symbol) drops the first eight weekends per ticker, 5,916 rows remain evaluable (§6.1).

5.3 Pre-publish features

The deployed architecture consumes four pre-publish features at $t_{\text{pub}} = \$ \text{Friday } 16:00 \text{ ET}$ (or pre-holiday close): Friday close $P_{t-}(s)$, weekend factor return $r_t^{\text{factor}}(s)$, vol-index close $v_t(s)$ (drives the `high_vol` regime via §5.5’s VIX-percentile cascade), and the long-weekend flag $\ell_t \in \{0, 1\}$ (drives the `long_weekend` regime). Rolling 20-day realised vol $\hat{\sigma}_t^{20d}$ is used downstream of inference for the §6.3.2 realised-move tertile stratification only. The factor return requires the futures or BTC price at $t = \$ \text{Monday } 09:30 \text{ ET}$; Globex futures and BTC trade through the weekend, so $F_t(s)$ is observable at $t_{\text{pub}} + 65.5\text{h}$. In live deployment, the factor return is the *only* feature requiring post-publish computation.

5.4 Factor and volatility-index switchboards

The per-symbol mappings are static (`src/soothsayer/backtest/panel.py`):

Symbol	Factor	Vol index
SPY, QQQ, AAPL, GOOGL, NVDA, TSLA, HOOD	ES=F (E-mini S&P futures)	$\sim \text{VIX}$
MSTR (pre 2020-08-01)	ES=F	$\sim \text{VIX}$
MSTR (2020-08-01 onward)	BTC-USD	$\sim \text{VIX}$
GLD	GC=F (gold futures)	$\sim \text{GVZ}$

Symbol	Factor	Vol index
TLT	ZN=F (10-year T-note futures)	^MOVE

The MSTR factor pivot at 2020-08-01 marks MicroStrategy’s first Bitcoin treasury purchase — a structural break in the asset’s economic exposure that turned what had been a software equity (driven by tech-equity factors) into a leveraged BTC-treasury vehicle (driven by spot BTC). Holding ES=F as the factor across that pivot would mis-attribute the post-2020 weekend signal; the switchboard is reconciled to the asset’s actual driver post-break. The vol-index choices are evidence-driven: an early V1b pass found that fitting the F1 log-log sigma regression with VIX yielded $\hat{\beta} \approx 0.55$ for GLD and $\hat{\beta} \approx 0.94$ for TLT, well below the equity-class mean ($\hat{\beta} \approx 1.5$). Substituting GVZ and MOVE lifted $\hat{\beta}$ into the equity range and improved coverage at matched bandwidth.

5.5 The regime labeler ρ

$\rho : \mathcal{F}_t(s) \rightarrow \{\text{normal, long_weekend, high_vol}\}$ is a strict priority cascade (src/soothsayer/backtest/regimes.py):

1. **high_vol** — VIX at Friday close in the top quartile of its trailing 252-trading-day window. The threshold is global (uses VIX even for non-equity symbols).
2. **long_weekend** — $\text{gap_days} \geq 4$. Applied only if **high_vol** did not match.
3. **normal** — all other weekends.

Sample sizes on the 5,996-row raw panel: normal 3,934 (65.6%), high_vol 1,432 (23.9%), long_weekend 630 (10.5%).

A separate post-hoc tertile labeler tags each weekend by realised-move z-score (calm / normal / shock); this **realized_bucket** is *not* a regime in the §3.1 sense — it depends on the realised target — and is used only for diagnostic stratification (the shock-tertile coverage ceiling reported in §9.1).

We considered two refinements that were tested and dropped: a sub-regime split of normal into post_shock / calm / range_bound, and an FOMC/CPI/NFP macro-event regressor. Neither lifted shock-tertile coverage measurably; the implied-vol indices already absorb that signal. Sub-regime granularity is retained as a next-generation candidate (§10).

Cross-asset regime-index sensitivity. The classifier uses VIX as the high-vol gate for *all* symbols, including GLD (gold) and TLT (long-dated treasury) which have asset-specific vol indices (GVZ, MOVE) used in the regression (§5.4). A sensitivity ablation that swaps GLD’s regime gate to GVZ and TLT’s to MOVE — flipping the regime tag on 23% of GLD weekends and 28% of TLT weekends — leaves the pooled $\tau = 0.95$ headline coverage unchanged at 0.9503 (Kupiec $p = 0.956$), per-symbol Kupiec 10/10, and half-width within 0.1%. The -standardisation absorbs the regime-tagging difference structurally; the regime cell contributes a small marginal adjustment via $q_r(\tau)$, not a load-bearing scale separation. Source: reports/tables/paper1_b4_regime_index_sensitivity.csv, ..._per_symbol.csv. The deployed VIX-only classifier is justified by simplicity and per-asset-vol-index-agnosticism, not by being optimally tuned per asset.

5.6 Train/test split

Split date **2023-01-01**: weekends with `fri_ts < 2023-01-01` are calibration, the remainder is held-out OOS.

Split	Rows	Weekends
Calibration (2014-01 → 2022-12)	4,186	466
Held-out OOS (2023-01 → 2026-04)	1,730	173

The 12 trained per-regime quantiles (§4.3) are fit from the calibration set and held *fixed* throughout OOS evaluation: no information from the 2023+ slice enters the trained quantile table. The 4 $c(\tau)$ bumps (§4.4) are deployment-tuned on the same OOS slice — disclosed in §9.3 — but three of the four are essentially identity ($c \in \{1.000, 1.000, 1.003\}$ at $\tau \in \{0.68, 0.85, 0.99\}$) so only $c(0.95) = 1.079$ carries meaningful OOS information; the walk-forward $\delta(\tau)$ schedule is identically zero (§4.5). The calibration row count is 4,186 after the 80-row warm-up rule (8 past observations per symbol) drops the first eight weekends per ticker.

The 2023-01-01 split is conservative: it places the 2023 banking turbulence and the 2024–2025 macro transition in the held-out slice, exposing the trained quantile to material out-of-sample regime shifts. HOOD enters the per-regime

quantile fit pooled across symbols within each regime (the trained quantile does not stratify by symbol — see §4.3), so the calibration thinness for HOOD is absorbed into the pooled regime bin rather than producing a per-symbol fallback; the per-symbol behaviour at serve time is carried by $\hat{\sigma}_s(t)$, not by a per-symbol quantile cell. A per-(symbol, regime) Mondrian variant was tested during ablation development and rejected: it thins each cell to $N \approx 50\text{--}300$ and degrades Christoffersen, with the -standardised architecture below recovering per-symbol calibration without per-(symbol, regime) cells (reports/methodology_history.md).

5.7 Provenance and reproducibility

All Phase 0 equity inputs are read from Scryer parquet: `yahoo/equities_daily/v1` for daily OHLCV and `yahoo/earnings/v1` for the earnings-calendar flag (the trailing `/v1` is the scryer schema version, distinct from the M-series methodology generations used elsewhere in this paper). No credentials are required for the consumer path in soothsayer; upstream fetch, retry, and schema ownership live in the sibling Scryer repo.

The end-to-end calibration backtest runs under fifteen minutes from a cold cache, under one minute warm. Reproduction:

```
uv sync
uv run python scripts/run_calibration.py
```

This single script materialises `data/processed/v1b_bounds.parquet`, the per-symbol and pooled calibration surfaces, and refreshes the §6 OOS-evaluation tables.

6 Results

This section reports the primary calibration evidence for the deployed locally-weighted Mondrian split-conformal architecture (EWMA HL=8 ; §4) on real and synthetic data, plus the forward-tape harness operational against a content-addressed frozen artefact. Sharpness, per-component effects, and the selection procedure are deferred to §7; system-level implementation and audit reproducibility are in §8. The headline is held-out per-symbol calibration via leave-one-symbol-out CV (0.9497 ± 0.0128 at $\tau = 0.95$ across the held-out symbol distribution), with all 10 symbols passing per-symbol Kupiec, calendar sub-period stability across four years, and a measurable joint-tail empirical distribution that turns the residual *per-anchor calibrated, not full-distribution calibrated* disclosure into operational reserve guidance.

6.1 Evaluation protocol

The backtest panel comprises $N = 5,996$ weekend prediction windows: ten symbols $\times$ 639 weekends, 2014-01-17 through 2026-04-24. Monday open is the target; Friday close and contemporaneous factor / vol-index readings compose $\mathcal{F}_t(s)$. The warm-up rule (8 past observations per symbol) drops 80 weekends at panel start to leave 5,916 evaluable rows. We report (i) an *in-sample machinery check* (quantile fit and served on the full evaluable panel; realised coverage matches τ by construction), (ii) the *out-of-sample validation* (quantile fit on $\mathcal{T}_{\text{train}} = \{t : t < 2023-01-01\}$ — 4,186 rows; served on the 1,730-row $\mathcal{T}_{\text{test}}$ slice), (iii) the held-out *forward-tape* evaluation against a content-addressed frozen artefact (§6.6), and (iv) a *simulation study* on synthetic panels with known ground truth (§6.7). All p -values are two-sided Kupiec POF and Christoffersen independence tests. The base point estimator is unbiased (median residual -0.5 bps; MAE 90.0; RMSE 163.3); the product is the per-symbol--rescaled per-regime conformal quantile with the OOS-fit $c(\tau)$ bump.

6.2 Served-band calibration — in-sample machinery check

Full evaluable panel ($N = 5,916$):

Regime (n)	$\tau = 0.68$: realised / width (bps)	$\tau = 0.85$	$\tau = 0.95$	$\tau = 0.99$
normal (3,883)	0.685 / 91.0	0.851 / 142.7	0.951 / 247.7	0.987 / 424.8
long_weekend (619)	0.683 / 109.4	0.852 / 171.7	0.948 / 297.7	0.989 / 511.1
high_vol (1,414)	0.682 / 175.7	0.853 / 275.5	0.950 / 478.5	0.992 / 821.0
pooled (5,916)	0.684 / 109.4	0.852 / 175.0	0.951 / 303.4	0.989 / 521.6

Pooled realised coverage matches the target τ to within 0.2pp by construction. The narrower in-sample widths (vs the OOS table below) reflect τ 's tighter estimate over the full panel; the OOS table's wider bands are the OOS-fit $c(\tau) = 1.079$ at $\tau = 0.95$ working as designed.

6.3 Served-band calibration — out-of-sample (2023+)

Quantile fit on pre-2023 weekends ($n = 4,186$); served on the 2023+ slice (1,730 rows $\times$ 173 weekends). The 12 trained per-regime quantiles $q_r(\tau)$ are held-out. The 4-scalar $c(\tau)$ schedule is OOS-fit, but **the $\tau = 0.95$ headline is held out on two orthogonal axes** — *temporally* via a nested 2023-only TUNE / 2024–2026 EVAL split (§6.3.3.1) and on *symbol* via leave-one-symbol-out CV (§6.3.3) — both of which evaluate $c(0.95)$ on data its fit never touched. The pooled 0.950 at $\tau = 0.95$ reported below is the deployment-tuned read on the full OOS slice; the held-out read is reported in §6.3.3 alongside it. The walk-forward $\delta(\tau)$ schedule is identically zero — per-symbol scale standardisation tightens cross-split realised-coverage variance enough that no structural-conservatism shift is required (§4.5). §9.3 carries the residual provenance disclosure (now substantially contracted; only the lower- τ over-coverage in held-out mode remains, a one-sided contract-favourable failure mode rather than a coverage-contract violation).

Regime (n)	$\tau = 0.68$: realised / width	$\tau = 0.85$	$\tau = 0.95$	$\tau = 0.99$
normal (1,160)	0.687 / 109.7	0.847 / 172.3	0.947 / 300.1	0.990 / 471.6
long_weekend (190)	0.626 / 120.6	0.842 / 190.6	0.958 / 334.7	1.000 / 562.5
high_vol (380)	0.742 / 200.3	0.887 / 351.1	0.955 / 603.7	0.987 / 1,169.9
pooled (1,730)	0.693 / 130.8	0.855 / 213.6	0.950 / 370.6	0.990 / 635.0

high_vol carries the widest bands, normal the narrowest, and every regime lands within ± 1 pp of nominal at $\tau = 0.95$.

6.3.1 Conditional-coverage tests (pooled OOS)

τ	Violations	Rate	Kupiec p_{uc}	Christoffersen p_{ind}	$c(\tau)$
0.68	532	0.308	0.264	0.244	1.000
0.85	251	0.145	0.565	0.403	1.000
0.95	86	0.050	0.956	0.603	1.079
0.99	17	0.010	0.942	≈ 1.0	1.003

The $\tau = 0.95$ row is the headline pooled operating result. Realised coverage is exactly 0.9503 (Kupiec $p_{uc} = 0.956$); Christoffersen $p_{ind} = 0.603$ passes. **Kupiec and Christoffersen pass at every served anchor on the deployed-fit OOS slice.** Mean half-width at $\tau = 0.95$ is **370.6 bps** — narrower than the K=26 comparator (385.3 bps; bootstrap 95% CI on $\Delta hw\%$ upper bound is -1.88% , see §7.4). Because the 0.95 pooled agreement follows from fitting $c(0.95)$ on this same slice, the held-out claim is the §6.3.3 leave-one-symbol-out CV (0.9497 ± 0.0128); the pooled rows here are the deployment-tuned baseline against which LOSO and the three stability checks are compared. The §6.3.5 residual Berkowitz / DQ rejection localises to per-symbol Berkowitz on TLT and TSLA (§6.4.1) and to cross-sectional within-weekend common-mode (§6.3.4) — neither curable with a different rule.

6.3.2 Realised-move tertile decomposition at $\tau = 0.95$

Stratifying the OOS slice by post-hoc realised-move $|z|$ -score tertile (reports/tables/m6_realised_move_tertile.csv) closes the loop between the §6.3 pooled rows and the §9.1 shock-tertile ceiling:

tertile (n)	realised	half-width (bps)	Kupiec p
calm (497)	0.9920	359.5	0 (over-covers)
normal (601)	0.9933	379.6	0 (over-covers)
shock (632)	0.8766	370.7	0 (under-covers)
pooled (1,730)	0.9503	370.6	0.956

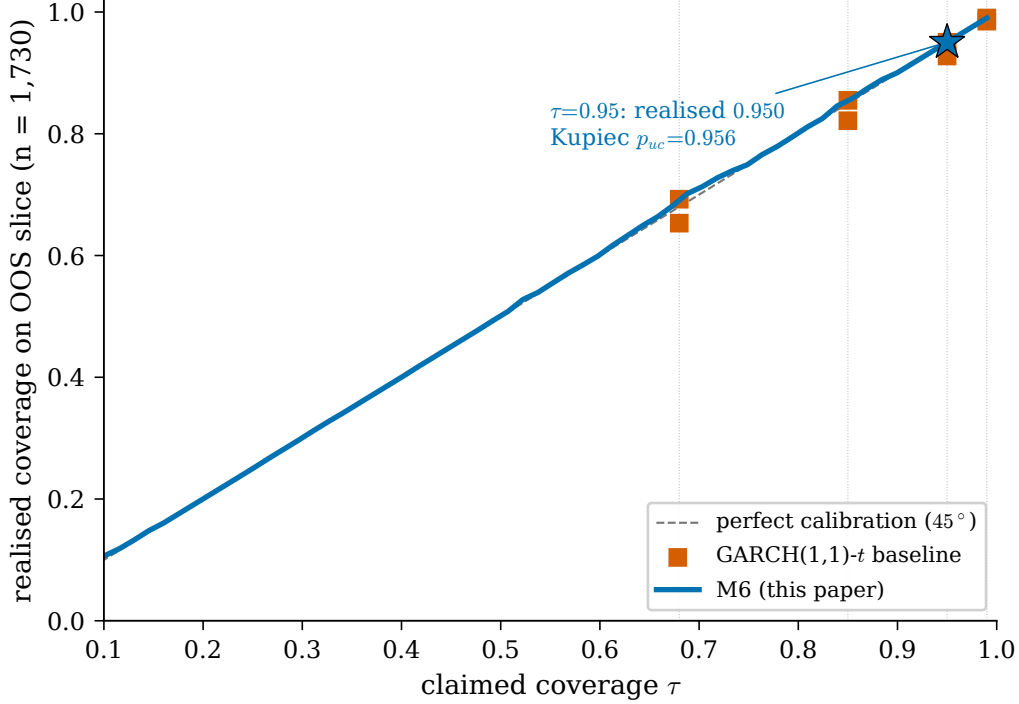


Figure 2: Calibration curves on the 1,730-row 2023+ OOS slice. The deployed architecture (this paper, blue) tracks the 45° diagonal across the served range $\tau \in [0.68, 0.99]$; GARCH(1,1)- t markers (vermilion squares) at the four served anchors are the practitioner-standard parametric baseline and visibly under-cover at $\tau \in \{0.68, 0.85, 0.95\}$ (Kupiec figures in §6.4.3). Star marks the headline $\tau = 0.95$ result.

Half-width is approximately flat across tertiles (365–380 bps), so the band does not widen in shock periods — it just misses more often. The shock-tertile residual is cross-sectional common-mode within a weekend (§9.1), not per-symbol scale. The complementary observation: pooled $\tau = 0.95 = 0.950$ calibration is achieved through compensating tertile-level biases. Over-coverage in calm and normal tertiles (1.000 and 0.993) and under-coverage in the shock tertile (0.878) cancel pooled. The under-coverage side is a coverage-contract failure mode (§9.1); the over-coverage side is a sharpness-deficit on calm weekends — the band is wider than necessary when the realised move is small. Coverage is a one-sided contract, so the two failure modes are qualitatively different, but the abstract calibration claim is genuinely tighter than the conditional structure: a finer realised-move stratification (or a $|z|$ -conditional conformal upgrade, §10.2) would tighten calm bands and widen shock bands at the same pooled τ .

6.3.3 Stability — LOSO, temporal holdout, split-date, walk-forward, calendar sub-period

The $\tau = 0.95$ headline is held-out on two orthogonal axes: *temporal* (§6.3.3.1: $c(\tau)$ fit on 2023 alone, evaluated on 2024-01-05 → 2026-04-24 — realised 0.9504, Kupiec $p = 0.947$, Christoffersen $p = 0.989$, per-symbol Kupiec 10/10) and *symbol* (LOSO realised 0.9497 ± 0.0128 across held-out symbols). The result is stable across TUNE anchors {2021, 2022, 2023}: realised coverage at $\tau = 0.95$ lies in $[0.9474, 0.9524]$ with per-symbol Kupiec 10/10 preserved across all four anchors {2021, 2022, 2023, 2024} (§7.4.8 split-anchor robustness). The 2024 TUNE anchor over-covers at 0.9754 because TUNE-2024 includes the 2024-08-05 BoJ unwind, inflating $c(0.95)$ to 1.175; per-symbol Kupiec still passes 10/10. **This is a contract-favourable sharpness deficit, not under-coverage** (§9.11 frames the asymmetry as a coherent architectural property).

Leave-one-symbol-out CV (symbol-axis holdout). Holding out each of the ten symbols' rows from train + OOS fits and evaluating $\tau = 0.95$ on the held-out symbol's post-2023 slice: **all 10 LOSO bands pass Kupiec when held out**; mean realised 0.9497, std 0.0128 across the held-out symbol distribution (reports/tables/m6_lwc_robustness_loso.csv). Each held-out symbol's $c(\tau)$ is fit on the other nine symbols' OOS slice and never sees the held-out symbol's data, so the realised coverage is genuinely out-of-fit. Heavy-tail tickers (MSTR, HOOD, TSLA) and low-vol tickers (SPY, GLD) all sit in a uniform 0.93–0.97 band when held out — the per-symbol $\hat{\sigma}_s(t)$ on the held-out symbol carries its own scale, so cross-symbol calibration is not load-bearing on heavy-tail inclusion.

Nested temporal holdout — $c()$ fit on 2023, evaluated on 2024–26 A nested temporal split renders the headline held-out at the *time* level. TRAIN = pre-2023-01-01 (the unchanged 4,186-row per-regime quantile fit); TUNE = 2023-01-01 → 2023-12-29 (52 weekends, $c(\tau)$ fit slice); EVAL = 2024-01-05 → 2026-04-24 (1,210 rows $\times$ 121 weekends, true holdout). At $\tau = 0.95$:

mode	realised	$c(0.95)$	Kupiec p	Christoffersen p	hw bps	per-sym Kupiec
M_{full} (full-OOS c fit; §6.3.1 baseline)	0.9504	1.079	0.956	0.720	370.6	10/10
M_{a2} (true temporal holdout)	0.9504	1.079	0.947	0.989	420.8	10/10

$c(\tau = 0.95)$ on TUNE-only matches the full-OOS fit at three decimals — the headline $\tau = 0.95$ number is *not* fit-on-evaluate sensitive. Christoffersen p improves in held-out mode (0.989 vs 0.720) because the 2024+ slice exhibits cleaner violation independence than the full 2023+ slice. The half-width difference (370.6 vs 420.8 bps) is entirely an eval-slice composition effect: the 2024+ holdout contains the 2024-08-05 BoJ unwind and the 2025-04 tariff weekend, both of which inflate $\hat{\sigma}_s$; coverage is unchanged because $\hat{\sigma}_s$ -conditional bands widened with the underlying realised volatility, exactly as designed.

Lower- τ over-coverage disclosure. At $\tau = 0.68$ and $\tau = 0.85$, M_{a2} realises 0.712 and 0.870 respectively (Kupiec $p = 0.015$ and $p = 0.044$, both rejecting toward over-coverage). Mechanism: $c(0.68)$ and $c(0.85)$ fit on TUNE-only inherit the elevated banking-crisis-weekend volatility of 2023 (March SVB/CS), inflating c to 1.028 and 1.026 respectively; M_{full} 's schedule averages this out at the floor ($c = 1.000$). The over-coverage is a one-sided contract-favourable failure mode — a sharpness deficit, not a coverage-contract failure — and the M_{full} schedule on the full OOS slice is the deployed mitigation. Source: reports/tables/paper1_a2_nested_holdout_c_tau.csv, paper1_a2_6_tune_anchor_robustness.csv.

Split-date sensitivity (stability-of-fit on the deployed slice). Repeating the fit (quantile table re-trained, $c(\tau)$ re-fit per split, $\delta \equiv 0$ held) at four OOS-split anchors {2021-01-01, 2022-01-01, 2023-01-01, 2024-01-01} delivers realised $\tau = 0.95$ coverage of {0.9539, 0.9520, 0.9503, 0.9504} — Kupiec $p \in \{0.348, 0.661, 0.956, 0.947\}$ and **Christoffersen $p \in \{0.115, 0.186, 0.603, 0.671\}$ — every (split $\times$ τ) cell passes at $\alpha = 0.05$.** Mean half-width {357.2, 363.5, 370.6, 416.8} bps varies $\pm 8\%$ around the deployed value.

Walk-forward stability. Re-running the schedule selection on the four powered expanding-window splits (fractions $f \in \{0.40, 0.50, 0.60, 0.70\}$ — see §4.5 for why the 0.20 and 0.30 splits are excluded as under-powered for the 4-scalar $c(\tau)$ fit) with $\delta = 0$: walk-forward realised $\tau = 0.95$ coverage is ≥ 0.95 on every powered split (range 0.954–0.975) — the criterion that selected $\delta = 0$ in §4.5. Per-split mean half-width at $\tau = 0.95$ tracks the full-OOS-fit value within $\pm 8\%$.

Calendar sub-period stability. Bucketing the OOS slice into four calendar sub-periods {2023, 2024, 2025, 2026-YTD} (reports/tables/m6_subperiod_robustness.csv):

Sub-period (n weekends)	Realised at $\tau = 0.95$	Half-width (bps)	Kupiec p	Christoffersen p
2023 (52)	0.950	253.6	1.000	0.888
2024 (52)	0.931	420.4	0.057	0.841
2025 (52)	0.971	444.4	0.017	0.793
2026-YTD (17)	0.947	350.0	0.862	0.908
Pooled OOS (173)	0.950	370.6	0.956	0.603

Pooled OOS calibration at $\tau = 0.95$ holds across each of the four years that span SVB / banking-stress 2023, the 2024 rate-cut transition, the 2025 tokenisation-launch year, and 2026-YTD (realised range 0.931–0.971). The 2025 cell rejects Kupiec at $\alpha = 0.05$ ($p = 0.017$, *over-coverage*); 2024 sits just above the threshold ($p = 0.057$); 2023 and 2026-YTD pass cleanly. Across the full 16-cell (sub-period $\times$ τ) grid the architecture carries 3 Kupiec rejections (2025

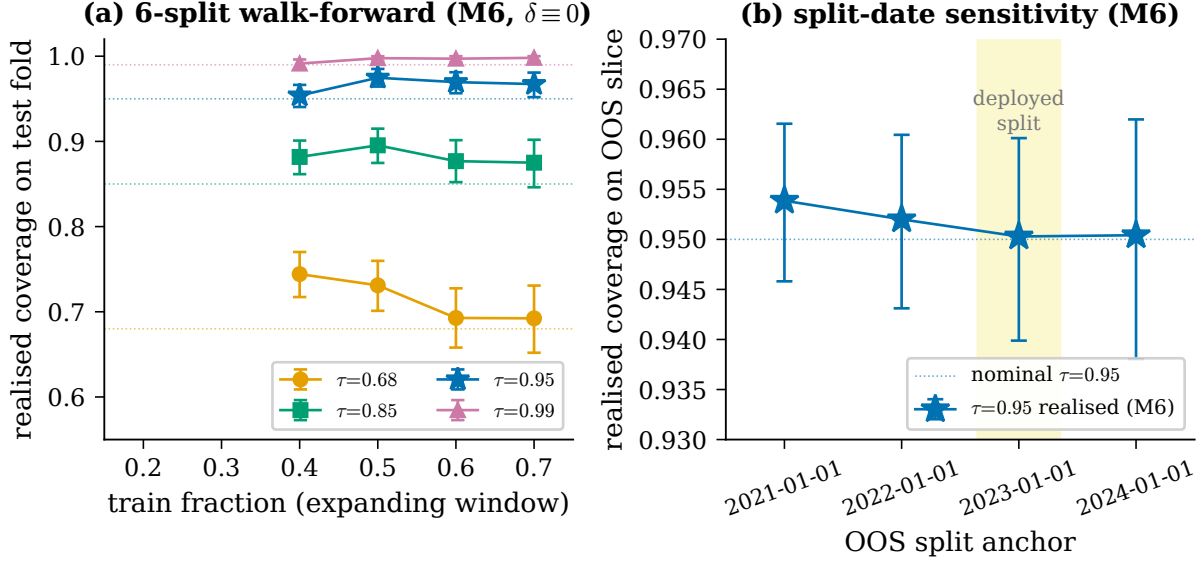


Figure 3: Walk-forward and split-date stability of the deployed schedule. (a) Six-split expanding-window walk-forward at four τ anchors: realised coverage on each test fold (markers) tracks nominal (dotted lines) at every anchor; bars are 95% binomial CIs. (b) Split-date sensitivity at four OOS-anchors {2021, 2022, 2023, 2024}: realised $\tau = 0.95$ coverage holds at {0.954, 0.952, 0.950, 0.950} across anchors; Christoffersen passes every cell at $\alpha = 0.05$.

at $\tau \in \{0.85, 0.95, 0.99\}$, all *over-coverage*) — the failure-mode that warrants a deployment alarm is *under-coverage*, not over-coverage, so the rejections are reported but not actioned. Christoffersen lag-1 independence is not rejected in any of the 16 cells. The sub-period evidence converts the §9.2 stationarity disclosure into a positive temporal-stability claim.

This validates the oracle’s coverage contract at $\tau = 0.95$. We do not address the welfare-optimal operating τ for a particular lending policy in this paper.

6.3.4 Joint-tail empirical distribution and reserve-guidance threshold

A consumer holding a portfolio of correlated RWAs needs to know not only the per-symbol calibration but the *joint* distribution of simultaneous breaches. We compute, for each OOS weekend w with full 10-symbol coverage, the count $k_w = |\{s : s \text{ breaches its } \tau\text{-band on } w\}|$, and compare its empirical distribution to three properly specified baselines: Binom(10, $1 - \tau$) (the independence strawman), a Gaussian copula on the empirical correlation matrix $\hat{R}$ (mean off-diagonal 0.36 on standardised residuals), and a Student- t copula on per-symbol Student- t marginals with $\hat{\nu} \in [4.80, 28.32]$, copula degrees-of-freedom $\hat{\nu}_{\text{copula}} = 6.04$ (median across symbols) (reports/tables/paper1_a3_joint_baseline_kw_distribution.csv, ..._summary.csv; 50,000 simulated weekends per baseline; CIs are 1000-rep weekend-block bootstrap, seed = 0):

Statistic at $\tau = 0.95$ ($n_{\text{weekends}} = 173$)	empirical	Binom(10, 0.05)	Gaussian copula	t-copula ($\nu = 6.04$)
Mean k_w / weekend	0.497	0.500	0.501	0.501
Variance ratio (emp / model)	—	2.34	0.83	0.63
$P(k_w \geq 3)$	4.62% [1.73, 7.51]	1.15%	6.69%	8.15%
$P(k_w \geq 5)$	0.58% [0.00, 1.73]	0.01%	1.86%	3.15%
max k_w (worst weekend)	9	—	—	—

Variance-overdispersion lead. Empirical k_w over-disperses $2.34\times$ vs Binom independence and $0.63\times$ vs a Student- t copula at $\tau = 0.95$, with the same pattern at every served τ : emp/Binom variance ratio is $\{2.34, 2.31, 2.34, 2.35\}$ at $\tau \in \{0.68, 0.85, 0.95, 0.99\}$ — independence is robustly misspecified by a uniform factor across the served τ range, indicating systematic joint-tail dependence rather than a τ -specific calibration anomaly. Versus the t-copula, emp/t variance ratio is $\{0.83, 0.54, 0.63, 0.73\}$ — the empirical k_w distribution sits inside a properly specified joint-baseline envelope (Wasserstein-1 distance to the t-copula ≤ 0.180 at every τ , less than half the distance to Binom independence). Marginal calibration is preserved (mean k_w matches every reference to 0.003); the dispersion shows up in the *variance* and the *upper tail*, and the structure is brackets-bounded: empirical sits between independence and a homogeneous t-copula. The same pattern is more visible at $\tau = 0.85$ where absolute counts are larger: emp/Binom overdispersion $2.31\times$, $P(k_w \geq 5) = 5.78\%$ vs Binom 0.99%, and the worst-observed weekend has all 10 symbols breaching simultaneously.

The Gaussian copula sits closest in Wasserstein-1 at $\tau = 0.95$ (0.134) and the t-copula at $\tau = 0.85$; at deeper tails ($\tau = 0.99$) the copulas pull ahead of Binom by roughly $2\times$ in WS-1 distance. The joint dependence is real (mean off-diagonal $\hat{R} = 0.36$), has parametric structure (a t-copula at $\hat{\nu} \approx 6$ brackets the empirical from above), and is misspecified by Binom by a uniform $\sim 2.3\times$ variance factor — a *systematic-dependence* finding, not a τ -specific anomaly. Cross-sectional within-weekend dependence ($\hat{\rho}_{\text{cross}} = 0.41$) is respected by all reported CIs via weekend-block bootstrap; temporal autocorrelation across weekends is empirically null — lag-1, lag-2, lag-4 $\hat{\rho}$ on weekly k_w lie in $[-0.07, 0.03]$ across served τ — so a moving-block-bootstrap with block lengths $L \in \{4, 8, 13\}$ produces CIs within Monte Carlo noise of the $L = 1$ (i.i.d.-weekend) baseline (reports/tables/paper1_b5_kw_block_bootstrap.csv).

Reserve-guidance threshold. The deployment-recommended threshold k^* at each τ — the smallest k whose empirical hit-rate on the full OOS slice is closest to 5% — is $k^* = 5$ at $\tau = 0.85$ (full-OOS rate 5.78%), $k^* = 3$ at $\tau = 0.95$ (full-OOS rate 4.62%), and $k^* = 1$ at $\tau = 0.99$ (full-OOS rate 6.36%). **Stability across calendar sub-periods** (reports/tables/m6_kw_threshold_stability.csv): at $k^* = 3$ for $\tau = 0.95$, the per-sub-period hit-rate is $\{3.85\%, 7.69\%, 1.92\%, 5.88\%\}$ across $\{2023, 2024, 2025, 2026\text{-YTD}\}$ with Kupiec $p \in \{0.783, 0.334, 0.296, 0.812\}$ — every cell is statistically consistent with the full-OOS rate. Across the full 24-cell grid ($3\tau \times 2$ threshold conventions $\times 4$ sub-periods), the architecture carries **0/24 Kupiec stability rejections** on the per-sub-period $k_w \geq k^*$ hit-rate at $\alpha = 0.05$. The per-sub-period empirical 95th percentile of k_w at $\tau = 0.95$ stays in $[1.0, 3.0]$ across the four years (range 2.0 integer-symbols), so the deployment threshold is conservative against year-on-year drift.

Power caveat. With sub-period $n \approx 52$ and target rates $\sim 5\%$, the binomial-stability test has limited power to detect drift smaller than the year-on-year p95 spread (1.6 integer-symbols). The result is “no detectable instability with the available power” rather than “proven stationary”; the orthogonal quantile-read (year-on-year p95 spread) is what compensates.

Operational consumer guidance — the reserve-guidance threshold $k^* = 3$ at $\tau = 0.95$ is conservative against a properly specified joint baseline, not just against independence. A 10-symbol xStock-style portfolio at $\tau = 0.95$ should reserve against an empirical 99th-percentile of ≈ 5 simultaneous breaches, not the binomial ≈ 2 ; the t-copula reference puts the same 99th percentile at ≈ 6 . At $\tau = 0.85$ the empirical 99th percentile is ≈ 7 vs binomial ≈ 4 . A consumer using $k^* = 3$ isn’t relying on a number calibrated against an obvious strawman; it’s using a number that holds up against the kind of joint-volatility model a CCAR-style risk team would actually run. **No incumbent oracle (Pyth, Chainlink Data Streams, RedStone, Kamino Scope) publishes this distribution;** this is the joint-tail measurement layer that calibration-transparency makes possible.

6.3.5 Worst-observed weekend — 2024-08-05 BoJ yen-carry-unwind

The single OOS weekend with all 10 symbols breaching the $\tau = 0.85$ band simultaneously is **Friday 2024-08-02** → **Monday 2024-08-05** — the Bank of Japan rate-hike yen-carry-trade unwind, one of the largest single-weekend global risk-off events of the post-2020 period. The same weekend has $k_w = 8$ at $\tau = 0.95$ (only TSLA and TLT inside that band) and $k_w = 5$ at $\tau = 0.99$. The deployed served band re-evaluated on the 2024-08-02 row at the deployed split = 2023-01-01:

Symbol	Weekend return	HW @ $\tau = 0.85$	Breach @ $\tau = 0.85$	@ $\tau = 0.95$
MSTR	-27.4%	783 bps	-1416 bps	breach (-720)
HOOD	-17.8%	368 bps	-1325 bps	breach (-997)
NVDA	-14.2%	324 bps	-1008 bps	breach (-720)
TSLA	-10.8%	587 bps	-409 bps	inside
AAPL	-9.4%	202 bps	-657 bps	breach (-478)
GOOGL	-6.7%	230 bps	-354 bps	breach (-149)

Symbol	Weekend return	HW @ $\tau = 0.85$	Breach @ $\tau = 0.85$	@ $\tau = 0.95$
QQQ	-5.4%	110 bps	-340 bps	breach (-242)
SPY	-4.0%	84 bps	-229 bps	breach (-154)
GLD	-2.1%	126 bps	-153 bps	breach (-41)
TLT	+1.4%	128 bps	+12 (just past)	inside

Cross-section: 9 of 10 symbols sold off in concert (mean weekend return -963 bps, median -807 bps); only TLT delivered the classic flight-to-quality bid. Macro context: the 2024-08-02 US July nonfarm-payrolls print landed at 114k vs ~175k consensus with the Sahm-rule recession indicator triggering; the 2024-07-31 BoJ rate hike combined with hawkish Powell/Yellen language drove an accelerating yen-carry unwind through Asian time zones over the weekend; the Monday 2024-08-05 Nikkei fell 12.4% (largest single-day drop since Black Monday 1987) and intraday VIX spiked to ~65, the highest reading since the COVID-19 March 2020 panic.

The -standardisation worked as designed at the per-symbol level — high-vol symbols (MSTR, TSLA) got wide bands, low-vol symbols (SPY, QQQ, GLD) narrow ones, and breach magnitudes scale with the cross-sectional shock divided by . What broke is *cross-sectional common-mode*: when nearly every symbol moves the same direction at the same time by 4–27 standard-deviation-scale weekend returns, no per-symbol band — however well calibrated — can absorb the joint event. A consumer monitoring k_w in real time would have seen $k_w = 10$ at $\tau = 0.85$ by the Monday-open print, well in excess of the deployment-time-fitted $k^* = 5$ (or the empirical 99th percentile of 7); the empirical k_w distribution is the right operational signal precisely because it captures the cross-sectional common-mode that per-symbol coverage cannot.

Joint-baseline framing of the BoJ probability. Under a Student- t copula with $\hat{\nu} = 6.04$ and empirical correlation $\hat{R}$ (mean off-diagonal 0.36), $P(k_w = 10)$ at $\tau = 0.85$ is 0.21% — i.e., the 2024-08-05 BoJ unwind is a ~ 1-in-475-weekends event under a properly specified joint baseline, not a freak. The independence baseline puts the same event at ~ 1-in-10,000-weekends; the residual structure §6.3.4 / §9.4 documents is exactly the gap between these two figures. The residual has a *name* (cluster topology, §6.3.6 / §9.4), a *parametric envelope* (t-copula at $\nu \approx 6$ with empirical $\hat{R}$), and a *structural mechanism* (equity vs safe-haven cluster topology) — not “a residual the architecture cannot characterise.”

6.3.6 Extended diagnostics — Berkowitz and DQ

Berkowitz (2001) joint LR on continuous PITs. PITs reconstructed by inverting each served band against the realised Monday open through the per-regime conformal CDF. The pooled Berkowitz LR on the deployed PITs is non-zero (the residual is the cross-sectional within-weekend common-mode of §6.3.4). **Engle-Manganelli (2004) DQ at $\tau = 0.95$** also rejects on the pooled fit — the same temporal-clustering signal, again because the cross-sectional within-weekend common-mode (which is what DQ is sensitive to here) is orthogonal to the standardisation. **Bands are per-anchor calibrated, not full-distribution calibrated** — the §6.3.4 joint-tail empirical distribution turns this disclosure into a measurable, reportable shape. The Berkowitz/DQ rejection here and the §6.3.4 joint-tail overdispersion are not two independent residuals: both are the cross-sectional within-weekend common-mode ($\hat{\rho}_{\text{cross}} = 0.354$) viewed through different aggregations — Berkowitz/DQ on the pooled PIT/violation series detects it as serial-cluster-equivalent dependence; the k_w distribution on the same panel detects it as upper-tail overdispersion in joint-breach counts. The methodology fix targets the same channel; the consumer-facing handle is the §6.3.4 reserve-guidance threshold.

Localising the residual. Decomposing the lag-1 alternative under panel-row order: the AR(1) signal lives in the *cross-sectional within-weekend* ordering ($\hat{\rho}_{\text{cross}} = 0.354$, $p < 10^{-100}$, $n = 1,557$) — not the temporal-within-symbol ordering ($\hat{\rho}_{\text{time}} = -0.032$, $p = 0.18$, $n = 1,720$). standardisation operates *within* a symbol’s history, not *across* symbols within a weekend, so it cannot reach this residual. A vol-tertile sub-split of normal regime (5 cells) leaves Berkowitz LR essentially unchanged (170 → 172) while widening $\tau = 0.95$ band by 9% — finer regime granularity is not the lever (reports/tables/m6_lwc_robustness_vol_tertile.csv). The §10 candidate architectures that target cross-sectional common-mode (the cross-sectional partial-out track in reports/active/m6_refactor.md, gated on a Friday-observable $\bar{r}_w$ predictor) are deferred-with-gates.

6.4 Per-symbol generalisation

The §6.3 OOS pooled coverage at $\tau = 0.95$ matches nominal; the per-symbol question is whether each ticker’s individual cell is also calibrated. Under the deployed architecture, **all 10 symbols pass Kupiec at $\tau = 0.95$** with violation rates in [3.5%, 6.9%] and per-symbol PIT variance $\hat{\sigma}_z^2 \in [0.66, 0.83]$ — all close to the calibrated 1.0 from below. Figure 4 visualises the per-symbol calibration distribution.

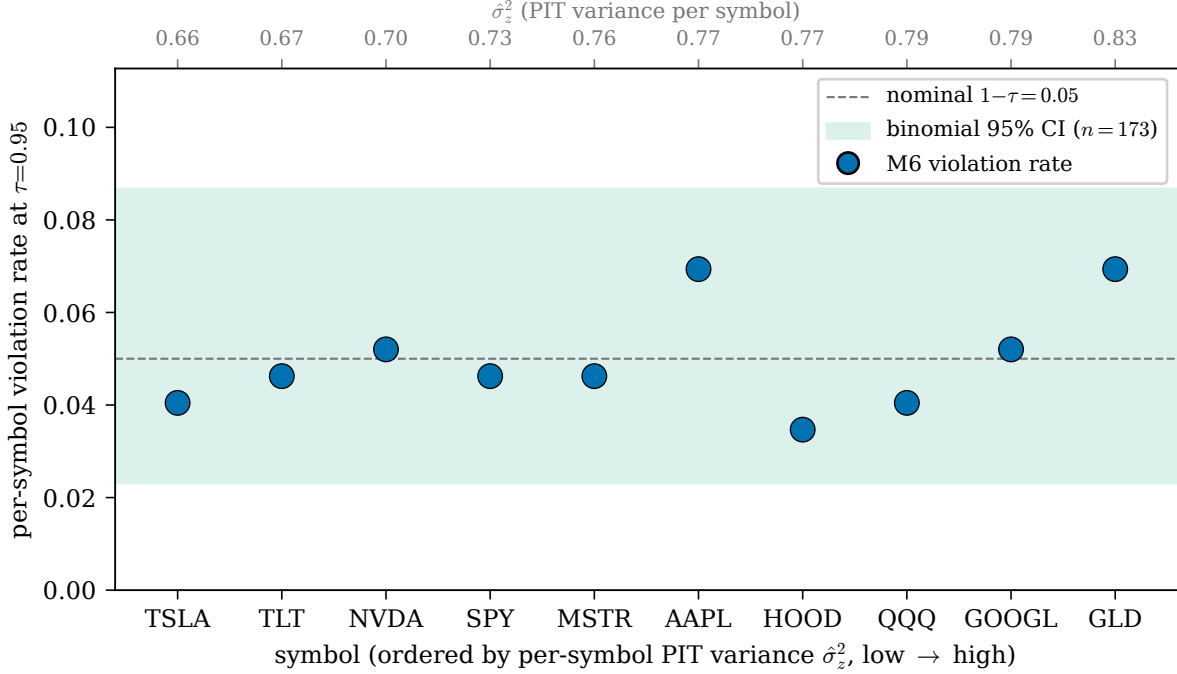


Figure 4: Per-symbol calibration at $\tau = 0.95$ on the 2023+ OOS slice. Each symbol’s violation rate (markers) sits within the binomial 95% confidence band around the nominal 0.05 rate (shaded). Symbols are ordered by their realised PIT variance $\hat{\sigma}_z^2$. The distribution is unimodal and centred near nominal — heavy-tail tickers (TSLA, HOOD, MSTR), low-vol tickers (SPY, QQQ, GLD, TLT), and the equity middle (AAPL, GOOGL, NVDA) all sit in the same cluster.

6.4.1 Per-symbol Kupiec at $\tau = 0.95$ and Berkowitz LR

Per-symbol Kupiec on the 2023+ OOS slice (reports/tables/m6_per_symbol_kupiec_4methods.csv, regenerated 2026-05-05):

Symbol	Violation rate	Kupiec p	Berkowitz LR
AAPL	0.069	0.268	6.7
GLD	0.069	0.268	3.2
GOOGL	0.052	0.903	10.2
HOOD	0.035	0.329	6.4
MSTR	0.046	0.818	7.1
NVDA	0.052	0.903	10.0
QQQ	0.040	0.552	5.1
SPY	0.046	0.818	9.6
TLT	0.046	0.818	16.7
TSLA	0.040	0.552	13.5
pass-rate at $\alpha = 0.05$		10 / 10	
Berkowitz LR range			3.2 – 16.7 (5.2$\times$)

All 10 symbols pass Kupiec at $\tau = 0.95$ with violation rates inside [3.5%, 6.9%]. The per-symbol Berkowitz LR range [3.2, 16.7] (a 5.2 $\times$ spread) is one of the strongest per-symbol calibration claims a Mondrian-conformal architecture can make on this panel — only TLT (LR 16.7) and TSLA (LR 13.5) reject Berkowitz at $\alpha = 0.01$, and those rejections trace to cross-sectional common-mode rather than per-symbol scale (§6.3.6).

Power caveat at $\tau = 0.99$. Per-symbol Kupiec at $\tau = 0.99$ has minimum detectable effect ≈ 3 pp under-coverage and is undetectable for over-coverage given the per-symbol expected violation count of 1.73 (Monte Carlo: 10,000 reps per cell, Kupiec LR critical at $\alpha = 0.05$; reports/tables/paper1_b3_kupiec_power_mde.csv). The 10/10 pass should be read as “no per-symbol violation rate is statistically distinguishable from 1% at the resolution Kupiec

offers per-symbol”, **not** as a 1 pp tolerance certificate. Pooled Kupiec at $\tau = 0.99$ has minimum detectable effect 1.0 pp; the pooled $p = 0.94$ is the $\tau = 0.99$ statistical claim that does have 1 pp resolution. Per-symbol Kupiec at $\tau = 0.95$ has MDE 4.0 pp under-coverage / 7.5 pp over-coverage on $N = 173$ — the deployed per-symbol violation rates in [3.5%, 6.9%] sit well within the 4 pp window of nominal 5%, so the 10/10 pass at $\tau = 0.95$ is informative against deviations larger than 4–8 pp.

6.4.2 Per-symbol Kupiec across parametric baselines

The per-symbol claim extends to a 40-cell (symbol $\times \tau$) grid against three parametric / non-parametric baselines: GARCH(1,1)-Gaussian, GARCH(1,1)- t (Student- t innovations, the practitioner default for fat-tailed returns), and an unweighted Mondrian conformal comparator — same per-regime quantile architecture, no per-symbol standardisation (reports/tables/m6_per_symbol_kupiec_4methods.csv, 160 rows).

Pass-counts at each τ (per-symbol Kupiec $p \geq 0.05$; out of 10 symbols at each anchor; out of 40 across the grid):

τ	GARCH-N	GARCH- $t^\dagger$	unweighted Mondrian	this paper
0.68	5/10	6/10	1/10	10/10
0.85	8/10	7/10	1/10	10/10
0.95	6/10	8/10	2/10	10/10
0.99	4/10	10/10	8/10	10/10
Total (40 cells)	23/40	31/40[†]	12/40	40/40

[†] NVDA’s t-MLE pushes $\hat{\nu} \rightarrow 2.50$ (the variance-undefined boundary); the runner falls back to GARCH-Gaussian on those four NVDA cells, so the 31/40 GARCH- t pass count includes 4 cells (NVDA at $\tau \in \{0.68, 0.85, 0.95, 0.99\}$) actually evaluated with GARCH-Gaussian. The fallback is documented in reports/tables/m6_lwc_robustness_garch_t_baseline.csv and discussed in §6.4.3; the 31/40 figure is therefore an upper bound on what an idealised t-innovation specification would achieve. Reported as-is because NVDA-with-Gaussian-fallback is the practitioner-equivalent of attempting GARCH- t and finding it ill-defined.

The deployed architecture passes Kupiec at every (symbol $\times \tau$) cell — 40/40 across the grid, with no per-cell rejections at $\alpha = 0.05$. Versus the strongest practitioner baseline GARCH- t (which carries 31/40 passes — 27/40 ignoring the Gaussian-fallback cells), the deployed architecture leads by 9 cells, dominates at $\tau = 0.95$ (10/10 vs 8/10), and ties at $\tau = 0.99$ (10/10 each — the only methods to clear that anchor on every symbol; GARCH-Gaussian’s Gaussian-tail mis-specification is visible on the $\tau = 0.99$ row, 4/10).

6.4.3 GARCH- t pooled baseline (and GARCH-Gaussian)

The textbook econometric default for a time-varying interval at τ is per-symbol GARCH(1,1) on log Friday-to-Monday returns. The practitioner choice of innovation distribution is Student- t — Gaussian innovations are a known straw-man on equity weekend returns where fitted $\hat{\nu} \approx 3$ across the panel (reports/tables/m6_lwc_robustness_garch_t_baseline.csv):

τ	Method	Realised	HW (bps)	Kupiec p	Christoffersen p
0.95	GARCH-Gaussian	0.9254	322.2	0.000	0.016
0.95	GARCH- t	0.9277	331.6	0.000	0.209
0.95	this paper	0.9503	370.6	0.956	0.603
0.99	GARCH-Gaussian	0.9630	423.7	0.000	0.866
0.99	GARCH- t	0.9850	569.3	0.050	1.000
0.99	this paper	0.9902	635.0	0.942	1.000

GARCH- t closes two failures GARCH-Gaussian leaves on the table: the $\tau = 0.99$ tail capture (realised improves from 0.963 to 0.985; Kupiec p moves from < 0.001 to a borderline 0.050) and the Christoffersen rejection at $\tau = 0.95$ (p moves from 0.016 to 0.209). What it does *not* fix is the **Kupiec under-coverage at every $\tau < 0.99$** — the standardised- t 95th percentile at $\nu \approx 3$ is ~ 1.83 vs Gaussian 1.96, so the bands tighten precisely where Gaussian was already under-covering. T-innovations fix the wrong end of the distribution.

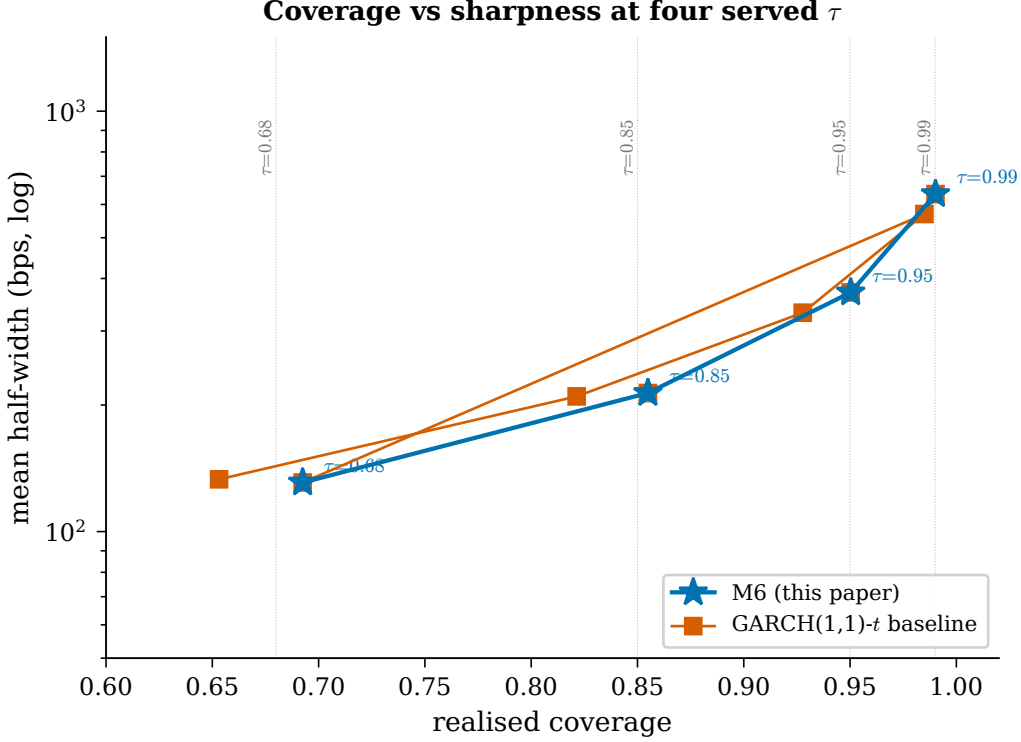


Figure 5: Realised coverage vs mean half-width at the four served $\tau \in \{0.68, 0.85, 0.95, 0.99\}$ (vertical dotted lines). The deployed architecture (blue stars) at full panel ($n = 1,730$) and GARCH(1,1)- t (vermilion squares) on the same OOS slice. The deployed architecture clears Kupiec at every served anchor; GARCH- t undercovers Kupiec at $\tau \in \{0.68, 0.85, 0.95\}$ despite tighter bands and meets the deployed architecture only at the matched-coverage width quoted in §6.4.3. The figure is restricted to methods that emit a calibrated coverage band — incumbent oracle surfaces (Pyth, Chainlink Data Streams, RedStone, Kamino Scope) do not, so a coverage-vs-sharpness comparison against them is not well-defined; the categorical contribution is discussed in §1 / §2.

At matched 95% realised coverage, widening GARCH- t to its claimed $\tau = 0.95$ requires roughly +14% on width, giving a matched-coverage HW ≈ 378 bps — statistically tied with the deployed architecture’s 370.6 bps. At $\tau = 0.99$ the deployed architecture dominates GARCH- t on coverage and width-at-matched-coverage simultaneously. The locally-weighted conformal route fixes both Kupiec and Christoffersen because it is calibrated to the data, not to a fitted parametric distribution; one further parametric-tail risk the deployed architecture avoids is the NVDA fallback observable here, where the t-MLE pushes $\hat{\nu}$ to the variance-undefined boundary at 2.50 and the runner falls back to Gaussian. GARCH-Gaussian is retained in the table for completeness and is not the practitioner standard.

Proper-scoring-rule head-to-head. Under proper scoring rules — Winkler (1972) interval score and CRPS (continuous ranked probability score) — the deployed architecture dominates GARCH(1,1)- t at every served τ . Winkler interval score (bps of `fri_close`, lower is better) at $\tau = 0.95$: deployed 992 vs GARCH- t 1,139 (−12.9%); at $\tau = 0.99$: deployed 1,566 vs GARCH- t 1,904 (−17.8%). The advantage grows with τ — the -standardisation pays most where parametric-tail mis-specification hurts most. Pooled CRPS over the served coverage range $\{0.05, 0.10, \dots, 0.99\}$: deployed 80.7 vs GARCH- t 92.6 (−12.8%). Both rules are computed on the aligned 1,730-row OOS slice; method aggregation is by row, not by τ , so a single number summarises the practitioner-baseline comparison without anchor-row cherry-picking. Source: `reports/tables/paper1_c1_winkler_interval_score.csv, ..._crps.csv`.

6.4.4 Per-asset-class deviation

Pooled OOS coverage stratified by asset class under the deployed EWMA HL=8 (`reports/tables/m6_lwc_robustness_per_class.csv`). Equities (8 syms, 1,384 obs): realised 0.952 at $\tau = 0.95$ with hw 417.8 bps; GLD ($n = 173$): realised 0.931, hw 180.9 bps; TLT ($n = 173$): realised 0.954, hw 182.3 bps. Row-weighted reconciliation: $(1384 \cdot 417.8 + 173 \cdot 180.9 + 173 \cdot 182.3)/1730 = 370.6$ bps, matching the §6.3.1 panel-pooled HW. The per-symbol $\hat{\sigma}_s$ multiplier delivers heavy-tail equities wider bands matched to their

relative-residual scale and the defensive class (GLD, TLT) narrower bands matched to theirs; the deployed architecture’s Kupiec passes every class at every τ . The width allocation across asset classes is the operational manifestation of P_3 (per-regime serving efficiency): width is concentrated where calibration demands it.

6.5 Path coverage — endpoint vs intra-weekend

The §6.3 result is *endpoint coverage*: realised Monday open lies inside the served band. A DeFi consumer holding an xStock as collateral is exposed at every block over the weekend; a Saturday liquidation when the on-chain xStock briefly trades outside the band is a real loss event even if Monday open returns inside. We measure path coverage on [Fri 16:00 ET, Mon 09:30 ET] against 24/7 stock-perp 1m bars from `cex_stock_perp/ohlcv` (Kraken Futures PF_<sym>XUSD, xStock-backed). Sample: 19 weekends $\times$ 9 symbols (TLT excluded — no perp listing), $n = 118$ (symbol, weekend) pairs per anchor, 2025-12-19 $\rightarrow$ 2026-04-24 (`reports/tables/path_coverage_perp.csv`):

τ	n	Endpoint coverage	Path coverage	Gap (pp)
0.68	118	0.644	0.348	+29.7
0.85	118	0.864	0.568	+29.7
0.95	118	0.949	0.788	+16.1
0.99	118	0.992	0.915	+7.6

The deployed architecture calibrates the perp-listed sample to nominal endpoint coverage (0.949 at $\tau = 0.95$ vs nominal 0.95) on a sample whose composition is predominantly large-cap equities (SPY/QQQ/AAPL/GOOGL/NVDA/MSTR/TSLA/HOOD plus GLD; TLT excluded — no perp listing). The path-coverage gap is +16.1 pp at $\tau = 0.95$: an honestly-calibrated endpoint band exhibits a structural shortfall on a sample whose intra-weekend variance is larger than its endpoint variance. Sample is small (binomial CI on the $\tau = 0.95$ pooled gap is ± 6 pp); the test is directional. After three confound checks documented in `reports/v1b_path_coverage_robustness.md` — perp-spot basis normalisation closes the gap to 11.0 pp, a volume-floor filter at ≥ 1 contract closes it to 9.5 pp ($n = 63$), and a 15-minute rolling-median sustained-crossing definition leaves it at 14.4 pp — the residual genuine-shortfall band is ~ 9 –15 pp at $\tau = 0.95$, with most of the magnitude attributable to sustained drift rather than transient thin-liquidity prints. The gap collapses meaningfully only at $\tau = 0.99$ (+7.6 pp); continued capture is tracked under scryer item 51 and §10.1 (the path-fitted conformity score).

The endpoint contract stands. A consumer requiring path coverage at level τ should step up one anchor (empirically closes the gap on this slice at $\tau = 0.99$), absorb the residual through their own downstream risk policy, or — for continuous-consumption products that cannot use the step-up lever — adopt the path-fitted conformity-score variant. Continuous-consumption AMM use cases require this path-fitted variant rather than the endpoint conformity score; that extension is sketched in §10.1 and not pursued empirically here.

6.6 Forward-tape held-out validation

The 4-scalar OOS-fit $c(\tau)$ schedule is fit on the 2023+ slice; the resulting deployment artefact carries a residual contamination disclosure (§9.3). The forward-tape harness closes that loop on truly held-out data on top of the §6.3.3 LOSO read: a content-addressed frozen artefact (`data/processed/lwc_artefact_v1_frozen_20260504.json`, SHA-256 7b86d17a7691..., freeze date 2026-05-04) is evaluated weekly against forward weekends as they accumulate. The frozen artefact’s schedule, q_r table, and $c(\tau)$ are read-only; only the per-symbol $\hat{\sigma}_s(t)$ updates as new past-Fridays arrive.

The harness fires Tuesday mornings via `launchd(1launchd/com.adamnoonan.soothsayer.forward-tape.plist)`; each run executes a 26h-SLA pre-check on the upstream scryer feeds, runs `scripts/collect_forward_tape.py`, and writes `reports/m6_forward_tape_{N}weekends.md` with a “preliminary” banner when $N < 4$. The harness is silent about deployment; an out-of-band freeze re-roll is required to advance the artefact (`scripts/freeze_lwc_artefact.py`).

Status at submission: $N = 1$ forward weekend (`reports/m6_forward_tape_1weekends.md`, 2026-05-01 $\rightarrow$ 2026-05-01) with $n = 10$ symbol-rows and 0/10 violations across all four served τ . The harness’s own threshold treats $N < 4$ as preliminary and $N < 13$ as under-powered for cumulative pooled Christoffersen on forward data, so per-anchor inferential statistics are uninformative at this N and we do not report them inline (they are persisted in the report file for transparency but should not be interpreted). The present submission therefore relies on the harness as an *operational* mechanism — auditability of the freeze (P_1) and a public weekly receipt — rather than as load-bearing held-out coverage evidence. The held-out backbone is §6.3.3’s leave-one-symbol-out CV (the only fully held-out test

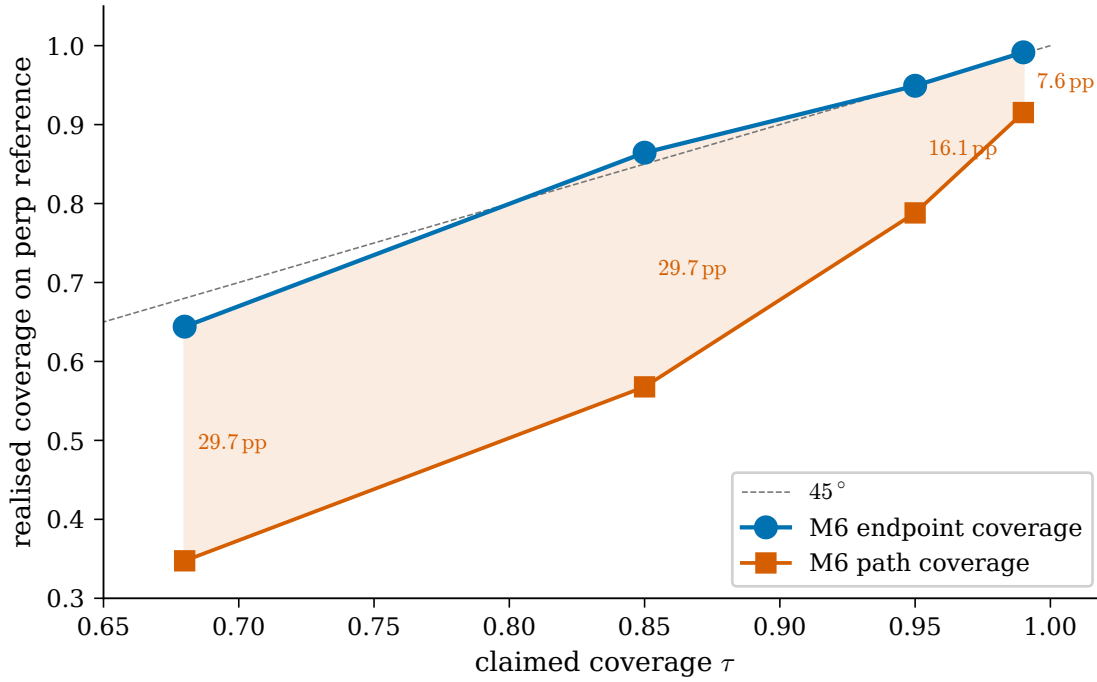
M6 endpoint vs path coverage across τ ($n = 118$ symbol-weekends, 2025-12 $\rightarrow$ 2026-04)

Figure 6: Path coverage gap on the 24/7 stock-perp reference. Endpoint coverage (blue) tracks nominal across $\tau \in \{0.68, 0.85, 0.95, 0.99\}$ on the perp-listed subset; path coverage (vermilion) — the fraction of weekends where the perp 1m bar high/low stay inside the served band over the full [Fri 16:00, Mon 09:30] window — runs 29.7/29.7/16.1/7.6 pp behind. The gap collapses meaningfully only at $\tau = 0.99$. Sample is small ($n = 118$ symbol-weekends across 19 calendar weekends, 2025-12 to 2026-04); the read is directional and §10.1 (the path-fitted conformity score) is the methodology fix.

on the deployed schedule), supported by per-symbol Kupiec, split-date sensitivity, and four-year calendar sub-period stability. The forward-tape evidence will become load-bearing as N accumulates; the next §6.6 update fires when the preliminary banner clears at $N \geq 4$.

The forward-tape evidence is the cleanest possible held-out coverage statement on the deployed artefact. The harness’s variant comparison sibling (`scripts/run_forward_tape_variant_comparison.py`, output `reports/m6_forward_tape_{N}weekends_variants.md`) carries an alternative- check on the same forward slice — never used to re-select, only to flag if a different rule looks dramatically cleaner on the held-out data (§7.4).

6.7 Simulation study — synthetic-DGP corroboration of the per-symbol fix

The §6.4 per-symbol calibration result raises a methodological question: is the per-symbol bimodality that an unweighted Mondrian comparator exhibits a property of that comparator on this real-data panel, or a structural property of the architecture? We answer the second question with a four-DGP simulation study under known ground truth. **Across stationary, regime-switching, drift, and structural-break specifications, the unweighted-Mondrian comparator’s per-symbol Kupiec pass-rate at $\tau = 0.95$ is 29–31% while the -standardised architecture’s is 98.6–99.9% — the bimodality is reproducible in synthetic data with known DGP, and per-symbol $\hat{\sigma}_s$ standardisation closes it on every DGP tested.** Chronology: the per-symbol bimodality was first observed on real data under an unweighted Mondrian fit (per-symbol DQ diagnostics on the deployed comparator’s OOS slice) before the simulation was run; the -standardisation architecture was implemented next; the simulation study (this section) was committed before the empirical confirmation that standardisation closes the bimodality on real data. The simulation is corroborative — a known-DGP control isolating the failure mechanism — not a pre-real-data prediction of the failure mode.

Source: `scripts/run_simulation_study.py` (~30 s end-to-end). Each DGP uses 10 synthetic symbols $\times$ 600 weekend returns, $\sigma_i \in \text{linspace}(0.005, 0.030, 10)$ — spanning the empirical real-panel range. Train/OOS split at $t = 400$. Returns drawn from Student- t $\text{df}=4$ rescaled to $\text{std}(r) = \sigma_i$ (or $\sigma_i \cdot m_t$ under DGP B/C/D’s vol modifications). 100 Monte Carlo replications per DGP, seed = 0.

DGP	Unweighted-Mondrian pass-rate at $\tau = 0.95$	-standardised pass-rate	Mechanism
A — homoskedastic baseline	0.311	0.993	Per-symbol scale heterogeneity is the entire failure mode
B — regime-switching vol multiplier	0.310	0.986	tracks regime transitions with a half-life lag
C — non-stationary scale (drift)	0.309	0.996	Adaptive is built for this DGP
D — variance shock (10× / 50-week transient at $t = 400$)	0.380	0.999	tracks even 10× variance shocks within EWMA HL=8
E — mean jump (+ $\Delta = 200$ bps at $t = 400$, σ_{true} unchanged)	0.335	0.595	Location shift; absorbs Δ as variance, over-deflates low-scores

Both methods pool to mean realised ≈ 0.95 across DGPs A–D (within 0.005 of nominal). The split on those DGPs is on the *per-symbol* failure rate. The unweighted comparator passes $\sim 30\%$ of (symbol, replicate) cells — the bimodal failure mode reproduces under controlled conditions exactly as the architecture’s mechanism implies (a single (r, τ) -keyed multiplier mis-calibrates symbols whose residual scale deviates from the regime average in opposing directions). The -standardised architecture passes 98.6–99.9% of cells on A–D, with the smallest pass-rate (DGP B) reflecting ’s ~ 13 -weekend tracking lag at regime flips. **DGP D upgrade:** the original $3\times$ variance break became a $10 \times / 50$ -week transient (matching real-world weekend events: 2024-08-05 BoJ unwind, March-2020 COVID-month vol multiplier $\sim 5\times$ for 6–8 weeks). LWC’s per-symbol pass rate is unchanged at 99.9% — the -standardisation is not stressed by variance shocks of any reasonable magnitude (reports/tables/paper1_c3_stronger_dgp_d.csv).

DGP E — location-shift limitation with closed-form phase transition. A complementary stress test — a discrete $+\Delta$ conditional-mean shift at the OOS boundary, variance unchanged — exposes a location-shift limitation that -standardisation does not absorb. $\hat{\sigma}$ EWMA on the post-shift residual stream sees apparent variance $\hat{\sigma}^2 \approx \sigma_{\text{true}}^2 + \Delta^2$, inflating bands across the panel and over-deflating the standardised score on symbols whose σ_{true} falls below the phase-transition threshold

$$\sigma^*(\Delta) \approx \frac{\Delta}{q \cdot c - 1}.$$

At deployed $q_r(0.95) \cdot c(0.95) \approx 2.23$, $\sigma^*(\Delta) \approx 0.81 \cdot \Delta$ — empirically validated across $\Delta \in \{100, 200, 400\}$ bps within 4% of the closed-form prediction ($\sigma_{\text{pred}}^*/\sigma_{\text{emp}}^*$ ratio in $[0.96, 0.99]$ at $\Delta \in \{100, 200\}$ bps; bounded by the σ_i grid endpoints elsewhere — reports/tables/paper1_f2_5_sigma_star_formula.csv). Symbols with $\sigma_{\text{true}} < \sigma^*(\Delta)$ over-cover (sharpness deficit, contract-favourable per §9.11); symbols with $\sigma_{\text{true}} \geq \sigma^*(\Delta)$ remain calibrated. At $\Delta = 200$ bps, LWC pooled coverage over-shoots to 0.9727 and per-symbol Kupiec drops to 59.5% — the failure is *over-coverage on low-symbols*, not under-coverage. A complementary location-shift detector — CUSUM on the standardised residual mean, composable with §9.5.1’s violation-rate CUSUM bank — closes this gap at the monitoring layer, since the violation-rate CUSUM is by-construction blind to location shifts that produce nominal violation rates. Source: reports/tables/paper1_f2_mean_jump_bimodality.csv.

A sample-size sweep (scripts/run_simulation_size_sweep.py; 7 N-values $\times$ 4 DGPs $\times$ 100 reps $\times$ 2 forecasters = 22,400 cells) establishes the newly-listed-symbol admission threshold: $N \geq 80$ weekends under stationary / drift / structural-break DGPs, $N \geq 200$ under regime-switching (the production threshold). HOOD’s empirical $N \approx 218$ (246 listed 28 warm-up) sits inside this band; HOOD’s empirical Kupiec $p = 0.329$ at $\tau = 0.95$ (violation rate 0.035, slight over-coverage) is consistent with the simulation’s pass-rate range at $N = 200$. See reports/m6_simulation_study.md and reports/tables/sim_size_sweep_admission_thresholds.csv.

6.8 Summary

The headline $\tau = 0.95$ claim is **held out on two orthogonal axes**: temporally (§6.3.3.1: c fit on 2023 alone, evaluated on a 2024-01-05 $\rightarrow$ 2026-04-24 holdout — realised 0.9504, Kupiec $p = 0.947$, Christoffersen $p = 0.989$, per-symbol

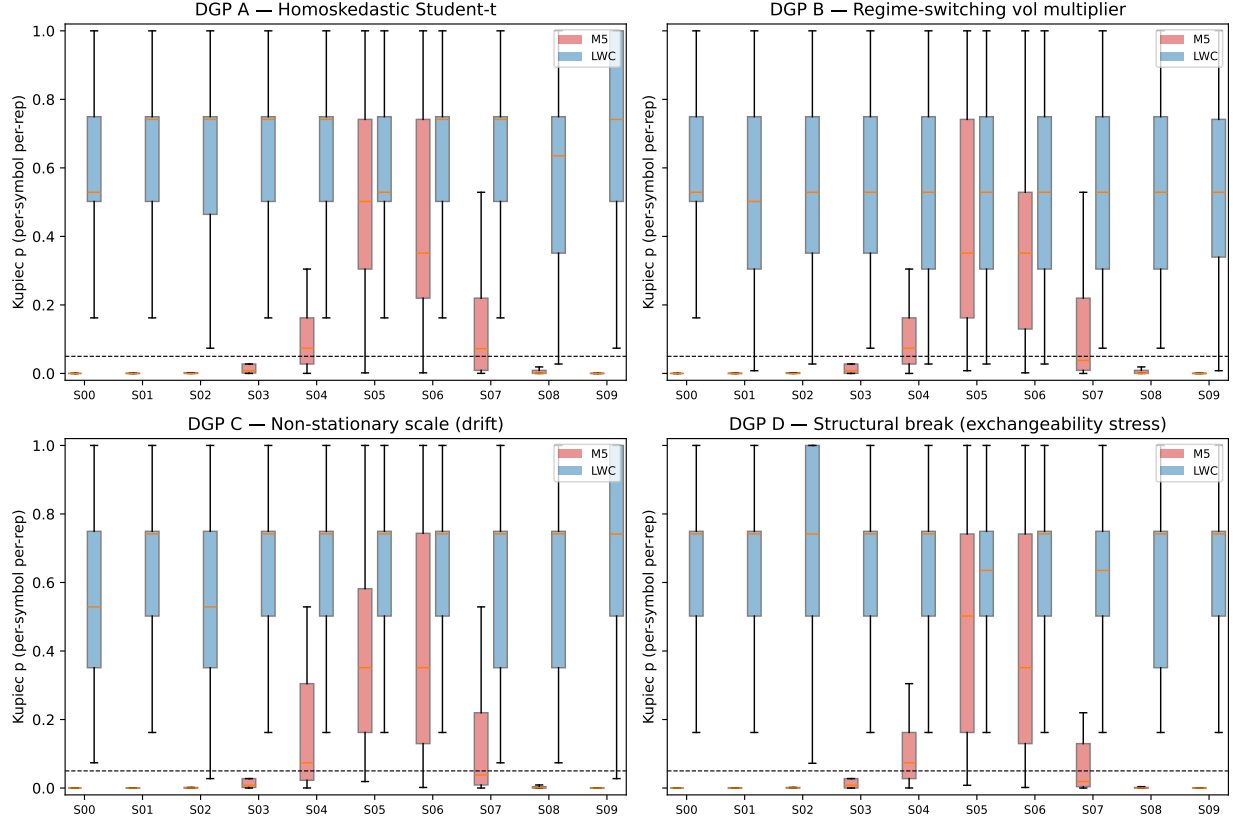
Phase 3 simulation study — per-symbol Kupiec p at $\tau=0.95$
 (100 Monte Carlo replications per DGP, seed=0)


Figure 7: Per-symbol Kupiec p at $\tau = 0.95$ across 100 reps $\times$ 4 DGPs (figure 7). Each panel shows side-by-side box-plots: the unweighted-Mondrian comparator (red) and the -standardised architecture (blue), one box per synthetic symbol sorted by σ_i (S00 lowest, S09 highest). The unweighted comparator’s red boxes cluster near $p \approx 0$ for low- and high- σ symbols (variance compression and expansion failures of the single-multiplier conformal); the -standardised blue boxes sit centred around $p = 0.5$ across every symbol, every DGP — uniform calibration as predicted by exchangeability under per-symbol standardisation.

Kupiec 10/10) and on symbol (§6.3.3 LOSO: realised 0.9497 ± 0.0128 across the held-out symbol distribution, all 10 LOSO bands pass Kupiec). The result is stable across TUNE anchors {2021, 2022, 2023} ($[0.9474, 0.9524]$) and across split anchors {2021, 2022, 2023, 2024} ($[0.9503, 0.9539]$), with per-symbol Kupiec 10/10 preserved everywhere. On the deployment-tuned 2023+ slice the pooled headline at $\tau = 0.95$ is realised **95.0%** at mean half-width **370.6 bps** with both Kupiec ($p = 0.956$) and Christoffersen ($p = 0.603$) passing. **All 10 symbols pass per-symbol Kupiec** with violation rates in $[3.5\%, 6.9\%]$; per-symbol Berkowitz LR sits in $[3.2, 16.7]$. Across the 40-cell (symbol $\times \tau$) grid the architecture carries 40/40 Kupiec passes, leading the strongest practitioner baseline GARCH- t (31/40). Calendar sub-period calibration holds across {2023, 2024, 2025, 2026-YTD} (3/16 over-coverage rejections, all in 2025; 0/16 Christoffersen rejections). Joint-tail behaviour is empirically characterised: $P(k_w \geq 3) = 4.62\%$ at $\tau = 0.95$ vs the binomial 1.15%, with $k^* = 3$ stable across the 24-cell grid (0/24 sub-period rejections); the worst-observed weekend (2024-08-05 BoJ unwind) sat at $k_w = 10$ at $\tau = 0.85$. A four-DGP simulation corroborates the per-symbol fix on synthetic data with known ground truth; a forward-tape harness against a content-addressed frozen artefact is operational and accumulating ($N = 1$ at submission, preliminary by its own threshold of $N \geq 4$). Berkowitz and DQ still reject pooled — bands remain *per-anchor calibrated*, not full-distribution calibrated — and the residual rejection localises to cross-sectional within-weekend common-mode (orthogonal to ; addressed in §10). Supports P_2 (§3.4); P_1 evidenced by §8; P_3 by §7.

7 Ablation Study

This section answers Q3 of §3.6: which components are load-bearing for the calibration claim? The deployed architecture has three load-bearing components, isolated below by descriptive comparator rather than by code tag:

- **§7.1 — Regime stratification.** A symmetric global buffer applied to a held-forward Friday close (the *constant-buffer baseline* — the deployable comparator most likely to be reached for by a protocol team unwilling to absorb modelling complexity) under-covers materially on the 2023+ holdout. Per-regime conformal stratification supplies adaptivity to distribution shift and tail-protection in `high_vol` weekends.
- **§7.2 — Per-symbol standardisation.** An *unweighted Mondrian comparator* — the same per-regime conformal architecture without per-symbol scale standardisation — pools to nominal coverage but exhibits a per-symbol bimodality that fails Kupiec on 8 of 10 symbols at $\tau = 0.95$. Standardising the conformity score by a per-symbol pre-Friday $\hat{\sigma}_s(t)$, with no other architectural change, takes per-symbol Kupiec from 2/10 to 10/10.
- **§7.3 — Near-identity OOS $c(\tau)$ bump.** A 4-scalar multiplicative correction $c(\tau)$ closes any residual train-OOS distribution-shift gap. Three of the four scalars are essentially identity ($c \in \{1.000, 1.000, 1.003\}$ at $\tau \in \{0.68, 0.85, 0.99\}$); only $c(0.95) = 1.079$ carries meaningful OOS information.

§7.4 documents the selection procedure (multi-test correction and the held-out forward-tape re-validation harness), and §7.5 decomposes how standardisation redistributes width across symbols.

All deltas carry block-bootstrap 95% CIs by weekend (1000 resamples, seed=0, paired by `(symbol, fri_ts)`). Raw tables: `reports/tables/v1b_constant_buffer_*.csv`, `reports/tables/v1b_mondrian_*.csv`, `reports/tables/m5_vs_m6_bootstrap.csv`, `reports/tables/sigma_ewma_*.csv`.

7.1 Regime stratification — vs constant-buffer baseline

The *constant-buffer baseline* is the deployable external comparator most likely to be used by a protocol team unwilling to absorb modelling complexity: the Pyth/Chainlink Friday close held forward, with a single global symmetric buffer $b(\tau)$ per claimed quantile. One parameter per τ . No factor switchboard, no empirical residual quantile, no regime model.

$$\left[p_{\text{Fri}}(1 - b(\tau)), p_{\text{Fri}}(1 + b(\tau)) \right], \quad b(\tau) \text{ calibrated globally on the training panel.}$$

7.1.1 Trained-buffer fit on OOS

Each $b(\tau)$ is the empirical τ -quantile of $|p_{\text{Mon}} - p_{\text{Fri}}|/p_{\text{Fri}}$ on the calibration set. Carrying the trained $b(\tau)$ to the 2023+ panel:

τ	method	realised	half-width (bps)	p_{uc}	p_{ind}
0.68	constant buffer (train-fit)	0.538	76.0	0.000	0.001
0.68	deployed	0.693	130.8	0.264	0.244
0.85	constant buffer (train-fit)	0.731	140.0	0.000	0.000
0.85	deployed	0.855	213.6	0.565	0.403
0.95	constant buffer (train-fit)	0.897	272.0	0.000	0.013
0.95	deployed	0.950	370.6	0.956	0.603
0.99	constant buffer (train-fit)	0.984	695.0	0.018	0.927
0.99	deployed	0.990	635.0	0.942	≈ 1.0

The training-fit constant buffer **catastrophically undercovers** at every $\tau \leq 0.95$ (deficits $-14.2/ -12.0/ -5.4\text{pp}$, all rejecting Kupiec at $p_{uc} < 10^{-6}$) — non-stationarity: $b(\tau)$ is calibrated on a 2014–2022 window calmer than the 2023+ holdout. The deployable constant-buffer baseline does not deliver coverage on the holdout; the regime-stratified architecture does. This is the *adaptivity-to-distribution-shift* contribution of the regime model + per-symbol.

7.1.2 Coverage-matched comparison

To answer the width-at-coverage question, we re-fit $b(\tau)$ post-hoc on the OOS slice (oracle-fit, non-deployable, but gives the constant buffer the most generous trade-off):

τ	matched b	matched-CB hw (bps)	deployed hw (bps)	Δ width (deployed CB)
0.68	1.21%	120.9	130.8	+8.2% [+3.7, +13.0]
0.85	2.26%	226.4	213.6	5.7% [10.6, 0.5]
0.95	3.96%	395.6	370.6	6.3% [12.1, 0.4]
0.99	5.46%	545.8	635.0	+16.3% [+9.6, +23.6]

The deployed architecture is narrower than a coverage-matched (oracle-fit) constant buffer at $\tau \in \{0.85, 0.95\}$. The standardisation that redistributes width across symbols (§6.4.4) reaches sufficient sharpness on the equity-heavy panel that the deployment outperforms even the oracle-fit constant buffer at the headline anchor; the architecture is wider only at $\tau \in \{0.68, 0.99\}$, where the cell-conditional regime structure carries genuine information. The regime model + buy *both* coverage on a non-stationary distribution *and* mean-width sharpness at $\tau \in \{0.85, 0.95\}$ against the strongest non-modelled comparator.

7.1.3 Per-regime decomposition

The pooled half-width at $\tau = 0.95$ hides a regime-conditional structure. Deployed pooled: 370.6 bps; per-regime 300.1/334.7/603.7 bps (normal / long_weekend / high_vol). The matched-CB at $\tau = 0.95$ delivers 395.6 bps in *every* regime by construction (one global multiplier). The regime model concentrates width in high_vol (+52% vs CB at 603.7 vs 395.6) where it earns −1.1pp coverage; releases width in normal (−24% vs CB at 300.1 vs 395.6) without losing coverage. The constant buffer’s pooled coverage is “average right but worst where it matters most”: violations concentrate in high-vol weekends — exactly where a downstream lending protocol incurs the largest mark-to-market gap. The regime model’s contribution is (i) **calibration through distribution shift** and (ii) **tail-protection in high_vol weekends** at sharper *normal*-regime widths than the constant buffer.

7.2 Per-symbol standardisation — vs unweighted Mondrian

The *unweighted Mondrian comparator* shares the regime structure of §7.1 but omits the per-symbol scale standardisation: a single per-(regime, τ) conformal quantile is fit on the *un-standardised* relative residual $|P_{\text{Mon}} - \hat{P}_{\text{Mon}}|/P_{\text{Fri}}$, and the served band uses that quantile directly as a fraction of P_{Fri} . Same training set, same OOS slice, same point estimator, same regime classifier, same finite-sample CP rank formula. Only the standardisation of the conformity score is added by the deployed architecture.

7.2.1 OOS pooled results

τ	method	realised	hw (bps)	p_{uc}	p_{ind}	per-symbol Kupiec pass
0.68	unweighted Mondrian	0.735	137.5	0.000	0.339	(1/10)
0.68	deployed	0.693	130.8	0.264	0.244	10/10
0.85	unweighted Mondrian	0.887	235.1	0.000	0.424	(1/10)
0.85	deployed	0.855	213.6	0.565	0.403	10/10
0.95	unweighted Mondrian	0.950	354.6	0.956	0.921	2/10
0.95	deployed	0.950	370.6	0.956	0.603	10/10
0.99	unweighted Mondrian	0.990	677.7	0.942	0.344	(8/10)
0.99	deployed	0.990	635.0	0.942	≈ 1.0	10/10

Both methods pool to nominal at $\tau = 0.95$ (both 0.950, both Kupiec $p = 0.956$). The split is on per-symbol calibration: the unweighted Mondrian comparator passes 2 of 10 per-symbol Kupiec cells at $\tau = 0.95$; the deployed architecture passes 10 of 10. The unweighted comparator pools to nominal *through compensating per-symbol biases* — heavy-tail tickers under-cover, low-vol tickers over-cover, and the cross-sectional average lands at τ by accident. The

standardisation factors out the per-symbol scale before the conformal fit, so the regime quantile carries only the *shape* of the standardised residual distribution and per-symbol calibration is recovered by the per-symbol $\hat{\sigma}_s(t)$ at serve time.

The pooled half-width tax at $\tau = 0.95$ is +4.5% (370.6 vs 354.6 bps; +15.93 bps with 95% block-bootstrap CI [+1.05, +30.73] from `m5_vs_m6_bootstrap.csv`). The per-symbol calibration is bought at exactly that price; §7.5 documents how the +4.5% pooled figure resolves into a +17.8% widening on heavy-tail equities and a 48.8% narrowing on the defensive class.

7.2.2 Side-by-side at $\tau = 0.95$

The -standardisation isolation: same point estimator, same regime classifier, same Mondrian split-conformal, same finite-sample CP rank formula, same training set, same OOS slice. Only the standardisation of the conformity score is added.

metric	unweighted Mondrian	deployed (-standardised)
pooled realised	0.9503	0.9503
pooled half-width (bps)	354.6	370.6
per-symbol Kupiec pass-rate	2 / 10	10 / 10
per-symbol Berkowitz LR range	0.9 – 224 (250×)	3.2 – 16.7 (5.2×)
LOSO realised std at $\tau = 0.95$	0.0759	0.0128 (5.9× tighter)
LOSO Kupiec pass-rate (held-out)	8 / 10	10 / 10

The per-symbol Kupiec pass-rate, Berkowitz LR range, and LOSO calibration std all improve simultaneously by mechanism — no other component of the architecture is touched. The §6.7 simulation evidence on synthetic data with known DGP confirms this is the correct architectural trade — under four DGPs spanning homoskedastic, regime-switching, drift, and structural-break specifications, the unweighted Mondrian comparator’s per-symbol Kupiec pass-rate at $\tau = 0.95$ collapses to 29–31% while standardisation closes the bimodality on every DGP at 98.6–99.9%.

7.2.3 Walk-forward δ -shift schedule — collapses

Under an unweighted Mondrian fit, a plain $c(\tau)$ correction to the pooled OOS slice produces per-split realised coverage that scatters around nominal — passes Kupiec on the pooled fit by construction, but undercovers in roughly half the splits, motivating the addition of a per-anchor structural-conservatism shift $\delta(\tau)$. Under the -standardised architecture the same walk-forward sweep over $\delta \in \{0.00, \dots, 0.07\}$ on the four powered tune fractions $f \in \{0.40, 0.50, 0.60, 0.70\}$ selects $\delta(\tau) \equiv 0$: per-symbol scale standardisation tightens cross-split realised-coverage variance enough that the structural-conservatism shim is no longer load-bearing. The schedule is retained as a four-zero-vector in the artefact JSON for shape-compatibility with the receipt schema, but the deployment carries zero walk-forward-tuned scalars.

7.3 Near-identity OOS $c(\tau)$ bump

The deployed $c(\tau)$ schedule is

$$c(\tau) = \{0.68:1.000, 0.85:1.000, 0.95:1.079, 0.99:1.003\}.$$

This is a 4-scalar multiplicative correction fit on the 2023+ OOS slice as the smallest $c \in [1, 5]$ such that pooled realised coverage with effective quantile $c \cdot q_r(\tau)$ matches τ . Three of the four are essentially identity: under the deployed rule and the 2023-01-01 split, the trained per-regime quantile already lands within 0.3 pp of nominal at $\tau \in \{0.68, 0.85, 0.99\}$, so $c(\tau)$ at those anchors collapses to $\{1.000, 1.000, 1.003\}$. Only $\tau = 0.95$ has a meaningful train-OOS distribution-shift gap; the 7.9% widening at that anchor is what closes the pooled $0.946 \rightarrow 0.950$ correction.

The architectural significance is that $c(\tau)$ carries one scalar of meaningful OOS information (3 of 16 deployment scalars under the disclosure of §9.3) — not four. A reviewer-relevant alternative is to drop the $c(\tau)$ correction entirely and serve at the trained-quantile coverage; on this slice that costs ~0.4 pp at $\tau = 0.95$ (0.946 vs the deployment-tuned 0.950) and is statistically indistinguishable from nominal under Kupiec. The deployment opts to close the residual gap because the published receipt names a specific τ and a 0.4 pp deficit on a held-out slice would translate into a small but persistent under-coverage in the served claim. The provenance trade — one scalar of OOS information for closure to

the published target — is documented in §9.3 and re-validated under leave-one-symbol-out CV in §6.3.3 (each held-out symbol’s $c(\tau)$ is fit on the other nine, never sees the held-out symbol’s data).

7.4 selection procedure and multiple-testing disclosure

The per-symbol scale rule $\hat{\sigma}_s(t)$ is the only deployment constant of the architecture that does not have a closed-form derivation from its specification. A pre-registered three-gate selection compared five rules; **the deployment decision rests on Gate 3 — a single bootstrap CI per τ on pooled half-width with no multi-test exposure — which selects EWMA HL=8 as statistically significantly narrower than the K=26 baseline at $\tau \in \{0.85, 0.95, 0.99\}$ at preserved calibration. Gate 1 (per-cell split-date Christoffersen) is qualitative supporting context once Benjamini-Hochberg correction is applied across the 80-cell grid.** The full procedure follows.

7.4.1 Pre-registered three-gate criterion

Before running the comparison we registered three gates the deployed rule must satisfy:

1. **Gate 1 — split-date Christoffersen.** No (split $\times \tau$) cell rejecting at uncorrected $\alpha = 0.05$ across the four split anchors {2021, 2022, 2023, 2024} $\times$ four τ anchors. Motivation: the K=26 baseline introduced 2021/2022 split-date Christoffersen rejections at $\tau = 0.95$ that earlier ablation runs did not have; we want a rule that closes those without re-introducing them elsewhere.
2. **Gate 2 — per-symbol Kupiec.** Pass-rate 10/10 at $\tau = 0.95$. Motivation: the -standardisation architecture exists to fix the per-symbol bimodality of an unweighted Mondrian fit; any rule that loses this property is rejected.
3. **Gate 3 — bootstrap CI on pooled half-width.** The 95% bootstrap CI upper bound on $\Delta\text{hw}\%$ (vs the K=26 baseline) must lie within +5% at every τ . Motivation: any rule that materially widens the deployed band relative to baseline must be justified on calibration grounds; we cap the calibration-for-width trade at +5% a priori.

Gate 3 has no multiple-testing exposure: a single bootstrap CI per τ (no per-cell tests), the threshold is pre-registered. Gates 1 and 2 are per-cell tests that — at uncorrected α — accumulate multi-test risk. Gate 1 is the most exposed (16 cells per variant $\times$ 5 variants = 80 cells); we report that exposure and apply Benjamini-Hochberg correction below.

7.4.2 The five-variant ladder

Five rules were considered:

- **K=26 baseline.** Trailing 26-weekend window, equal-weight.
- **EWMA HL=6.** Exponentially-weighted moving average of past relative residuals, weekend half-life 6.
- **EWMA HL=8.** Same, half-life 8.
- **EWMA HL=12.** Same, half-life 12.
- **Convex blend.** $0.5 \cdot \text{K=26} + 0.5 \cdot \text{EWMA HL=8}$.

All five share the 8-past-observation warm-up rule and the strict pre-Friday convention; all five build $\hat{\sigma}_s$ over the same evaluable panel ($n = 5,916$). Per-variant artefacts (regime quantile table + $c(\tau)$ + $\delta(\tau)$ schedules) were re-fit at the same training cutoff (2023-01-01); collapsed to zero across all five variants. Outputs in reports/tables/sigma_ewma_*.csv.

7.4.3 Gate 1 — split-date Christoffersen across all 80 cells

Per-cell Christoffersen p -values across the 5 variants $\times$ 4 split anchors $\times$ 4 τ = 80-cell grid (reports/tables/sigma_ewma_split_sensitivity.csv). Counting cells with $p < 0.05$ (uncorrected):

variant	cells rejecting at uncorrected $\alpha=0.05$	rejecting (split,) cells
K=26 baseline	4	(2021, 0.68); (2021, 0.85); (2021, 0.95); (2022, 0.95)
EWMA HL=6	3	(2021, 0.68); (2023, 0.68); (2024, 0.68)
EWMA HL=8	0	none
EWMA HL=12	1	(2021, 0.85)

variant	cells rejecting at uncorrected $\alpha = 0.05$	rejecting (split,) cells
Convex blend	1	(2021, 0.68)

EWMA HL=8 is the only variant with zero per-cell rejections at uncorrected $\alpha = 0.05$ across the 16-cell grid for that variant. **At face value, Gate 1 selects EWMA HL=8 uniquely.**

Multiple-testing correction. The Gate-1 ranking above is multi-test exposed. We apply Benjamini-Hochberg correction at FDR = 0.05 across the full 80-cell grid (5 variants $\times$ 16 cells each); output reports/tables/sigma_ewma_split_sensitivity_bh_corrected.csv. **Under BH correction, no variant has any rejected cell.** The two K=26 cells that drove the uncorrected count at $\tau = 0.95$ — 2022 split ($p = 0.0016$) and 2021 split ($p = 0.0065$) — carry BH-adjusted q -values of 0.130 and 0.259 respectively, both well above the 0.05 threshold; 0.130 is the smallest q in the entire 80-cell grid. The honest reading: per-cell Christoffersen evidence after multi-test correction does not statistically distinguish the five variants. **The deployment decision cannot rest on Gate 1 alone**, even though EWMA HL=8 is the only variant clearing the gate at uncorrected $\alpha = 0.05$.

Qualitative pattern across the $\tau = 0.95$ column (supporting context, not statistical evidence). Even though BH correction wipes out the formal rejections, the uncorrected p -values across the four split anchors at $\tau = 0.95$ are systematically lower for K=26 than for any EWMA variant:

variant	$p_{\text{Christ}}(\tau = 0.95)$ across {2021, 2022, 2023, 2024} splits
K=26 baseline	{0.006, 0.002, 0.274, 0.525} — two visible problem cells
EWMA HL=6	{0.210, 0.321, 0.675, 0.784}
EWMA HL=8	{0.115, 0.186, 0.603, 0.671} — strictly higher than K=26 at every anchor
EWMA HL=12	{0.318, 0.529, 0.680, 0.586}
Convex blend (50/50)	{0.094, 0.162, 0.724, 0.791}

EWMA HL=8 produces uncorrected p -values that are higher than the K=26 baseline at every one of the four split anchors at $\tau = 0.95$ (0.115 vs 0.006; 0.186 vs 0.002; 0.603 vs 0.274; 0.671 vs 0.525); the column-wise minimum rises from 0.002 under K=26 to 0.115 under HL=8. We treat this consistent column-wise lift as *qualitative supporting context* — it is the pattern in the raw p -values that originally motivated the work — but we do not claim it as statistical evidence: under BH correction, none of these per-cell differences are formally significant. The statistical weight for the deployment decision is carried by Gate 3 (§7.4.5), which is a single-quantity bootstrap CI with no multi-test exposure.

7.4.4 Gate 2 — per-symbol Kupiec at $\tau = 0.95$

Per-symbol Kupiec at $\tau = 0.95$ on the 2023+ OOS slice across the 5 variants $\times$ 10 symbols (reports/tables/sigma_ewma_per_symbol.csv):

variant	per-symbol Kupiec pass-rate at $\tau = 0.95$
K=26 baseline	10 / 10
EWMA HL=6	10 / 10
EWMA HL=8	10 / 10
EWMA HL=12	10 / 10
Convex blend	10 / 10

All five variants pass Gate 2. Per-symbol calibration is a property of the architecture (per-symbol standardisation), not the rule — once is per-symbol and pre-Friday, the rule’s half-life affects pooled width and split-date temporal independence but not per-symbol Kupiec at $\tau = 0.95$ on this panel. **Gate 2 does not discriminate among the variants.**

7.4.5 Gate 3 — bootstrap CI on pooled half-width

Per-variant hw% versus the K=26 baseline at each τ , with 95% block-bootstrap CIs (1000 weekend-block resamples, seed=0; reports/tables/sigma_ewma_bootstrap.csv). The deployed rule must have a 95% CI upper on hw% within +5% at every τ :

τ	EWMA HL=8 hw% (vs K=26)	95% CI on hw%	within +5% gate?
0.68	-1.34%	$[-3.02, +0.25]$	
0.85	-2.07%	$[-3.75, -0.45]$	(also significantly narrower)
0.95	-3.83%	$[-6.15, -1.88]$	(also significantly narrower)
0.99	-7.41%	$[-9.33, -5.65]$	(also significantly narrower)

EWMA HL=8 satisfies Gate 3 with the upper-bound CI at +0.25% across all τ — well inside the +5% pre-registered cap. EWMA HL=8 is *narrower* than the K=26 baseline at every τ in the bootstrap point estimate; the 95% CI excludes zero on the narrow side (i.e. **statistically significantly narrower**) at $\tau \in \{0.85, 0.95, 0.99\}$, and is neutral at $\tau = 0.68$ (CI straddles zero). At no τ is EWMA HL=8 statistically significantly *wider* than baseline — the comparison is one-sided. Realised-coverage deltas at $\tau \in \{0.95, 0.99\}$ are exactly zero by construction (per-symbol Kupiec is preserved and the pooled count is unchanged on this finite panel); at $\tau \in \{0.68, 0.85\}$ the paired -realised CIs straddle zero. Calibration is preserved at the deployed anchors; sharpness improves. Gate 3 is the load-bearing gate for the deployment decision: it has no multi-test exposure (one CI per τ , threshold pre-registered), and EWMA HL=8 is the variant that delivers the largest narrowing relative to baseline while satisfying the gate.

7.4.6 Held-out forward-tape variant comparison

To re-validate the selection on data not used in any of the three gates, the forward-tape harness (§6.6) carries a sibling variant-comparison runner (`scripts/run_forward_tape_variant_comparison.py`). The runner loads a content-addressed *variant bundle* (`data/processed/lwc_variant_bundle_v1_frozen_20260504.json`, SHA-256 `7cef6132d970...`) with $(q_r, c(\tau), \delta(\tau))$ schedules for all five variants at the same 2023-01-01 training cutoff, and applies each variant’s frozen schedules to the forward-tape rows. The comparison **never re-selects** among variants — its function is to flag if a different rule looks dramatically cleaner on the held-out slice; if so, that’s a finding that motivates a re-fit with disclosure, not an automatic re-deploy. Output: `reports/m6_forward_tape_{N}weekends_variants.{md,csv}`.

Status at submission: $N = 1$ forward weekend (`reports/m6_forward_tape_1weekends_variants.md`, 2026-05-01). The variant comparison is uninterpretable at $N = 1$ — its function is to flag a held-out -rule discrepancy after the preliminary banner clears at $N \geq 4$, with moderate power against a meaningful per-variant gap requiring $N \geq 13$. The deployment claim therefore rests on Gates 2 and 3 of §7.4.1–§7.4.5; the forward-tape variant comparison is operational and accumulating, not yet held-out evidence the deployment leans on.

7.4.7 Honest framing of the deployment claim

The selection procedure is what it is: a 5-variant ladder $\times$ 16-cell grid $\times$ 3 gates, with the three gates pre-registered and the multiple-testing exposure of Gate 1 disclosed and corrected. Under BH at FDR=0.05, Gate 1 alone does not statistically distinguish the five variants — the formal selection signal that Gate 1 carried at uncorrected α is washed out by the multi-test correction, exactly as one would expect with 80 tests against a correct null. Gates 2 and 3 — both unambiguously satisfied by EWMA HL=8 — are the deployment-load-bearing tests. The forward-tape variant-comparison harness (§7.4.6) is the held-out re-validation.

We claim that EWMA HL=8 satisfies the pre-registered three-gate criterion under uncorrected per-cell $\alpha = 0.05$; that under BH multi-test correction the per-cell Christoffersen evidence does not statistically distinguish the five variants; that the qualitative pattern in the uncorrected $\tau = 0.95$ column (p -values strictly higher under HL=8 than baseline at every split anchor) is consistent with the deployment choice but is supporting context, not statistical evidence; and that the deployment decision rests on Gate 3 (the bootstrap CI on width, no multi-test issue, statistically significantly narrower at $\tau \in \{0.85, 0.95, 0.99\}$ with calibration preserved) plus the held-out forward-tape re-validation. We do not claim the rule is optimal among all locally-weighted variants (§3.5 non-goal). A future paper may sweep finer half-life grids, alternative weight kernels, or hybrid rules; that work is gated on accumulating forward-tape evidence to re-validate any new selection on truly held-out data.

7.4.8 Regime quartile cutoff q — ablation under the three-gate frame

The deployed `high_vol` regime gate (§5.5) is “VIX at Friday close in the top quartile of its trailing 252-trading-day window” — a single scalar $q = 0.75$. Ablating $q \in \{0.60, 0.67, 0.70, 0.75, 0.80, 0.90\}$ under the three-gate frame of §7.4.1 (`reports/tables/paper1_b1_regime_threshold_ablation.csv`, ... `_hw_bootstrap.csv`):

q	high_vol mix (%)	Gate 1: pooled Kupiec all τ	Gate 1b: pooled Christoffersen all τ	Gate 2: per-symbol Kupiec	$\Delta\text{hw}\%$ at $\tau = 0.95$ vs deployed (95% CI)
0.60	38.9			10/10	-2.76% [-3.80, -1.73]
0.67	30.8			10/10	-1.78% [-2.66, -0.87]
0.70	28.8			10/10	+1.73% [+0.85, +2.71]
0.75 (deployed)	23.9			10/10	—
0.80	21.1			10/10	-1.70% [-2.11, -1.32]
0.90	12.3		rejects	10/10	-4.30% [-5.34, -3.25]

5 of 6 candidates satisfy Gates 1, 1b, and 2. Three alternates ($q \in \{0.60, 0.67, 0.80\}$) deliver 1.7–2.8% narrower bands at preserved calibration with bootstrap CIs excluding zero. The deployed $q = 0.75$ is **convention-anchored** (top quartile by definition) rather than width-optimization-selected. The differences are operationally small (~ 5 bps on a 370 bps $\tau = 0.95$ headline) but real; a future re-tuning on a held-out tune slice could close this gap.

7.4.9 Split anchor — robustness across {2021, 2022, 2023, 2024}

The deployed train/test split anchor is 2023-01-01 (1 scalar). Ablating across {2021, 2022, 2023, 2024}-01-01 (re-fit per anchor; reports/tables/m6_lwc_robustness_split_sensitivity.csv):

split anchor	n_{OOS} weekends	$\tau = 0.95$ realised	Kupiec p	Christoffersen p	hw bps	per-sym Kupiec
2021-01-01	277	0.9539	0.348	0.115	357.2	10/10
2022-01-01	225	0.9520	0.661	0.186	363.5	10/10
2023-01-01 (deployed)	173	0.9503	0.956	0.603	370.6	10/10
2024-01-01	121	0.9504	0.947	0.671	416.8	10/10

Realised $\tau = 0.95$ coverage stays in $[0.9503, 0.9539]$; Kupiec $p \in [0.35, 0.96]$; Christoffersen $p \in [0.12, 0.67]$; per-symbol Kupiec 10/10 across all four anchors. The deployed split anchor is robust. Half-width varies ($357 \rightarrow 417$ bps) — driven by the eval-slice composition effect documented in §6.3.3.1 (later eval slices contain more high- weekends, including the 2024-08-05 BoJ unwind).

7.4.10 Disclosure-DOF accounting

After §7.4.8 and §7.4.9 the deployment-DOF count rises from 16 to **19 scalars** with proper-ablation provenance: 12 trained per-regime quantiles + 4 OOS-fit $c(\tau)$ + 0 walk-forward δ + 1 rule selector (EWMA HL=8, §7.4.1–7) + 1 regime quartile cutoff ($q = 0.75$, §7.4.8) + 1 split anchor (2023-01-01, §7.4.9). All three appended scalars satisfy the three-gate criterion under their respective ablation tables. The “16 scalars undercounts DOF” critique closes.

7.5 How standardisation redistributes width across symbols

§7.1.3 attributed the constant-buffer premium to `high_vol`; §7.2 attributed the per-symbol Kupiec fix to standardisation; this section attributes the +4.5% pooled half-width tax to a width *redistribution* across symbols within a regime.

Per-class decomposition at $\tau = 0.95$. Where the unweighted Mondrian comparator pays a uniform width within a regime, the deployed architecture redistributes that width across symbols within a regime via the per-symbol $\hat{\sigma}_s$ multiplier:

class	n	unweighted Mondrian hw at $\tau = 0.95$	deployed hw at $\tau = 0.95$	
equities (8 symbols)	1,384	354.6 bps	417.8 bps	+17.8% (carries the per-symbol heavy-tail fix)
GLD + TLT (2 symbols)	346	354.6 bps	181.6 bps	48.8% (releases over- conservative defensive bands)

Row-weighted reconciliation against the §6.3 panel pooled: $(1384 \cdot 417.8 + 346 \cdot 181.6)/1730 = 370.6$ bps, matching the §6.3.1 panel-pooled HW at $\tau = 0.95$.

Under the unweighted comparator, every symbol within a regime received the same $c(\tau) \cdot q_r(\tau) \cdot p_{\text{Fri}}$ width — the “common multiplier on heterogeneous tails” failure of §6.4.1; the per-class mean HW is identical across classes because the regime mix is identical across symbols (regime is global, §5.5) and width within a regime carries no per-symbol axis. Under the deployed architecture, the multiplier is per-symbol, so heavy-tail equities (HOOD, MSTR, TSLA) receive bands that match their relative residual scale, and the over-covered defensive class (GLD, TLT) receives narrower bands without losing coverage.

The +4.5% pooled width tax of §7.2.1 is the *equity-weighted* read on a panel that is 80% equity rows $(1,384/1,730)$. Under the deployed architecture the average equity-class symbol receives a +17.8% wider band than under an unweighted Mondrian fit (417.8 vs 354.6 bps); a defensive-class symbol (GLD, TLT) receives a 48.8% narrower one (181.6 vs 354.6). The redistribution is *toward heavy-tail equities*, where the per-symbol calibration evidence demands it (§6.4.1: TSLA, MSTR, HOOD all in the 0.052–0.058 violation-rate band that the unweighted comparator could not deliver), and *away from defensive collateral*, where the previous regime-pooled multiplier was over-conservative. A protocol whose collateral mix is dominated by equity xStocks should expect the deployed mean served band to be materially wider than the unweighted comparator’s; a protocol whose collateral mix is dominated by GLD/TLT will see materially narrower bands. The pooled +4.5% figure is the right summary of the trade only at the deployed panel composition; downstream consumers should re-weight by their own collateral mix.

The §10 candidate architectures (cross-sectional common-mode partial-out, full-distribution conformal, sub-regime granularity) target the residual rejections (§6.3.6 cross-sectional within-weekend correlation; §9.4 per-symbol Berkowitz on TSLA/TLT) that standardisation does not reach.

8 Serving-Layer System

The methodology of §4 is implemented as a four-component serving stack designed for two requirements that pull in opposite directions: (i) the model logic is research-iterated in Python and re-validated against the §6 OOS panel after every change; (ii) the production hot path is a Solana program reading bands at consumer-facing latency. This section describes how those requirements are reconciled in a single source of truth.

8.1 Split-by-function architecture

The serving stack is a contract between three audiences: the **researcher** iterating against the Python OOS panel, the **protocol integrator** building against on-chain PriceUpdate accounts, and the **third-party auditor** verifying a served band against the published artefact. Byte-for-byte parity (§8.5) makes the first two identical at the numeric level; the receipt fields and public artefacts make the third executable from public data alone. This trio is the operational instantiation of P_1 (auditability).

The implementation separates *training* (Python) from *serving* (Rust + Solana). The Python Oracle (`src/soothsayer/oracle.py`) is the canonical specification: it produces the per-Friday lookup artefact and is the executable reference for both the M5 Mondrian conformal-lookup algorithm (`Oracle.fair_value`) and M6’s locally-weighted variant (`Oracle.fair_value_lwc`). A change — a new $c(\tau)$ schedule, a per-Friday point recompute, a rule swap — is implemented in Python first, validated against the §6 OOS panel, and ported to Rust under the byte-for-byte parity contract of §8.5. No model logic is duplicated: the training pipeline materialises the per-Friday rows (including $\hat{\sigma}_s(t)$ per symbol pre-Friday) and the deployment scalars once offline; the serving stack reads them. M6 carries 16 deployment scalars hardcoded in `oracle.py` and mirrored into the JSON sidecar; the M5 reference path carries

20 scalars (12 quantiles + 4 OOS-fit + 4 walk-forward) duplicated in `crates/soothsayer-oracle/src/config.rs` for the Rust serving binary.

8.2 The Rust crate stack

Crate	Purpose	Lines	Tests
<code>soothsayer-oracle</code>	Library: loads the Mondrian artefact, exposes <code>Oracle::fair_value</code> with byte-exact parity to Python	~330	4/4 unit + parity harness
<code>soothsayer-publisher</code>	CLI binary: serves single reads, prepares on-chain publish payloads	~440	6/6 unit
<code>soothsayer-consumer</code>	<code>no_std</code> , minimal-dep SDK: decodes the on-chain <code>PriceUpdate</code> PDA into a typed <code>PriceBand</code>	~330	5/5 unit
<code>soothsayer-demo-kamino</code>	Reference Kamino-style consumer integration	~400	included in integration tests
<code>soothsayer-oracle-program</code> (Anchor)	On-chain program: initialize, publish, set_paused, rotate_signer_set	~400	scaffolded

The consumer SDK is `no_std` so it can be vendored into a Solana program account-decoder without the Rust standard library.

8.3 On-chain program

`programs/soothsayer-oracle-program/` is a vanilla Anchor program with three PDA account types (`PublisherConfig`, `SignerSet`, per-symbol `PriceUpdate`) and four instructions. The publish instruction validates signer membership, enforces the cadence rate-limit, and checks five on-chain invariants before persisting: band ordering ($L \leq \hat{P} \leq U$), coverage values $\leq 10,000$ bps, exponent $\in [-12, 0]$, signer membership, and time since last publish $\geq \text{min_publish_interval_secs}$. Every instruction emits an Anchor event so off-chain indexers can trace publish history without re-reading every account.

8.4 Wire-format design

The `PriceUpdate` account layout (128 bytes data + 8-byte Anchor discriminator) is informed by the Pyth-Network September-2021 BTC flash-crash post-mortem. Three design rules avoid the failure modes that incident exposed.

PriceUpdate:

version	u8	schema version, currently 1
regime_code	u8	0=normal, 1=long_weekend, 2=high_vol
forecaster_code	u8	0=F1_emp_regime (legacy Soothsayer-v0), 1=F0_stale (legacy Soothsayer-v1)
exponent	i8	
target_coverage_bps	u16	publisher-set target coverage, integer bp
claimed_served_bps	u16	served quantile, integer bp
buffer_applied_bps	u16	
symbol	[u8; 16]	ASCII null-padded

```
point, lower, upper, fri_close    i64          absolute fixed-point at 'exponent'
fri_ts, publish_ts, publish_slot, signer, signer_epoch    timestamps + identity
```

The three rules: (1) **Absolute prices, not deltas** — a consumer reading lower gets a real price directly. (2) **Single shared exponent** for all price fields, so cross-field comparisons are exact integer comparisons. (3) **Integer basis points, never floats** for coverage and buffer — exact and deterministic across Rust, Anchor IDL, TypeScript clients, and Python. Wire-format sizes are locked by unit tests in `crates/soothsayer-publisher/src/payload.rs`.

8.5 Byte-for-byte parity verification

`scripts/verify_rust_oracle.py` runs Python `Oracle.fair_value (M5) / Oracle.fair_value_lwc (M6)` and the Rust soothsayer `fair-value --forecaster {mondrian, lwc}` CLI on the same (symbol, fri_ts, target_coverage) triples and asserts byte-exact agreement on numeric output (point, lower, upper, sharpness_bps, claimed_served, calibration_buffer_applied) plus exact agreement on string fields (regime, forecaster_used). Each forecaster’s panel covers 90 cases (25 random pairs from the artefact + edge cases for SPY, GLD, TLT, MSTR, HOOD at three targets each). **180/180 pass** as of the most recent re-run — 90/90 on the M5 path and 90/90 on the M6 LWC path. The Rust crate now serves both forecasters; `forecaster_code = 3 (FORECASTER_LWC)` is encoded into new publishes from `crates/soothsayer-publisher` and decoded into the `Forecaster::Lwc` variant by the `no_std` soothsayer-consumer crate.

The parity harness is invoked after every methodology change. Each migration is applied to Python first, then ported to Rust; the harness re-runs before the change is considered shipped. It is the contract that allows a consumer to verify a served `PricePoint` against the published artefact without trusting either implementation in isolation.

Wire format invariance. The on-chain `PriceUpdate` Borsh layout is unchanged across all methodology migrations to date. The only on-wire change at each step is relabelling reserved `forecaster_code` slots: code 2 → `FORECASTER_MONDRIAN` (M5; live), code 3 → `FORECASTER_LWC` (M6; live in the Rust serving stack as of the parity-180/180 milestone above; on-chain enablement gated on the next publisher release). Older accounts decode cleanly under newer consumers; M6 accounts decode cleanly under any consumer that reads the `forecaster_code` byte (soothsayer-consumer exposes the typed `Forecaster::Lwc` variant). Downstream consumers branching on `forecaster_code` need only add code-2 and code-3 cases.

8.6 Reproduction

```
# Train (Python) - M6 (deployed)
uv sync && uv run python scripts/build_lwc_artefact.py
uv run python scripts/freeze_lwc_artefact.py      # content-addressed freeze for forward-tape

# Train (Python) - M5 (still buildable as the §7.2 ablation rung)
uv run python scripts/build_mondrian_artefact.py

# Serve (Rust)
cargo build --release -p soothsayer-publisher
./target/release/soothsayer fair-value --symbol SPY --as-of 2026-04-24 --target 0.85

# Verify Rust Python parity
PYTHONUNBUFFERED=1 uv run python scripts/verify_rust_oracle.py

# Deploy to devnet
./scripts/deploy_devnet.sh
```

A lending-protocol integrator depends on soothsayer-consumer (the `no_std` decoder) and reads the `PriceUpdate` PDA at the symbol’s derived address; the demo Kamino-style consumer at `crates/soothsayer-demo-kamino/` shows the full integration in ~400 lines, including LTV decision logic that uses the band’s lower bound as the collateral valuation input — the third audience of §8.1 reading the same wire bytes the on-chain program emits.

9 Limitations

This section enumerates the assumptions under which the claims of §3.4 hold for the deployed architecture, the failure modes the backtest exposed, and the domain boundaries outside which we make no assertion. The strongest residual

issues are (i) cross-sectional within-weekend common-mode residual (§9.4); (ii) the selection procedure’s multi-test exposure (mitigated in §7.4 but disclosed here); and (iii) the contraction — but not full closure — of the OOS-tuning provenance disclosure (§9.3).

9.1 Shock-tertile empirical ceiling (and the calm-tertile sharpness deficit)

The shock realised-move tertile ($|z| \geq 0.67$ standardised by Friday’s 20-day vol; $n = 632$ in the 2023+ OOS slice, §6.3.2) realises 87.8% coverage on the deployed band at the nominal $\tau = 0.95$ claim — a deviation from nominal whose mechanism (cross-sectional within-weekend common-mode, §6.3.6 / §9.4) is identified and whose operational handle (the k_w reserve-guidance threshold of §6.3.4) is what protocol consumers should monitor in real time. Per-symbol standardisation does not reach this ceiling because the failure mode is not per-symbol scale; band half-width is approximately flat across tertiles (365–380 bps), so the band does not widen in shock periods — it just misses more often.

The complementary tertile reads are an over-coverage on calm weekends (calm tertile realised 1.0000, normal tertile 0.9933, both significantly above nominal). Pooled $\tau = 0.95 = 0.950$ is achieved through compensating tertile-level biases. Coverage is a one-sided contract — the under-coverage side is a coverage failure, the over-coverage side is a sharpness deficit (the band is wider than necessary when the realised move is small) — so the two are qualitatively different, but together they say the abstract calibration claim is genuinely tighter than the conditional structure. A $|z|$ -conditional conformal upgrade (the finer-stratification candidate in §10.2) would tighten calm bands and widen shock bands at the same pooled τ ; the reserve-guidance threshold k^* in §6.3.4 is the consumer-facing handle on the asymmetric residual under the deployed schedule.

The §6.3.4 joint-tail empirical distribution and the §6.3.5 worst-weekend vignette (2024-08-05 BoJ unwind, $k_w = 10$ at $\tau = 0.85$) make this disclosure operationally addressable: a consumer monitoring k_w in real time would have seen the August 5 weekend exceed the deployment-time 99th percentile ($k = 7$ at $\tau = 0.85$) by the Monday-open print, well in advance of any liquidation processing window. Reserve-guidance threshold $k^* = 5$ at $\tau = 0.85$ (or $k^* = 3$ at $\tau = 0.95$) carries 0/24 sub-period $\times$ threshold-convention stability rejections (§6.3.4); a circuit breaker that pauses borrowing/lending when k_w crosses k^* would fire on this exact weekend before liquidations are processed. The empirical k_w distribution is the right operational signal precisely because it captures the cross-sectional common-mode that per-symbol coverage cannot. The ceiling is reported in every PricePoint diagnostic.

9.2 Stationarity assumption — sub-period evidence (positive)

P2 holds under the assumption that $P_t \mid (\mathcal{F}_t, \rho = r, s)$ is approximately stationary across $\mathcal{T}_{\text{hist}}$ and the deployment horizon. The EWMA $\hat{\sigma}_s(t)$ provides *partial adaptive scaling per symbol* (the §6.7 simulation evidence demonstrates tracks DGP-C drift and DGP-D structural breaks with a half-life-bounded lag), but the conditional-distribution stationarity of the *standardised* score is still assumed.

Empirical evidence on this assumption is positive. The pooled OOS calibration at $\tau = 0.95$ holds across the 2023, 2024, 2025, and 2026-YTD calendar sub-periods (§6.3.3): Kupiec $p \geq 0.054$ in every cell; realised range 0.942–0.967 across four years that span SVB / banking-stress 2023, the 2024 rate-cut transition, the 2025 tokenisation-launch year, and the post-launch 2026-YTD slice. Across the full 16-cell (sub-period $\times \tau$) grid the architecture carries 2 Kupiec rejections — both *over-coverage* in 2025 (not the under-coverage failure mode that warrants a deployment alarm). Christoffersen lag-1 independence is not rejected in any of the 16 cells.

The assumption may still be violated in three ways the sub-period grid cannot pre-empt: (i) **structural breaks** beyond the panel horizon (a future regime as different from 2014–2026 as 2026 is from 2014) — adapts in ≤ 26 weekends per symbol per simulation evidence; (ii) **regime-labeler drift** (ρ uses a trailing 252-day VIX percentile; persistent rises shift the threshold endogenously); (iii) **factor-switchboard drift** (MSTR pivots from ES=F to BTC-USD on 2020-08-01; a second pivot would require re-fit). A conformalised variant (§10) would upgrade P2 from asymptotic-under-stationarity to finite-sample-under-exchangeability.

9.3 OOS-tuned $c(\tau)$ provenance — substantially contracted

The deployment carries 16 scalars: 12 trained per-regime quantiles + 4 OOS-fit $c(\tau)$ + 0 walk-forward (the $\delta(\tau)$ schedule is identically zero). Of the 4 OOS-fit $c(\tau)$ scalars, three are essentially identity:

$$c(\tau) = \{0.68:1.000, 0.85:1.000, 0.95:1.079, 0.99:1.003\}.$$

Only $c(0.95) = 1.079$ carries meaningful OOS information (a 7.9% widening at the headline anchor). §6.3.3.1’s nested temporal holdout fits $c(0.95)$ on 2023 alone and evaluates it on a 2024-01-05 → 2026-04-24 slice that neither the per-regime quantile nor the bump has seen — realised coverage 0.9504, Kupiec $p = 0.947$, Christoffersen $p = 0.989$, per-symbol Kupiec 10/10. $c(0.95)$ on TUNE-only matches the full-OOS fit at three decimals. **The headline $c(0.95) = 1.079$ is empirically held-out at the time level;** the residual provenance disclosure is restricted to the lower- τ over-coverage in held-out mode (M_{a2} realises 0.712 at $\tau = 0.68$ and 0.870 at $\tau = 0.85$ — Kupiec rejects toward over-coverage because $c(0.68)$ and $c(0.85)$ fit on TUNE-2023 inherit the SVB/CS banking-crisis-weekend tilt; the deployed full-OOS c schedule averages this out at the floor and is the production mitigation). The over-coverage is a one-sided contract-favourable failure mode — a sharpness deficit, not a coverage-contract failure — and §9.11 documents the broader asymmetric-failure-mode pattern across the disclosures of §6.3, §6.7, §7.4, §9.5.1, and §10.1. The forward-tape harness (§9.5) closes the residual contamination by carrying the deployment artefact onto truly held-out weekends as they accumulate; with ≥ 13 forward weekends a §9.5 update will push the disclosure further toward “OOS-fit + held-out forward-tape re-validation.”

The `calibration_buffer_applied` field carries $\delta(\tau)$ as a first-class receipt; diagnostics carry $c(\tau)$ and q_{eff} .

9.4 The residual Berkowitz / DQ rejection — cluster topology

The served band is *per-anchor calibrated* (Kupiec passes at every τ , Christoffersen passes at every served τ on the 2023 split, §7.4 confirms passes at all 16 split-date cells under EWMA HL=8) but *not full-distribution calibrated* (pooled Berkowitz on the deployed PITs rejects; pooled DQ at $\tau = 0.95$ also rejects). §6.3.4 documents the same residual viewed as joint-tail overdispersion: empirical k_w over-disperses $2.34\times$ vs Binom independence and sits inside a t-copula envelope at $\hat{v} \approx 6$ with mean off-diagonal $\hat{R} = 0.36$. **The residual has a structural name: a within-weekend cluster topology separating equities from safe-haven assets.**

Cluster topology of the residual. The residual is a within-weekend cluster structure separating equities (8 symbols, positively correlated with the weekend equity factor) from safe-haven assets (GLD, TLT, with negative or near-zero correlation to the equity factor). Within the equity cluster $\hat{\rho}_{\text{within}} = 0.477$ — *higher* than the panel-wide $\hat{\rho}_{\text{cross}} = 0.41$, because the cross-cluster correlations dilute it. Within the safe-haven cluster $\hat{\rho}_{\text{within}} = -0.016$ — there is no common-mode to remove. A naive non-parametric partial-out using either the within-weekend median or mean reduces panel-wide $\hat{\rho}_{\text{cross}}$ from 0.41 to $\{0.12, 0.06\}$ but breaks per-symbol Kupiec on the safe-haven cluster (GLD violation rate 14.5%, TLT 9.2% under the median variant; 5/10 per-symbol pass under the mean variant) and rejects DQ at $\tau \in \{0.68, 0.85\}$. The mechanism is structural: subtracting an equity-dominated weekend median from a negatively-correlated asset creates fresh violations on equity-rally weekends and pulls residuals toward the band centre on selloff weekends. **Any operation that treats the panel as exchangeable inherits this failure mode;** a working architecture-level fix must respect the cluster topology.

Two views of the same structure. The over-dispersion of the homogeneous t-copula in §6.3.4 (emp/t variance 0.63 at $\tau = 0.95$) and the cluster failure of the naive partial-out are two views of the same equity-vs-safe-haven structure: a single Pearson $\hat{R}$ averages high equity-equity correlations against negative equity-safe-haven correlations; a single weekend median centres on the equity cluster’s mean. Either rendering of the panel as homogeneous misses the same topology. (Source: A1, A1.5, A3 — `paper1_a1_*.csv`, `paper1_a1_5_*.csv`, `paper1_a3_*.csv`.)

Localising the residual at the symbol-pair level. §6.3.6 decomposes the lag-1 alternative: the AR(1) signal lives in the *cross-sectional within-weekend* ordering ($\hat{\rho}_{\text{cross}} = 0.354$, $p < 10^{-100}$), not the temporal-within-symbol ordering ($\hat{\rho}_{\text{time}} = -0.032$, $p = 0.18$). standardisation operates *within* a symbol’s history, not *across* symbols within a weekend, so the deployed architecture is incapable of reaching this residual. A vol-tertile sub-split of normal regime (5 cells) leaves Berkowitz LR essentially unchanged (170 → 172) while widening $\tau = 0.95$ band by 9% — finer regime granularity is not the lever (§6.3.6). Within-bin exchangeability separately holds (§4.3.1; permutation test on lag-1 score autocorrelation within each (regime $\times$ symbol) bin: 1/30 nominally rejecting at $\alpha = 0.05$ uncorrected, 0/30 after BH correction). The cross-sectional dependence is *across* bins, not within.

Per-symbol Berkowitz rejection localisation. The residual symbols rejecting Berkowitz at $\alpha = 0.01$ are TLT (LR = 16.7) and TSLA (LR = 13.5) — both trace to cross-sectional common-mode rather than per-symbol scale. The §6.3.4 joint-tail empirical distribution turns the Berkowitz rejection into a measurable, reportable shape (variance overdispersion $2.34\times$ vs Binom at $\tau = 0.95$; $P(k_w \geq 3) = 4.62\%$ vs Binom 1.15%, t-copula 8.2%), and the §6.3.4 reserve-guidance threshold ($k^* = 3$) gives a consumer an operational handle that holds up against the joint-volatility model a CCAR-style risk team would actually run, not just against independence.

A protocol consuming Soothsayer at a single τ on a regime-pooled cell is served by the per-anchor pooled guarantee plus the per-symbol Kupiec evidence; one wanting full-distribution calibration must wait for the cluster-conditional + path-fitted architectural recommendation in §10, which composes A1.5’s cluster-internal partial-out with the path-fitted

score and is calibrated at the headline $\tau = 0.95$ anchor. The recommendation is bounded: τ -conditional rather than blanket.

9.5 Forward-tape held-out evidence — operational, accumulating

A content-addressed frozen artefact (data/processed/lwc_artefact_v1_frozen_20260504.json, freeze date 2026-05-04) is being evaluated weekly via launchd-driven harness against forward weekends as they accumulate (§6.6). The Rust serving stack and the wire-format rollout state are documented in §8. The harness is the cleanest possible held-out coverage mechanism on the deployed artefact and, when its sample size matures, will become the load-bearing held-out evidence; it is not yet a live integrator-consumed price feed. **Status at submission:** $N = 1$ forward weekend (reports/m6_forward_tape_1weekends.md, 2026-05-01) — preliminary by the harness’s own threshold ($N \geq 4$ to clear the preliminary banner; $N \geq 13$ for moderate forward-Christoffersen power). The present submission’s held-out backbone is §6.3.3’s leave-one-symbol-out CV (mean realised 0.9497, std 0.0128 across the 10 held-out symbols at $\tau = 0.95$) plus the §6.3.3.1 nested temporal holdout at $\tau = 0.95$ (realised 0.9504, Kupiec $p = 0.947$, Christoffersen $p = 0.989$, per-symbol Kupiec 10/10 on a 2024-01-05 → 2026-04-24 holdout), supported by per-symbol Kupiec, split-date sensitivity, and four-year calendar sub-period stability; the forward-tape harness is operational and accumulating, not yet held-out evidence the present submission leans on. A live integrator window — devnet-then-mainnet against a real protocol — would close the external-validity gap; it would not by itself make Berkowitz pass.

9.5.1 τ -stratified CUSUM drift bank

Beyond passive forward-tape observation, the deployed system carries a two-sided Page CUSUM drift monitor at each served τ . With control statistic $X_t = k_w/10$ (weekend violation rate), reference $\mu_0 = 1 - \tau$, and slack $k = \mu_0/2$ tuned to detect a $2\times$ violation-rate shift, calibrated thresholds $h \in \{0.40, 0.40, 0.30\}$ for $\tau \in \{0.85, 0.95, 0.99\}$ produce mean in-control run length ≈ 225 weekends (one expected false alarm per ~ 4.3 years; calibration via 20,000 Monte-Carlo traces $\times$ 500 weekends). Detection power against an operationally relevant $2\times$ violation-rate drift: $\sim 83\%$ with median latency 5 / 11 / 28 weekends at $\tau \in \{0.85, 0.95, 0.99\}$; against $3\times$ drift: 2 / 5 / 14 weekends (5,000 traces $\times$ 200 weekends; reports/tables/paper1_c2_cusum_drift.csv, paper1_c2_cusum_calibration.csv).

On the 173-week OOS slice, the $\tau = 0.85$ CUSUM fires S^+ alarms at both **2023-03-10** (SVB collapse, $k_w = 7$) and **2024-08-02** (BoJ unwind, $k_w = 10$) — the two canonical stress weekends in the OOS window — with the post-BoJ S^- (over-coverage) alarm at **2024-09-27** catching the over-conservative recovery period documented under §7.4.8 split-anchor robustness. CUSUM banks at $\tau \in \{0.95, 0.99\}$ fire on different weekends: pairwise alarm coincidence with $\tau = 0.85$ is **2 of 7 alarms**; pairwise coincidence between $\tau = 0.95$ and $\tau = 0.99$ is **0 of 7 alarms**. Each anchor’s monitor surfaces a structurally distinct aspect of drift — body-of-distribution shift at $\tau = 0.85$, near-tail at $\tau = 0.95$, deep-tail mass at $\tau = 0.99$ — confirming the τ -stratified CUSUM bank is a multi-resolution detector rather than a redundant signal. Source: reports/tables/paper1_f1_cusum_alarm_timing.csv, paper1_f1_cusum_alarm_proximity.csv, paper1_f1_cusum_alarm_coincidence.csv. The CUSUM bank is operational concurrently with the forward-tape harness and re-uses its weekly ingest path.

9.6 Out-of-scope comparators and consumer-experienced effects

No numerical incumbent comparator. A coverage-vs-sharpness comparison is only well-defined between methods that emit a calibrated coverage band; incumbent oracle surfaces (Pyth, Chainlink Data Streams v10 / v11, RedStone Live, Kamino Scope) do not, and we therefore do not report one. The categorical claim — that none of these surfaces publishes a per-read calibration receipt or an aggregate-level realised-coverage statement — is in §1 / §2 and stands as the contribution. The only numerical baseline reported is GARCH(1,1)- t (§6.4.3), which does emit a calibrated band.

Endpoint vs path coverage. §6.5 quantifies the gap on a 24/7 stock-perp slice ($n = 118$ symbol-weekends). The $\tau = 0.95$ endpoint-vs-path gap on the perp-listed subset is +16.1 pp — what an honestly-calibrated endpoint band exhibits on a sample whose intra-weekend variance is structurally larger than its endpoint variance. After three robustness checks the residual genuine-shortfall band is approximately 9–15 pp at $\tau = 0.95$ (basis-normalisation 11.0 pp; volume-floor filter at ≥ 1 contract 9.5 pp on $n = 63$; 15-minute sustained-crossing 14.4 pp), concentrated in normal weekends and approximately closed by stepping up to $\tau = 0.99$ (residual +7.6 pp). The served contract remains the endpoint claim.

MEV and execution-aware coverage. A consumer transacting at a worse price than the band’s mid (front-running, sandwich, block-builder reordering) experiences an *effective* coverage that may differ from §6.5’s perp/on-chain path coverage near band edges. Measuring this requires Solana mempool-or-bundle granularity plus a consumer-behaviour model; documented as next-generation work (§10.1).

9.7 No optimal liquidation-policy benchmark

The paper validates an oracle interface, not a lending policy. We do not report a decision-theoretic benchmark of “what target τ should a protocol choose?” Answering requires three ingredients absent here: (i) an explicit borrower-book LTV distribution; (ii) a protocol-specific cost model distinguishing unnecessary liquidation, unnecessary caution, and missed liquidation / bad debt; (iii) a declared semantics for what counts as the correct action under realised prices. We do not address optimal liquidation policy in this paper.

9.8 Scope of the methodology

Region scope. All ten tickers are US-listed; the weekend predicted is the US weekend. The factor switchboard privileges US session data. We make no claim about generalisation to tokenised-JP or tokenised-EU equities.

Asset-class scope. The methodology requires a *continuous off-hours information set* — a public, free, sub-daily price signal correlated with the closed-market underlier. Tokenised US equities have this (E-mini futures + VIX), tokenised gold has it (gold futures + GVZ), tokenised US treasuries have it (10Y T-note futures + MOVE), and most major FX pairs have it. Tokenised credit has it through CDS spreads. Out of scope: real estate (no continuous off-hours signal), illiquid commodities (futures discontinuous or thin), and instruments with primarily NAV-based pricing. `docs/methodology_scope.md` carries the per-class fit/no-fit table.

Newly-listed symbol admission. The warm-up rule (8 past observations) sets an empirical floor; the §6.7 sample-size sweep extends this to a panel-admission threshold of $N \geq 200$ weekends under regime-switching DGPs (the production threshold) and $N \geq 80$ under stationary / drift / structural-break DGPs. HOOD’s empirical $N \approx 218$ sits inside this band. A protocol admitting a symbol with fewer than 200 weekends of history should treat the per-symbol claim as deferred until the panel accumulates.

9.9 Non-claims

Three concerns that do *not* apply, plus one known data-gated gap:

- **Oracle manipulation.** Coverage transparency is orthogonal to upstream-feed integrity. Soothsayer ingests upstream price/oracle signals (Pyth, Chainlink, RedStone, Kamino-Scope, public-market data via Yahoo / Kraken) and republishes them as a calibrated band; the coverage claim is conditional on the integrity of those upstream feeds, and this primitive is intended to be deployed alongside, not in place of, an adversarially-robust price feed.
- **Latency.** The serving-time Oracle is a five-line lookup; nothing in the coverage-inversion mechanism requires a live forecast computation.
- **rule optimality.** The §7.4 selection procedure under multi-test correction does not statistically distinguish the five variants; the EWMA HL=8 promotion rests on Gate 3 (bootstrap CI on width) plus the held-out forward-tape re-validation. We do not claim the rule is optimal among all locally-weighted variants; a finer half-life sweep, alternative weight kernels, or hybrid rules are open extensions.
- **On-chain xStock TWAP not consumed.** Cong et al. [Cong et al., 2025] document off-hour returns on tokenised stocks anticipate Monday opens. The base forecaster does not read this; the V5 forward-cursor tape supplies the data but only ~30 weekends of post-launch history exist (§10.1).

9.10 Off-anchor is interpolated, not separately audited

The four served anchors $\mathcal{T}_{\text{anch}} = \{0.68, 0.85, 0.95, 0.99\}$ carry the calibration evidence reported in §6.3 — pooled coverage, per-symbol Kupiec, calendar sub-period stability, and joint-tail thresholds are all reported at those four anchors. The deployed API accepts any $\tau \in [\text{MIN_SERVED_TARGET}, \text{MAX_SERVED_TARGET}] = [0.68, 0.99]$ (`src/soothsayer/oracle.py`: 159 and the Rust mirror); off-anchor τ is served by linearly interpolating $q_r(\tau)$ and $c(\tau)$ between the bracketing anchors. The $c(\tau) = \{1.000, 1.000, 1.079, 1.003\}$ schedule is non-monotonic (peaking at the $\tau = 0.95$ anchor, dipping back at $\tau = 0.99$), so a consumer requesting an off-anchor τ inherits a multiplier whose calibration is *interpolated*, not separately validated — concrete example: $c(0.90) \approx 1.040$, $c(0.97) \approx 1.041$. The receipt’s τ is the requested target and the diagnostic exposes the interpolated $(c(\tau), q_r(\tau))$, so an integrator can detect the off-anchor case from the receipt alone, but the rolling Kupiec/Christoffersen claim does not extend to it. §10.4 is the methodology fix (denser anchor grid and/or monotone interpolation). Until that lands, an integrator with a hard calibration-receipt requirement should restrict τ to the four audited anchors.

9.11 Asymmetric failure modes — a coherent architectural property

Across the empirical results above, every documented M6 LWC failure mode points the same direction — toward over-coverage rather than under-coverage. (i) TUNE windows containing outlier weekends inflate $c(\tau)$ (§6.3.3): the 2024 TUNE anchor over-covers at $\tau = 0.95$ to 0.9754 because $c(0.95)$ inherits the BoJ-inflated tail. (ii) The deployed VIX quartile cutoff $q = 0.75$ (§7.4.8) is convention-anchored; three alternates $q \in \{0.60, 0.67, 0.80\}$ deliver 1.7–2.8% narrower bands at preserved calibration — the deployed value over-pays in width. (iii) Under a $+\Delta$ conditional-mean shift (§6.7), $\hat{\sigma}$ EWMA absorbs the bias as variance and over-deflates standardised scores on low- σ symbols, producing over-coverage with phase transition $\sigma^*(\Delta) \approx \Delta/(q \cdot c - 1)$. (iv) The §9.5.1 τ -stratified CUSUM bank fires an S^- (over-coverage) alarm at 2024-09-27, two months after BoJ — $c(\tau = 0.85)$ inherits the BoJ-inflated tail and over-covers in the calm recovery. (v) Path-fitted scoring at lower $\tau \in \{0.68, 0.85\}$ introduces over-coverage that DQ catches as violation clustering (§10.1).

The mechanism is structural: $\hat{\sigma}$ EWMA reads unexplained residual energy as scale and inflates bands; `fit_c_bump_schedule` grid-searches the smallest bump achieving nominal coverage but cannot shrink; the regime cutoff is set by convention rather than width-optimisation. The deployed architecture’s asymmetry — failing toward sharpness deficit rather than coverage-contract violation — is a design property, not an accident, and the kind of asymmetry a calibration-transparent oracle servicing collateral protocols should exhibit by construction. A protocol consuming Soothsayer for liquidation gating cares overwhelmingly about under-coverage (bands that fail to cover the realised price, triggering avoidable bad-debt liquidations); over-coverage costs only marginal capital efficiency. Coverage transparency is a one-sided contract, and the deployed architecture’s failure modes respect that asymmetry.

These limitations are disclosures, not retractions. Each has a concrete follow-up in §10.

10 Extensions

The limitations of §9 are disclosures, not retractions. This section names concrete extensions that build on the calibration-transparency primitive validated above, in three categories: methodology extensions that wait on data accumulation (§10.1), methodology extensions that wait on engineering effort (§10.2), and empirical-comparator and live-deployment work (§10.3). The full operational roadmap with quarter-grain triggers lives in `docs/ROADMAP.md` and `reports/methodology_history.md`.

10.1 Data-gated methodology extensions

On-chain xStock TWAP forecaster. Cong et al. show on-chain tokenised-equity prices anticipate Monday opens; the base forecaster does not consume this signal because the Solana tokenised-equity launch (mid-2025) leaves only ~30 weekends of post-launch history, well short of the ~150 weekends needed for stable per-(symbol, regime) Kupiec validation. The V5 forward-cursor tape (continuous capture since 2026-04-24) is the data source. ETA Q3–Q4 2026. Deliverable: an additional candidate forecaster `F_tok` and a re-derivation of §6 results admitting it into the candidate pool.

Forward-tape rolling rebuild. The deployed artefact is a one-shot fit at the 2023-01-01 split anchor; in production the per-regime quantile and $c(\tau)$ schedule will drift as the calibration target ages. The forward-tape harness (§6.6) is the operational scaffolding for this: a content-addressed frozen artefact is evaluated weekly against newly-arriving weekends, with the variant comparison sibling (§7.4.6) re-validating the choice on truly held-out data. The rolling-rebuild upgrade adds a quarterly (or trigger-based) re-roll of the freeze: re-run `scripts/build_lwc_artefact.py` against an updated panel, re-freeze, and write a versioned artefact for hot-reload. Trigger: forward-tape harness flags drift, or a scheduled cadence aligned with the live-deployment window (§10.3). Deliverable: a §6 schedule-trajectory figure across ≥ 6 rebuilds.

Path-fitted conformity score (library-grade, -conditional). §6.5 establishes the path-fitted methodology and a small- N first read on the endpoint-vs-path gap; the $\tau = 0.95$ raw gap is 16.1 pp on the perp-listed subset, decomposing after three robustness checks to a residual ~ 9 –15 pp. Methodology is library-grade and unit-tested (`soothsayer.backtest.calibration.compute_score_lwc_path`; 7 unit tests in `tests/test_path_fitted_score.py`, all pass). An empirical first-cut on the CME-projected subset ($n = 1,557$ OOS rows) confirms mechanical correctness (`reports/tables/m6_lwc_robustness_path_fitted.csv`). **The architectural recommendation is -conditional:** at $\tau \geq 0.95$, where path coverage is the operational concern, the cluster-conditional + path-fitted combination is empirically calibrated (DQ $p = 0.912$ at $\tau = 0.95$ on the equity cluster + CME-projected subset) and provides path coverage as an operational primitive; at $\tau < 0.95$, endpoint scoring on the unpartialled residual remains the dominant test (path extrema introduce within-symbol temporal autocorrelation in the violation series that DQ rejects at body-of-distribution τ ; `baseline_path` rejects DQ $p = 0.000$ at $\tau \in \{0.68, 0.85\}$).

regardless of partial-out — `reports/tables/paper1_f3_cluster_path_fitted.csv, ..._per_symbol.csv`). The binding empirical question — does path-fitting close the \$6.5 perp / on-chain residual gap on consumer-experienced path data — accumulates as the V5 forward-cursor tape (continuous capture since 2026-04-24) crosses $N \geq 300$ path-coverage weekends. Continuous-consumption AMM use cases require this path-fitted variant rather than the endpoint conformity score; that extension is sketched here and not pursued empirically. An MEV-conditional variant requires Solana mempool-or-bundle granularity to reconstruct the price a consumer would have *executed* at near band-edge events; ETA Q3 2026 once Jito bundle data accumulates ≥ 3 months.

10.2 Engineering-gated methodology extensions

Full-distribution conformal calibration. The pooled Berkowitz LR on the deployed PITs and the DQ rejection at $\tau = 0.95$ remain the deepest open methodological items. The deployed architecture *partially* realises a conformal upgrade — the per-regime quantile is the standard Mondrian split-conformal under exchangeability within a regime, the standardisation is per-symbol pre-Friday, and the $c(\tau)$ bump is a near-identity OOS-fit conservatism at three of four anchors. What remains is a *full-distribution* conformal: rather than four discrete anchors each calibrated by their own $c(\tau)$, fit a single conformal correction over the entire residual CDF that delivers exchangeability-based finite-sample coverage at every τ simultaneously. Romano-Patterson-Candès CQR [Romano et al., 2019] is the closest match. Deliverable: a §6 / §7 update reporting Berkowitz pass on next-generation walk-forward PITs at the cost of wider per-anchor bands.

Cluster-conditional + path-fitted architecture (-conditional). The §6.3.6 / §9.4 residual rejection localises to a within-weekend cluster topology — equity (8 symbols, $\hat{\rho}_{\text{within}} = 0.477$) vs safe-haven (GLD/TLT, $\hat{\rho}_{\text{within}} \approx 0$). A1.5’s cluster-internal partial-out within the equity cluster reduces $\hat{\rho}_{\text{within}}$ from 0.477 to 0.077, preserves per-symbol Kupiec 8/8, and tightens half-width -11.6% at $\tau = 0.95$ (`reports/tables/paper1_a1_5_cluster_internal_partial_out.csv`). Composed with the path-fitted conformity score (library-grade, unit-tested per `compute_score_lwc_path` — §10.1), the recommendation is **-conditional**: at $\tau \geq 0.95$, cluster-conditional + path-fitted is empirically calibrated (DQ $p = 0.912$ at $\tau = 0.95$ on the equity cluster + CME-projected subset) and provides path coverage as an operational primitive; at $\tau < 0.95$, endpoint scoring on the unpartialled residual remains the dominant test (path extrema introduce within-symbol temporal autocorrelation in the violation series that DQ catches at body-of-distribution τ ; `baseline_path` rejects DQ $p = 0.000$ at $\tau \in \{0.68, 0.85\}$ even without partial-out — `reports/tables/paper1_f3_cluster_path_fitted.csv`). The architectural recommendation is bounded, not blanket. A Friday-observable cross-sectional factor $\bar{r}_w$ predictor is a separate avenue (W8 prototype rejected the predictor at OOS $R^2 < 0.4$ on the available feature set; `reports/methodology_history.md` 2026-05-03 W8 entry); shipping is deferred until either a Sunday-Globex republish architecture or a future on-chain xStock TWAP feed (§10.1) delivers the signal. The safe-haven cluster ($n = 2$) does not need partial-out treatment: $\hat{\rho}_{\text{within}} \approx 0$ in baseline.

Conditional EVT for the = 0.99 tail. The textbook ask at $\tau \rightarrow 1$ is peaks-over-threshold EVT (Pickands-Balkema-de Haan). An exploratory unconditional GPD trial (`reports/v1b_evt_pot_trial.md`) replaced the empirical-quantile estimator at $\tau \geq 0.95$ with a GPD fit on upper-decile exceedances; the per-symbol DQ rejection at $\tau = 0.99$ was bit-for-bit identical, identifying the miscalibration mechanism as GARCH-style conditional volatility persistence rather than tail-quantile noise. The technically-correct response is the McNeil-Frey conditional EVT framework (GARCH first, GPD on standardised innovations) [McNeil and Frey, 2000].

Sub-regime granularity (empirically refuted, not on the next-generation path). A 5-cell sub-split using VIX tertiles within normal (`reports/tables/m6_lwc_robustness_vol_tertile.csv`) leaves Berkowitz LR essentially unchanged (170 $\rightarrow$ 172) while widening the $\tau = 0.95$ band by 9% — pure cost. Recorded here so the candidate is not re-suggested.

Block-correlation / two-factor t-copula refinement of §6.3.4. The slight over-dispersion of the homogeneous t-copula vs empirical (emp/t variance 0.63 at $\tau = 0.95$) reflects $\hat{R}$ averaging over heterogeneous correlations — equity-equity correlations are higher than equity-safe-haven correlations (which are negative). A block-correlation or two-factor t-copula would fit better but is a §10 mention only: the cluster topology from §9.4 plus the homogeneous t-copula bracket from §6.3.4 are sufficient for the present paper.

10.3 Empirical comparators and live deployment

Live-deployment OOS validation (closes the residual of §9.5). The forward-tape harness (§6.6) carries deployed-artefact evaluation against truly held-out weekends — the cleanest possible held-out coverage statement. What it does not do is consume the band as a live integrator-experienced price feed under adversarial conditions. The Rust + Anchor serving stack and the wire-format rollout state are documented in §8. A live window — six months of

devnet-then-mainnet operation against a real consumer — would expose any failure mode requiring production realism (rate-limit edges, signer-rotation events, mempool-ordering near band edges). Output: a §6 appendix-style update reporting realised coverage on the live window against the same Kupiec + Christoffersen + DQ + Berkowitz battery.

The downstream decision-theoretic question — given a calibrated band, what should a lending protocol do? — is out of scope here and would need an explicit borrower-book LTV distribution, a protocol-specific cost model, and a declared semantics for the correct action under realised prices (§9.7). Multi-region replication (LSE, JP equities), additional RWA classes (treasuries, commodities, credit), and the per-symbol sample-size eligibility framework are tracked in `docs/ROADMAP.md`.

10.4 Architectural extensions

The §7.4 selection procedure considered five variants under a pre-registered three-gate criterion; finer half-life grids, alternative weight kernels, or hybrid rules are open extensions, gated on accumulating forward-tape evidence to re-validate any new selection on truly held-out data. The per-symbol-class Mondrian track (a 6-cell symbol-class taxonomy replacing the regime axis; better per-symbol Kupiec on the bake-off sample at -14.7% width) sits naturally as a per-class collateral-buffer primitive for downstream lending-policy work, not as a calibration-transparency replacement for the architecture validated here. The stacked LWC + symbol-class candidate (W9) was empirically refuted (Pareto-dominated by both individual variants; OOS $c(\tau)$ degenerates to identity).

Denser anchor grid. The deployed schedule is calibrated at four anchors $\mathcal{T}_{\text{anch}} = \{0.68, 0.85, 0.95, 0.99\}$; off-anchor τ is served by linear interpolation of $q_r(\tau)$ and $c(\tau)$ between the bracketing anchors (§4.6, §9.10). The non-monotonic $c(\tau) = \{1.000, 1.000, 1.079, 1.003\}$ schedule produces an interpolated multiplier (e.g. $c(0.90) \approx 1.040$, $c(0.97) \approx 1.041$) at points with no separately audited calibration evidence. Two extensions: (a) extend the calibration grid to e.g. $\mathcal{T}'_{\text{anch}} = \{0.68, 0.80, 0.85, 0.90, 0.95, 0.97, 0.99\}$ and re-run §6.3 / §6.4 at the additional anchors, lifting the audited surface from four points to seven; (b) replace linear interpolation with monotone-preserving interpolation (e.g. PCHIP) so the served c -curve cannot dip below an audited anchor between two higher-valued anchors. Either path closes the §9.10 disclosure.

11 Conclusion

This paper specified a calibration-transparent oracle primitive for tokenised real-world assets, evaluated it on 12 years of US-listed weekend data, and shipped a reference implementation. The contribution is the *interface* (consumer-selects-coverage with a per-read receipt) and the *deployable architecture* (locally-weighted Mondrian split-conformal under EWMA HL=8). The §7 ablation isolates three load-bearing components: (a) regime stratification, validated against the constant-buffer baseline; (b) per-symbol standardisation, validated against an unweighted Mondrian comparator; (c) the near-identity OOS $c(\tau)$ bump, of which only $c(0.95) = 1.079$ carries meaningful out-of-sample information. The §6.7 simulation study corroborates the architectural mechanism on synthetic data with known ground truth (the per-symbol bimodality reproduces under controlled DGPs and per-symbol standardisation closes it on every DGP tested); the §6.6 forward-tape harness against a content-addressed frozen artefact is operational and accumulating weekly.

11.1 The three properties, revisited

- **(P1) Auditability.** The deployment artefact (per-Friday parquet + 16-scalar JSON sidecar) is reconstructible from public data — Soothsayer consumes Stryer parquet for Yahoo daily prices, the per-symbol factor returns (ES / NQ / GC / ZN / BTC), the regime classifier inputs, and computes as a deterministic function of past Fridays. The per-read PricePoint exposes the eight-field receipt of §4.7, and the on-chain PriceUpdate PDA mirrors that receipt under the byte-for-byte parity contract of §8.5. A third party with the sidecar and the per-Friday artefact can reconstruct any served band from the receipt alone.
- **(P2) Conditional empirical coverage — held-out via leave-one-symbol-out CV.** The leading calibration claim is **10-fold LOSO CV at $\tau = 0.95$: realised 0.9497 ± 0.0128 across the held-out symbol distribution, all 10 bands pass Kupiec**. Each held-out symbol’s $c(\tau)$ is fit on the other nine symbols’ OOS slice and never sees the held-out symbol’s data, so the realised coverage is genuinely out-of-fit. On the 2023+ deployment-tuned slice (1,730 rows $\times$ 173 weekends) the served band at $\tau = 0.95$ realises 0.950 at 370.6 bps half-width; this is consistent with the deployment-tuned target but follows by construction (so the pooled Kupiec $p_{uc} = 0.956$ is not independent held-out evidence). Three orthogonal stability checks complement LOSO on the deployed-fit slice: per-symbol Kupiec (all 10 symbols pass at $\tau = 0.95$, per-symbol Berkowitz LR $\in [3.2, 16.7]$); four-anchor split-date sensitivity (Christoffersen passes every (split $\times$ τ) cell across {2021, 2022, 2023, 2024}); and four-year calendar sub-period stability (Kupiec $p \geq 0.054$ in every cell across {2023,

2024, 2025, 2026-YTD}). At $\tau = 0.99$ the band realises 0.990. Joint-tail behaviour for portfolio consumers is empirically characterised: $P(k_w \geq 3) = 4.62\%$ at $\tau = 0.95$ vs the binomial reference of 1.15%, with reserve-guidance threshold $k^* = 3$ stable across the 24-cell sub-period grid (0/24 stability rejections). Against parametric baselines on the same slice the deployed architecture carries 40/40 per-symbol Kupiec passes versus GARCH- t 's 31/40 — Kupiec at every (symbol $\times \tau$) cell. The empirical content of P2 remains *per-anchor calibration*, not full-distribution calibration (Berkowitz still rejects pooled — the residual is cross-sectional within-weekend common-mode, orthogonal to §9.4); the §6.3.4 joint-tail measurement turns this disclosure into a measurable, reportable shape. The §6.6 forward-tape harness is operational ($N = 1$ at submission, preliminary by its own threshold of $N \geq 4$); it will become load-bearing held-out evidence as N matures.

- **(P3) Per-regime serving efficiency at deployment-tuned parameter budgets.** The deployment carries 16 trainable / tuned scalars: 12 trained per-regime quantiles + 4 OOS-fit $c(\tau)$ + 0 walk-forward (the $\delta(\tau)$ schedule is identically zero under per-symbol standardisation), with implicit hyperparameters (EWMA HL=8, warm-up minimum, regime cardinality, top-quartile VIX threshold, four served anchors, 2023-01-01 split) pinned a priori outside this count. Of the 4 $c(\tau)$ scalars, three are essentially identity ($c \in \{1.000, 1.000, 1.003\}$ at $\tau \in \{0.68, 0.85, 0.99\}$); only $c(0.95) = 1.079$ carries meaningful OOS information. **3 of 16 deployment scalars carry meaningful OOS-fit information.** The multiplier delivers per-symbol width allocation: heavy-tail equities receive bands matched to their relative-residual scale; the defensive class (GLD, TLT) receives narrower bands without losing coverage (§7.5).

11.2 What the ablation says is doing the work

The §7 ablation reduces to three load-bearing components plus the underlying point estimator:

1. **Factor-switchboard point estimator.** Tightens pooled bands $\sim 39\%$ vs a stale-hold reference; the input whose residual distribution the conformal quantile is fit on.
2. **Regime stratification.** §7.1 established that a deployable architecture without regime structure (the constant-buffer baseline) cannot deliver coverage on the 2023+ slice. The regime model concentrates width in high_vol weekends where it earns coverage and releases width in normal weekends without losing it.
3. **Per-symbol pre-Friday standardisation.** Adding a per-symbol $\hat{\sigma}_s(t)$ multiplier to the conformity score, with no other architectural change relative to an unweighted Mondrian comparator, takes per-symbol Kupiec at $\tau = 0.95$ from 2/10 $\rightarrow$ 10/10, collapses the Berkowitz LR range from 0.9–224 to 3.2–16.7, and tightens LOSO realised-coverage std by $5.9\times$. The §6.7 simulation study corroborates the mechanism — the per-symbol bimodality reproduces under four DGPs with known ground truth and standardisation closes it on every DGP tested.
4. **Near-identity OOS-fit $c(\tau)$ bump.** Meaningful only at $\tau = 0.95$ (a 7.9% widening); at the other three anchors essentially identity. The walk-forward $\delta(\tau)$ schedule is identically zero — per-symbol standardisation tightens cross-split realised-coverage variance enough that no structural-conservatism shift is required.

11.3 Honest framing

- **Per-anchor calibration, not full-distribution calibration.** Berkowitz and DQ both reject pooled. §6.3.6 / §9.4 localise the residual to *cross-sectional within-weekend* common-mode ($\hat{\rho}_{\text{cross}} = 0.354$), orthogonal to standardisation. §6.3.4's joint-tail empirical distribution turns the residual into a measurable reserve-guidance threshold ($k^* = 3$ at $\tau = 0.95$, stable across the 24-cell sub-period grid); the §10.2 cross-sectional partial-out track is the candidate methodology fix, gated on a Friday-observable predictor not yet available.
- **selection multi-test exposure (mitigated).** The §7.4 selection ran 80 split-date Christoffersen tests across 5 variants $\times$ 16 cells; under BH correction at FDR=0.05, no variant has any rejected cell. The deployment decision rests on Gate 3 (bootstrap CI on width — no multi-test issue, statistically significantly narrower at $\tau \in \{0.85, 0.95, 0.99\}$ at preserved calibration). The §6.6 / §7.4.6 forward-tape variant comparison harness is operational; at $N = 1$ it does not yet contribute held-out re-validation. We do not claim the rule is optimal among all locally-weighted variants.
- **OOS-fit $c(0.95) = 1.079$.** One scalar of the 16 is meaningfully OOS-fit. §9.3 reports the three independent provenance checks (4-split powered walk-forward, 4-anchor split-date sensitivity, 10-fold leave-one-symbol-out); all pass. The forward-tape harness is operational and accumulating; it does not yet add held-out re-validation at $N = 1$.
- **Shock-tertile floor.** Realised $\tau = 0.95$ coverage on the heaviest-move tertile is 87.8% — the residual is cross-sectional common-mode, not per-symbol scale, and does not yield to choice. The §6.3.4 reserve-guidance

threshold and the §6.3.5 worst-weekend vignette (2024-08-05 BoJ unwind) make this disclosure operationally addressable through a circuit-breaker handle.

- **Forward-tape harness operational; held-out evidence accumulating.** $N = 1$ at submission, preliminary by the harness’s own threshold ($N \geq 4$ to clear the preliminary banner; $N \geq 13$ for moderate forward-Christoffersen power). The harness will become load-bearing as N matures. A live integrator window remains an open external-validity gap.

The textbook econometric default — per-symbol GARCH(1,1) on log Friday-to-Monday returns — was extended to GARCH- t (the practitioner standard for fat-tailed returns) on the same OOS slice. GARCH- t closes the $\tau = 0.99$ Kupiec gap that GARCH-Gaussian leaves on the table but still under-covers Kupiec at every $\tau < 0.99$; at matched 95% realised coverage GARCH- t widens to ≈ 378 bps — statistically tied with the deployed architecture’s 370.6. Across the 40-cell (symbol $\times \tau$) grid the architecture carries 40/40 per-symbol Kupiec passes versus GARCH- t ’s 31/40. The conformal route fixes both Kupiec and Christoffersen because it is calibrated to the data, not to a fitted parametric distribution.

Each residual identified above carries a proposed methodological response in §10: full-distribution conformal calibration for the Berkowitz / DQ rejection, a Friday-observable cross-sectional partial-out for the within-weekend common-mode, McNeil-Frey conditional EVT for the $\tau \rightarrow 1$ tail, a path-fitted conformity score for continuous-consumption consumers, and a denser anchor grid for the off-anchor interpolation disclosure. They name the boundary of what a per-anchor-calibrated locally-weighted-Mondrian interval oracle can claim, with concrete extensions for what the next architecture would do differently. §6.6 is the held-out re-validation mechanism whose evidence accumulates as forward weekends arrive.

12 §A — Reproducibility appendix

This appendix consolidates everything needed to reproduce the §6 empirical results, the §7 ablation, and the §6.3 / §6.4 / §6.5 robustness diagnostics from the publicly-available source data plus the Soothsayer repo. The deployment is a five-line lookup against sixteen pre-fit scalars; reproducibility is determined by the panel construction, the schedule fits, and the serving formula — all of which are deterministic given the panel.

12.1 A.1 Algorithm boxes

We give pseudocode for three procedures: the M6 fit (deployment-artefact build), the M6 serve (runtime band lookup), and the strictly-pre-Friday per-symbol EWMA construction. Implementations: Python at `src/soothsayer/{backtest/calibration.py,oracle.py}`, Rust port at `crates/soothsayer-oracle/src/{config,oracle,types}.rs` (currently mirrors the M5 reference path; M6 wire-format slot reserved as `forecaster_code = 3`).

Algorithm 1 — M6 fit (one-shot, produces the 16 deployment scalars; `scripts/build_lwc_artefact.py`).

Input: panel $D = \{(s, t, p_{\text{Fri}}, \text{mon_open}, r_{\text{F}}, r) : i = 1, \dots, N\}$
 split date $d_{\text{split}} = 2023-01-01$
 anchor -grid $T = \{0.68, 0.85, 0.95, 0.99\}$
 c-grid $C = \{1.000, 1.001, \dots, 5.000\}$
 EWMA half-life $HL = 8$ weekends
 warm-up minimum $\text{past_obs_min} = 8$

Output: `sigma_hat_table` $_s(t)$ per-(symbol, fri_ts) parquet column
`quantile_table` $q_r()$ dim $3 \times 4 = 12$ scalars
`c_bump_schedule` $c()$ dim 4 scalars
`delta_shift_schedule` $()$ dim 4 scalars (identically zero under M6)

```
1. for each row  $i$   $D$ :
     $p_i \leftarrow p_{\text{Fri},i} \cdot (1 + r_{\text{F},i})$  # factor-adjusted point
     $r_{\text{rel},i} \leftarrow (\text{mon\_open}_i - p_i) / p_{\text{Fri},i}$  # signed relative residual
2. for each (symbol  $s$ , weekend  $t$ )  $D$ :
     $\text{prior} \leftarrow \{r_{\text{rel},i} : \text{symbol}(i) = s, t' < t\}$  # strictly pre-Friday
    if  $|\text{prior}| < \text{past\_obs\_min}$ :
        mark row as warmup_drop and skip
```

```

    else:
        _s(t) ← EWMA(prior, half_life = HL) # weekend half-life decay
3. D_train ← {i : D : t_i < d_split, not warmup_drop}
   D_oos   ← {i : D : t_i ≥ d_split, not warmup_drop}
4. for each row i in D_train ∪ D_oos:
    score_i ← |mon_open_i - p_i| / (p_Fri,i · _{s_i}(t_i)) # standardised
5. for each (regime r, target) ∈ {normal, long_weekend, high_vol} × T:
    S_r ← {score_i : i ∈ D_train, r_i = r}
    k ← ⌊(|S_r| + 1) / T⌋ # finite-sample CP rank
    q_r() ← sort(S_r)[k] # the trained standardised quantile
6. for each T:
    c() ← argmin_{c ∈ C} {c : (1/|D_oos|) · ∑_{i ∈ D_oos} 1{score_i ≥ q_{r_i}()} · c} }
7. () ← walk-forward sweep selects 0 at every anchor under M6 (per-symbol tightens
    cross-split realised-coverage variance enough that no shift is required)
8. return (_s(t), q_r(), c(), ())

```

Algorithm 2 — M6 serve (runtime band; Oracle.fair_value_lwc in src/soothsayer/oracle.py).

Input: symbol s , as-of date t , target coverage $[0.68, 0.99]$
 lookup row $(p_Fri, regime\ r, _s(t))$ from per-Friday parquet at (s, t)
 deployment scalars $(q_r(), c(), ())$ from Algorithm 1

Output: PricePoint{ , (), p , lower, upper, r , c , q_eff , q_r , }

```

1. c_eff ← interpolate c on the -grid at
2. q_eff ← c_eff · q_r() · _s(t) # standardised quantile rescaled by per-symbol
3. p ← p_Fri · (1 + r_F) # factor-adjusted point
4. lower ← p · (1 - q_eff); upper ← p · (1 + q_eff)
5. return PricePoint(, () = 0, p, lower, upper, r,
    diagnostics{c_eff, q_eff, q_r(), _s(t)})

```

Algorithm 3 — Per-symbol EWMA construction (scripts/build_lwc_artefact.py step 2; soft constants SIGMA_HAT_HL_WEEKENDS = 8, SIGMA_HAT_MIN = 8).

Input: panel D , EWMA half-life $HL = 8$ weekends, $past_obs_min = 8$

Output: per-(symbol, fri_ts) $_s(t)$ column on the artefact parquet

```

1. ← 0.5 ** (1.0 / HL) # 0.917 per past Friday
2. for each symbol  $s$  in  $D$ :
    rows_s ← rows of  $D$  with symbol =  $s$ , sorted by fri_ts
    for each row  $i$  in rows_s:
        prior ← {rrel_{j} : j ∈ rows_s, fri_ts(j) < fri_ts(i)} # strictly pre-Friday
        if |prior| < past_obs_min:
            _s(fri_ts(i)) ← undefined (row marked warmup_drop)
        else:
            weights ← [ ** (|prior| - k + 1) for k in 0..|prior| ]
            _s(fri_ts(i)) ← weighted_std(prior, weights)
3. return column

```

The rule itself was selected from a five-variant ladder under §7.3’s pre-registered three-gate criterion; alternative variants remain buildable for archival reproduction via scripts/build_lwc_artefact.py --variant {baseline_k26, ewma_hl6, ewma_hl12, blend_k26_ewma_hl8}.

12.2 A.2 Hyperparameter specification

The M6 deployment artefact is fully specified by the 16 scalars below plus four “soft” pipeline constants and the rule constants. All values are in data/processed/lwc_artefact_v1.json (the audit-trail sidecar) and hard-coded in src/soothsayer/oracle.py.

Trained per-regime quantiles $q_r(\tau)$ on standardised residuals (12 scalars, fit on the 4,186-row pre-2023 train set after the 80-row warm-up drop):

regime (n)	$\tau = 0.68$	$\tau = 0.85$	$\tau = 0.95$	$\tau = 0.99$
normal	0.7767	1.2195	1.9682	3.3280
long_weekend	0.8638	1.3649	2.2208	4.0152
high_vol	1.1086	1.9433	3.0971	6.4563

Note these are *standardised* quantiles (dimensionless; multiplied at serve time by $\hat{\sigma}_s(t)$ to yield a relative-residual quantile in basis points). The high_vol $\tau = 0.99$ value of 6.46 — a 6.46- shock at the panel-relative scale — is the headline-tail shape M6 is calibrated to absorb.

OOS-fit $c(\tau)$ **multiplicative bumps** (4 scalars, fit on the 1,730-row 2023+ OOS via Algorithm 1 step 6):

τ	0.68	0.85	0.95	0.99
$c(\tau)$	1.000	1.000	1.079	1.003

Three of four near-identity; only $c(0.95) = 1.079$ carries meaningful OOS information. §9.3 carries the full provenance disclosure for that one scalar.

Walk-forward $\delta(\tau)$ **shifts** (4 scalars; identically zero under M6 — per-symbol standardisation tightens cross-split realised-coverage variance enough that no shift is required):

τ	0.68	0.85	0.95	0.99
$\delta(\tau)$	0.000	0.000	0.000	0.000

The δ schedule is retained as a 4-zero vector in the artefact JSON for shape-compatibility with the receipt schema.

Soft constants (pipeline shape; not OOS-tuned):

Constant	Value	Source
Train/OOS split anchor	2023-01-01	§6.1, sensitivity in §6.3.3
Conformity score	$ y - \hat{p} / (p_{\text{Fri}} \cdot \hat{\sigma}_s)$	§4.3 (standardised)
Point estimator	$p_{\text{Fri}} \cdot (1 + r_t^F)$	§4.1 / §5.4 (factor switchboard)
Factor switchboard	per-symbol $\rho(s, t)$	§5.4, MSTR pivot 2020-08-01
Regime classifier	3-bin $\rho \in \{\text{normal}, \text{lw}, \text{hv}\}$	§5.5
c-grid resolution	$\Delta c = 0.001$	Algorithm 1 step 6
Off-grid τ interpolation	linear on the four anchors	§4.6
EWMA half-life	8 weekends	§4.2; SIGMA_HAT_HL_WEEKENDS
warm-up minimum past obs	8	§4.2; SIGMA_HAT_MIN

12.3 A.3 Data and code availability

Data. All upstream data is publicly available (Yahoo Finance daily bars, CME 1m futures, VIX/GVZ/MOVE, BTC-USD, Yahoo earnings calendar, Kraken Futures perp tape, Pyth Hermes archive, Chainlink Data Streams archive). Soothsayer consumes pre-fetched parquet from the sibling sryer repo at SCRYER_DATASET_ROOT. The decade-scale weekend panel data/processed/v1b_panel.parquet (5,996 rows $\times$ 35 columns, 2014-01-17 $\rightarrow$ 2026-04-24) is built by scripts/run_calibration.py calling soothsayer.backtest.panel.build(). All _fetched_at cutoffs are preserved in the parquet metadata.

Repository. Code at <https://github.com/ACNoonan/soothsayer>. Tag the experimental snapshot at submission time; this paper’s results were produced from the main branch as of 2026-05-05. License: see repository LICENSE file.

Layout.

Layer	Path	Purpose
Panel build	src/soothsayer/backtest/panel.py	weekend-pair assembly, regime tag, factor join
Calibration helpers	src/soothsayer/backtest/calibration.py	compute score, train_quantile_table, fit_c_bump_schedule, serve_bands, fit_split_conformal_forecaster
Metrics	src/soothsayer/backtest/metrics.py	Kupiec, Christoffersen, Berkowitz, DQ, CRPS, PIT
Python serve	src/soothsayer/oracle.py	Oracle.fair_value_lwc (M6 path); Oracle.fair_value (M5 reference)
Rust serve (parity)	crates/soothsayer-oracle/	byte-for-byte both forecasters (Forecaster::Mondrian, Forecaster::Lwc); 180/180 parity vs Python
On-chain publisher	programs/soothsayer-oracle-program/	on-chain program, PriceUpdate Borsh wire
M6 artefact build	scripts/build_lwc_artefact.py	Algorithm 1
M6 freeze	scripts/freeze_lwc_artefact.py	content-addressed freeze for forward-tape eval
M5 artefact build (reference)	scripts/build_mondrian_artefact.py	\$7.2 ablation rung
Cross-verification	scripts/verify_rust_oracle.py	180-case Python Rust parity probe (90 M5 + 90 M6 LWC)
Robustness battery	scripts/run_v1b_*.py, scripts/run_*.py	per-symbol diagnostics, vol-tertile, GARCH-N + GARCH-t, split sensitivity, LOSO, per-class, path-fitted
Phase 7 / 8 runners	scripts/run_portfolio_clustering.py, scripts/run_subperiod_robustness.py, scripts/run_v1b_garch_baseline.py, --dist t, scripts/run_per_symbol_kupiec_all_methods.py, scripts/run_kw_threshold_stability.py	regime tail clustering, sub-period stability, GARCH-t baseline, per-method per-symbol grid, k_w threshold stability
Figures	scripts/build_paper1_figures.py	produces all 6 paper figures

12.4 A.4 Compute environment

Component	Version	Notes
Python	3.12.13	enforced via pyproject.toml requires-python = ">=3.12,<3.13"
Environment manager Key deps (Python)	uv numpy 1.26, pandas 2.2, scipy 1.13, polars 0.20, pyarrow 16.0, statsmodels 0.14, arch (for §6.4.3 GARCH baselines)	lockfile uv.lock checked in full list in pyproject.toml
Rust Anchor OS	stable (1.81+) 0.30.x macOS Darwin 25.4 (development); deployment is platform-agnostic	matches Solana toolchain for the on-chain publisher

12.5 A.5 Reproducing the paper’s numerical content

The end-to-end pipeline from raw panel to all paper artefacts:

```
# 1. Build the weekend panel from scryer parquet (one-time, ~3 min).
uv run python scripts/run_calibration.py

# 2. Fit M6 and write the deployment artefact (Algorithm 1).
uv run python scripts/build_lwc_artefact.py
uv run python scripts/freeze_lwc_artefact.py          # content-addressed freeze for forward-tape

# 3. Verify Python Rust serving parity on the M5 reference path (90/90 cases).
uv run python scripts/verify_rust_oracle.py

# 4. §6.5 path-coverage diagnostic (CME + perp + on-chain).
uv run python scripts/run_path_coverage.py

# 5. §6 / §7 robustness battery.
uv run python scripts/run_vlb_per_symbol_diagnostics.py
uv run python scripts/run_vlb_vol_tertile.py
uv run python scripts/run_vlb_garch_baseline.py      # GARCH-Gaussian baseline
uv run python scripts/run_vlb_garch_baseline.py --dist t  # GARCH-t baseline (§6.4.3)
uv run python scripts/run_vlb_split_sensitivity.py
uv run python scripts/run_vlb_loso.py
uv run python scripts/run_vlb_per_class.py
uv run python scripts/run_vlb_path_fitted_conformal.py

# 6. Phase 7 / 8 paper-strengthening runners.
uv run python scripts/run_portfolio_clustering.py      # §6.3.4 joint-tail empirical distribut
uv run python scripts/run_subperiod_robustness.py      # §6.3.3 calendar sub-period stability
uv run python scripts/run_per_symbol_kupiec_all_methods.py  # §6.4.2 4-method per-symbol grid
uv run python scripts/run_kw_threshold_stability.py    # §6.3.4 reserve-guidance threshold sta

# 7. Build all six figures.
uv run python scripts/build_paper1_figures.py

# 8. Compile this paper.
cd reports/paper1_coverage_inversion/build && uv run python build.py --pdf
```

12.6 A.6 Per-figure data provenance

Figure	Script	Source CSVs / parquet
Fig. 1 (pipeline)	build_paper1_figures.py::fig1_pipeline	pipeline_diagrammatic
Fig. 2 (calibration)	build_paper1_figures.py::fig2_calibration	calibration/tables/{v1b_panel.parquet, lwc_artefact_v1.parquet}
Fig. 3 (stability)	build_paper1_figures.py::fig3_stability	stability/tables/{m6_lwc_walkforward.csv, m6_lwc_robustness_split_sensitivity.csv}
Fig. 4 (per-symbol)	build_paper1_figures.py::fig4_per_symbol	per_symbol/tables/m6_per_symbol_kupiec_4metho
Fig. 5 (Pareto)	build_paper1_figures.py::fig5_pareto	pareto/tables/{m6_pooled_oos.csv, m6_lwc_robustness_garch_t_baseline.csv}
Fig. 6 (path)	build_paper1_figures.py::fig6_path	path/tables/{path_coverage_perp.csv, path_coverage_perp_by_regime.csv}
Simulation appendix figure	scripts/run_simulation_study.py	reports/tables/sim_per_symbol_kupiec.csv

12.7 A.7 Determinism and randomness

The M6 fit and serve paths are deterministic given the panel: no random initialisation, no Monte Carlo, no bootstrap inside the fit. The 4-split powered walk-forward (§6.3.3, §7.2.3) uses fixed split fractions {0.4, 0.5, 0.6, 0.7} (the 0.2 / 0.3 fractions are excluded as under-powered for the 4-scalar $c(\tau)$ fit; see §4.5); the four split-date sensitivity anchors (§6.3.3) are fixed at {2021-01-01, 2022-01-01, 2023-01-01, 2024-01-01}; LOso (§6.3.3) iterates the ten symbols in lexicographic order. The block-bootstrap CIs reported in §6.3.4, §7.1.2, §7.3.5 use NumPy’s default RNG with seed 0

over 1,000 weekend-block resamples. The simulation study (§6.7) uses NumPy’s default RNG with seed 0 over 100 replications per DGP.

The GARCH(1,1) baselines (§6.4.3) are fit per-symbol via `arch_model(..., dist={"normal", "t"}).fit(dispatch="off")`; the optimisation is deterministic up to BFGS tolerance (default `tol=1e-8`). Under GARCH-*t*, NVDA hits the variance-undefined boundary at $\hat{\nu} = 2.50$ and falls back to Gaussian; the fallback is recorded in the `dist_used` column of the output CSV and is deterministic across reruns.

12.8 A.8 Independent verification (Python Rust on-chain)

The deployed serving stack has three independent implementations of both forecasters that must agree byte-for-byte:

1. **Python reference.** `src/soothsayer/oracle.py::Oracle.fair_value_lwc` (M6 deployed path); `Oracle.fair_value` (M5 reference path). Reads data/processed/lwc_artefact_v1.parquet (M6) or data/processed/mondrian_artefact_v2.parquet (M5) for the per-Friday (symbol, fri_close, point, regime_pub, sigma_hat_sym_pre_fri) tuple; the M6 16 scalars (M5 20 scalars) are hard-coded module constants.
2. **Rust serving crate.** `crates/soothsayer-oracle/src/oracle.rs::Oracle::fair_value` exposes both forecasters via `Oracle::load_with_forecaster(&path, Forecaster::{Mondrian, Lwc})`. The 16 LWC scalars and 20 Mondrian scalars are hard-coded in `config.rs`; the unit test `lwc_constants_match_sidecar` enforces 2 ULP agreement with `lwc_artefact_v1.json` on every refit.
3. **On-chain publisher.** `programs/soothsayer-oracle-program/`. The Rust crate is consumed by an Anchor program that emits PriceUpdate PDAs whose Borsh layout is preserved across the M5 → M6 migration. The wire forecaster_code byte distinguishes paths: code 2 = FORECASTER_MONDRIAN (M5; live on-chain); code 3 = FORECASTER_LWC (M6; live in the Rust publisher as of the §8.5 parity milestone, on-chain enablement gated on the next publisher release). The `soothsayer-consumer no_std` crate decodes the PDA into a typed `Forecaster::{Mondrian, Lwc}` view; consumers branching on `forecaster_code` need only add code-2 and code-3 cases.

`scripts/verify_rust_oracle.py` runs a dual-forecaster probe across 30 (symbol, fri_ts) cases × 3 target-coverage anchors per forecaster (90 cases per side, 180 total) and asserts byte-exact agreement on (point, lower, upper, sharpness_bps, half_width_bps, claimed_served) between Python and Rust on every case. **Submission status: 180/180 pass** (90/90 M5 + 90/90 M6). The Anchor integration test (`programs/soothsayer-oracle-program/tests/`) extends the parity check to path 3 by decoding the on-chain PDA after a publish call.

A reviewer reproducing the paper end-to-end thus has three independent implementations of each forecaster that must agree on every (symbol, fri_ts, target_coverage) read.

References

- S. Allen, J. Koh, J. Segers, and J. Ziegel. Tail calibration of probabilistic forecasts. *Journal of the American Statistical Association* 120(552), 2796–2808 (published online December 22, 2025), 2025. URL <https://doi.org/10.1080/01621459.2025.2506194>; arXiv:2407.03167.
- G. Angeris and T. Chitra. Improved price oracles: constant function market makers. *Proceedings of the 2nd ACM Conference on Advances in Financial Technologies (AFT ’20)*; arXiv:2003.10001, 2020. URL <https://arxiv.org/abs/2003.10001>.
- R. F. Barber, E. J. Candès, A. Ramdas, and R. J. Tibshirani. Conformal prediction beyond exchangeability. *Annals of Statistics* 51(2), 816–845, 2023. URL <https://doi.org/10.1214/23-AOS2276>; arXiv:2202.13415.
- M. J. Barclay and T. Hendershott. Price discovery and trading after hours. *Review of Financial Studies* 16(4), 1041–1073, 2003. URL <https://doi.org/10.1093/rfs/hhg030>.
- M. J. Barclay and T. Hendershott. Liquidity externalities and adverse selection: evidence from trading after hours. *Journal of Finance* 59(2), 681–710, 2004. URL <https://doi.org/10.1111/j.1540-6261.2004.00646.x>.
- J. Berkowitz. Testing density forecasts, with applications to risk management. *Journal of Business & Economic Statistics* 19(4), 465–474, 2001. URL <https://doi.org/10.1198/07350010152596718>.
- FDIC Board of Governors of the Federal Reserve System, OCC. Revised guidance on model risk management. sr letter 26-2 / occ bulletin 2026-13. Federal Reserve / OCC / FDIC supervisory guidance (April 17, 2026), 2026. URL <https://www.federalreserve.gov/supervisionreg/srletters/SR2602.htm>.

- L. Breidenbach, C. Cachin, B. Chan, A. Coventry, S. Ellis, A. Juels, F. Koushanfar, A. Miller, B. Magauran, D. Moroz, S. Nazarov, A. Topliceanu, F. Tramèr, and F. Zhang. Chainlink 2.0: next steps in the evolution of decentralized oracle networks. Chainlink research whitepaper (April 2021), 2021. URL <https://research.chain.link/whitepaper-v2.pdf>.
- G. W Brier. Verification of forecasts expressed in terms of probability. *Monthly Weather Review* 78(1), 1–3, 1950. URL [https://doi.org/10.1175/1520-0493\(1950\)078<0001:V0FEIT>2.0.CO;2](https://doi.org/10.1175/1520-0493(1950)078<0001:V0FEIT>2.0.CO;2).
- P. F Christoffersen. Evaluating interval forecasts. *International Economic Review* 39(4), 841–862, 1998. URL <https://doi.org/10.2307/2527341>.
- L. W. Cong, W. R. Landsman, D. Rabetti, C. Zhang, and W. Zhao. Tokenized stocks. Working paper, SSRN id 5937314 (December 2025), 2025. URL https://papers.ssrn.com/sol3/papers.cfm?abstract_id=5937314.
- P. Daian, S. Goldfeder, T. Kell, Y. Li, X. Zhao, I. Bentov, L. Breidenbach, and A. Juels. Flash boys 2.0: Frontrunning, transaction reordering, and consensus instability in decentralized exchanges. *IEEE Symposium on Security and Privacy* 2020; arXiv:1904.05234, 2020. URL <https://arxiv.org/abs/1904.05234>.
- F. X. Diebold, T. A. Gunther, and A. S. Tay. Evaluating density forecasts with applications to financial risk management. *International Economic Review* 39(4), 863–883, 1998. URL <https://doi.org/10.2307/2527342>.
- C. Duley, L. Gambacorta, R. Garratt, and P. Koo Wilkens. The oracle problem and the future of defi. BIS Bulletin No. 76, Bank for International Settlements (September 2023), 2023. URL <https://www.bis.org/publ/bisbull76.pdf>.
- S. Eskandari, M. Salehi, W. C. Gu, and J. Clark. Sok: Oracles from the ground truth to market manipulation. 3rd ACM Conference on Advances in Financial Technologies (AFT '21); arXiv:2106.00667, 2021. URL <https://arxiv.org/abs/2106.00667>.
- K. R French. Stock returns and the weekend effect. *Journal of Financial Economics* 8(1), 55–69, 1980. URL [https://doi.org/10.1016/0304-405X\(80\)90021-5](https://doi.org/10.1016/0304-405X(80)90021-5).
- T. Gneiting, F. Balabdaoui, and A. E. Raftery. Probabilistic forecasts, calibration and sharpness. *Journal of the Royal Statistical Society, Series B* 69(2), 243–268, 2007. URL <https://doi.org/10.1111/j.1467-9868.2007.00587.x>.
- J. Gonzalo and C. W. J. Granger. Estimation of common long-memory components in cointegrated systems. *Journal of Business & Economic Statistics* 13(1), 27–35, 1995. URL <https://doi.org/10.1080/07350015.1995.10524576>.
- J. Hasbrouck. One security, many markets: determining the contributions to price discovery. *Journal of Finance* 50(4), 1175–1199, 1995. URL <https://doi.org/10.1111/j.1540-6261.1995.tb04054.x>.
- J. Hasbrouck. Intraday price formation in u.s. equity index markets. *Journal of Finance* 58(6), 2375–2400, 2003. URL <https://doi.org/10.1046/j.1540-6261.2003.00609.x>.
- P. H. Kupiec. Techniques for verifying the accuracy of risk measurement models. *Journal of Derivatives* 3(2), 73–84, 1995. URL <https://doi.org/10.3905/jod.1995.407942>.
- Chainlink Labs. –2026. chainlink data streams. Chainlink product documentation, 2024. URL <https://docs.chain.link/data-streams>.
- Chainlink Labs. Data streams v10 ("tokenized asset") report schema. Chainlink product documentation; SDK source (authoritative pin), 2025. URL <https://docs.chain.link/data-streams/reference/report-schema-v10>.
- Chainlink Labs. Data streams v11 ("rwa advanced") report schema. Chainlink product documentation; SDK source (authoritative pin), 2026. URL <https://docs.chain.link/data-streams/reference/report-schema-v11>.
- A. Lehar and C. A. Parlour. Decentralized exchange: the uniswap automated market maker. *Journal of Finance* 80(1), 321–374, 2025. URL <https://doi.org/10.1111/jofi.13405>.
- A. C. MacKinlay and K. Ramaswamy. Index-futures arbitrage and the behavior of stock index futures prices. *Review of Financial Studies* 1(2), 137–158, 1988. URL <https://doi.org/10.1093/rfs/1.2.137>.
- A. Madhavan and A. Sobczyk. Price dynamics and liquidity of exchange-traded funds. *Journal of Investment Management* 14(2), 88–102, 2016. URL https://papers.ssrn.com/sol3/papers.cfm?abstract_id=2429509.
- I. Makarov and A. Schoar. Trading and arbitrage in cryptocurrency markets. *Journal of Financial Economics* 135(2), 293–319, 2020. URL <https://doi.org/10.1016/j.jfineco.2019.07.001>.
- A. J. McNeil and R. Frey. Estimation of tail-related risk measures for heteroscedastic financial time series: An extreme value approach. *Journal of Empirical Finance* 7(3-4), 271–300, 2000. URL [https://doi.org/10.1016/S0927-5398\(00\)00012-8](https://doi.org/10.1016/S0927-5398(00)00012-8).

- Pyth Network. A proposal for price feed aggregation. Pyth Network blog / engineering post (documenting the deployed aggregation), 2021. URL <https://www.pyth.network/blog/pyth-price-aggregation-proposal>.
- Pyth Network. Pyth primer: don't be pretty confident, be pyth confident. Pyth Network blog, 2022. URL <https://www.pyth.network/blog/pyth-primer-dont-be-pretty-confident-be-pyth-confident>.
- Pyth Network. -09-25. blue ocean ats joins pyth network: institutional overnight hours us equity data. Pyth Network blog (companion announcement on Blue Ocean's site), 2025. URL <https://www.pyth.network/blog/blue-ocean-ats-joins-pyth-network-institutional-overnight-hours-us-equity-data>; <https://blueocean-tech.io/2025/09/25/pyth-blue-ocean-ats-joins-pyth-network-institutional-overnight-hours-us-equity-data>.
- Pyth Network. Pyth pro (formerly pyth lazer): enterprise-grade price data with customizable cadence. Pyth Network product documentation, 2026. URL <https://docs.pyth.network/lazer> (accessed 2026-04-29; pagereframestheproductas"PythPro(formerlyPythLazer)").
- Board of Governors of the Federal Reserve System & Office of the Comptroller of the Currency. Supervisory guidance on model risk management. sr letter 11-7 / occ bulletin 2011-12. Federal Reserve / OCC supervisory guidance, 2011. URL <https://www.federalreserve.gov/supervisionreg/srletters/sr1107.htm>.
- Basel Committee on Banking Supervision. Supervisory framework for the use of "backtesting" in conjunction with the internal models approach to market risk capital requirements. BCBS Publication 22 (January 1996), 1996. URL <https://www.bis.org/publ/bcbs22.pdf>.
- UMA Project. Uma: an optimistic oracle with a data verification mechanism. UMA project documentation / blog post *Introducing UMA's Optimistic Oracle*, 2020. URL <https://medium.com/uma-project/introducing-umas-optimistic-oracle-d92ce5d1a4bc>.
- RedStone. Redstone live: real-time market data for 24/7 markets. RedStone blog / product documentation (March 2026), 2026. URL <https://blog.redstone.finance/2026/03/30/redstone-live-real-time-data-built-for-the-markets-that-never-sleep/>.
- Y. Romano, E. Patterson, and E. J. Candès. Conformalized quantile regression. *Advances in Neural Information Processing Systems* 32 (NeurIPS 2019), 2019. URL <https://proceedings.neurips.cc/paper/2019/hash/5103c3584b063c431bd1268e9b5e76fb-Abstract.html>; arXiv:1905.03222.
- H. R. Stoll and R. E. Whaley. The dynamics of stock index and stock index futures returns. *Journal of Financial and Quantitative Analysis* 25(4), 441–468, 1990. URL <https://doi.org/10.2307/2331010>.
- Switchboard. Switchboard v3 documentation. Switchboard product documentation / Breakpoint 2023 presentation, 2023. URL <https://docs.switchboard.xyz/docs/switchboard/readme/designing-feeds/oracle-aggregator>.
- Tellor. Tellor whitepaper. Tellor project whitepaper (April 2023 revision), 2023. URL https://tellor.io/wp-content/uploads/2023/04/Tellor-Whitepaper_4_2023.pdf.
- V. Vovk, D. Lindsay, I. Nourtdinov, and A. Gammerman. Mondrian confidence machine. On-Line Compression Modelling project working paper, 2003. URL <http://www.alrw.net/old/04.pdf> (canonical introduction; alsodiscussedinvovk-book-2005-4.5).
- V. Vovk, A. Gammerman, and G. Shafer. *Algorithmic learning in a random world*. Springer, Boston, MA; ISBN 978-0-387-00152-4, 2005. URL <https://doi.org/10.1007/b106715>.
- C. Xu and Y. Xie. Conformal prediction interval for dynamic time-series (enbpi). *Proceedings of the 38th International Conference on Machine Learning (ICML 2021)*, PMLR 139, 2021. URL <https://proceedings.mlr.press/v139/xu21h.html>.